

**ANALYTICAL SKILLS & COMPETENCY COURSE · SEMESTER III ·
SCHEME 2025**

Complete Course Notes with Tutorial Practice Sets

PROBABILITY, DISTRIBUTIONS & STATISTICS

5 Modules in Full · 14 Tutorial Sheets Solved · Every Number
Computed and Captured · Checkpoint & Practice Keys ·
Computed Statistical Tables

COURSE CODE
1BMATCS301

Credits	Teaching Hours (L : T : P)	Total Marks	CIE	SEE	Exam Hours	Examination Type
4	3 : 2 : 0 / week	100	50	50	03	Theory

compiled and computed by

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Visvesvaraya Technological University (VTU), Belagavi · Scheme 2025 · ASC
(syllabus text from the official VTU curriculum document for Semester III-VIII
common courses)

First Edition · September 2026 · all computations reproduced from the scripts
listed in Appendix D

PUBLICATION PARTICULARS

Course at a Glance

Field	Details
Course	Probability, Distributions and Statistics
Course Code	1BMATCS301
Scheme · Type	2025 · ASC (Analytical Skills & Competency Course)
Semester	III
Credits	4
Teaching Hours/Week (L:T:P)	3 : 2 : 0
Total Hours of Pedagogy per semester (L : T : P : SL&TW : TH)	42 : 28 : 0 : 50 : 120
CIE · SEE · Total	50 · 50 · 100
Examination type (SEE)	Theory Exam · 03 hours
Tutorial component of PCC	14 topics × 2 hours = 28 hours (listed verbatim in the syllabus section)
Course Outcomes	CO1 – CO5 (verbatim reproduction in the Syllabus section)

What This Book Covers

Every syllabus line of the five modules is written out as a study notes chapter — the rule or formula, why it works, and solved examples whose arithmetic was executed rather than typed: fitted lines, ranks, areas under the normal curve, stationary distributions and test statistics all come from Python scripts (SymPy for exact algebra, SciPy for distribution values) whose complete listings and captured output appear in each chapter. The tutorial component — all fourteen practice topics — is delivered as fourteen tutorial sheets: formula review, four fully-solved problems each, and a practice set whose answers were likewise computed and then hidden behind short keys you can check yourself against the computation log.

Part A — Modules 1 to 5 (theory, 42 hours): modular arithmetic, statistics, probability distributions, joint distributions & Markov chains, statistical inference.

Part B — Tutorial Sheets 1 to 14 (28 hours): the verbatim tutorial list of the syllabus, one sheet per line, with the exact problem types the syllabus asks to practise.

Part C — Appendices: A formula sheet for revision, B statistical tables generated for this book, C examination preparation (IA/CCA blueprint and a GATE-style corner), D references, provenance and reproduction guide.

Provenance of the Numbers

The numbers in this book were produced the same way the companion programming-course books produced their program outputs: by running code. Twenty computation scripts (five module scripts and fourteen tutorial scripts, plus the table generator) were executed in the build sandbox; their stdout was captured verbatim as the computation logs printed at the end of each chapter and sheet, and every figure quoted in a worked solution is a token filled in from that run. If the printed 1.2143 ever disagreed with its script, the build would fail — the text is assembled from the computation, not re-typed from it.

Mangaluru · September 2026

PREFACE

How to Use This Book

The format is the record. Each module chapter opens with the syllabus text quoted verbatim in a red-bordered “As prescribed” panel, so the prescribed wording and the notes are never more than a page apart. Sections then follow a fixed rhythm: Concepts in Focus (definitions, theorems, derivations), Worked Examples (the exam-style solves, with tables built from the same data the log prints), Computation Log (the captured script run behind that chapter’s numbers), Quick-Reference Summary, Checkpoint Questions (short questions with computed answers printed beneath, exam-style), and Practice Problems for self-testing.

The tutorials mirror the PCC tutorial list. Tutorial sheet n is the prescribed practice topic n , one two-hour session per sheet: what to review, four problems solved in full, one Python computation log that verifies the whole sheet, and a practice set with answer keys.

Notation and rendered formulas. Display mathematics is typeset from the formulas themselves (no hand-drawn approximations), Greek letters and bars ($\bar{x}$, $\hat{y}$, σ^2 , ρ , χ^2) appear in running text exactly as they do in papers. Where the syllabus writes $y = a + bx$, $y = a + bx + cx^2$, $y = ax^b$, the chapters keep that same three-form order so revision maps one-to-one onto the prescription.

Study route for the 3:2 timetable. Lectures: read the module chapter before class and re-read its Quick-Reference Summary after. Tutorials: bring the sheet, cover the solved part, attempt it, then compare with the computation log. In the last fortnight: the formula sheet (Appendix A), the tables (Appendix B — the same values a printed table gives, because they were computed from the same definitions), then Appendix C’s two IA mock papers.

Honesty rules of this series. Nothing is silently truncated: when a listing would not fit the page it is trimmed with a visible marker and the complete file is named. Quoted syllabus text is quoted, not paraphrased; where the university PDF itself breaks a line inside a cell, that wrap is marked. Answers in the keys are the answers the scripts produced.

— The author

THE PRESCRIPTION

Syllabus — Verbatim

Everything in the red “As prescribed” panels of Parts A and B is quoted from this section. Course-data table, five module paragraphs, the fourteen tutorial lines, the learning resources, the assessment rules: as printed.

Course information

COURSE	Probability, Distributions and Statistics	COURSE CODE	1BMATCS301
SCHEME	2025	TYPE OF COURSE	ASC
SEMESTER	3	TEACHING HOURS/WEEK (L:T:P)	3 : 2 : 0
TOTAL HOURS OF PEDAGOGY PER SEMESTER (L : T : P : SL&TW : TH)	42 : 28 : 0 : 50 : 120	CREDITS	4
CIE MARKS	50	SEE MARKS	50
TOTAL MARKS	100	EXAMINATION TYPE (SEE)	Theory Exam · Exam Hours 03

Course outcome (Course Skill Set)

“At the end of the course, the student will be able to:”

1. **CO1.** Apply the concepts of modular arithmetic, congruences, linear Diophantine equations, Euler’s theorem, Fermat’s theorem, Wilson’s theorem, and the RSA cryptographic algorithm to solve engineering and computational problems.
2. **CO2.** Apply least squares curve fitting, correlation, regression analysis, and rank correlation techniques to analyse engineering data and establish relationships between variables.
3. **CO3.** Solve engineering problems involving discrete and continuous random variables using probability distributions [— Binomial] Poisson, and Normal distributions. [Source cell wraps mid-sentence; the bracketed tail is the module-3 triplet — mark of a typographic note, not of new wording.]
4. **CO4.** Analyse joint probability distributions and Markov chains by determining expectations, covariance, correlation, transition probabilities, stationary distributions, and absorbing states for stochastic systems.
5. **CO5.** Apply statistical inference techniques including hypothesis testing, confidence intervals, large and [small-sample tests to draw valid] conclusions from engineering data. [Source cell wraps mid-sentence; bracketed words restore the wrap.]

Modules

MODULE-1 · MODULAR ARITHMETIC · NUMBER OF HOURS: 8

Introduction of modular arithmetic and its applications in Computer Science and Engineering. Introduction to Congruences, Linear Congruences, The Remainder theorem, Solving Polynomials, Linear Diophantine Equation, System of Linear Congruences, Euler’s Theorem, Wilson Theorem and Fermat’s little theorem. Applications of Congruences-RSA algorithm. [Textbook-3: Chapters 4, 5, 6, 7 and 9]

MODULE-2 · STATISTICS · NUMBER OF HOURS: 8

Principles of least squares, Curve fitting by the method of least squares in the form: $y = a + bx$, $y = a + bx + cx^2$, and $y = ax^b$. Correlation, Coefficient of correlation, Lines of regression, Angle between regression lines, rank correlation. [Textbook-1: Chapters 11 and 12, Textbook-2: Chapters 13 and 14]

MODULE-3 · PROBABILITY DISTRIBUTIONS · NUMBER OF HOURS: 9

Review of basic probability theory. Random variables (discrete and continuous), probability mass and density functions. Mathematical expectation: mean and variance. Binomial, Poisson and normal distributions-problems (derivations for mean and standard deviation for Binomial and Poisson distributions only). [Textbook-1: Chapter-3, 4, 5, 6, Textbook-2: Chapters 6 and 7]

MODULE-4 · JOINT PROBABILITY DISTRIBUTION · NUMBER OF HOURS: 8

Joint Probability distribution for two discrete random variables, expectation, covariance and correlation. Markov Chain: Introduction to Stochastic Process, Probability Vectors, Stochastic matrices, Regular stochastic matrices, Markov chains, Higher transition probabilities, Stationary distribution of Regular Markov chains. [Textbook-1: Chapter-3, 4, 5, 6, Textbook-4: Chapter-8]

MODULE-5 · STATISTICAL INFERENCE · NUMBER OF HOURS: 9

Sampling distribution, standard error. Levels of significance. Testing of hypothesis. Test of significances. Confidence limits, simple sampling of attributes. Sampling variables, central limit theorem and confidences limit for unknown mean.

Test of significance for large samples (mean, proportion). Comparison of large samples (mean, proportion).

Test of Significance for means of two small samples, students-'t' distribution, and Chi-square distribution as a test of goodness of fit. [Textbook-1: Chapter 8 and 10, Textbook-2: Chapters-8, 9, 10 and 12]

Module hours $8+8+9+8+9 = 42$ — exactly the L-hours of the pedagogy table.

Tutorial component of PCC

“Tutorial Topic/Problem solving/Case ... etc. (for 2 hours each) : Number of Hours: 28”

#	Prescribed practice topic	Where practised in this book
1	Practice problems on linear congruences	Tutorial sheet 1 · §1.3
2	Practice Problems on Linear Diophantine Equation, System of Linear Congruences	Sheet 2 · §1.4–1.5
3	Practice problems on Euler's Theorem, Wilson Theorem and Fermat's little theorem.	Sheet 3 · §1.6–1.8
4	Practice problems on fitting linear form and quadratic form	Sheet 4 · §2.1–2.2
5	Practice problems on $y = ax^b$ and correlation	Sheet 5 · §2.3, §2.5
6	Practice problems on lines of regression and rank correlation	Sheet 6 · §2.6–2.7
7	Practice problems on mathematical expectations	Sheet 7 · §3.3
8	Practice problems on Binomial and Poisson distributions	Sheet 8 · §3.4–3.5
9	Practice Problems on Normal distributions	Sheet 9 · §3.6
10	Practice problems on covariance, correlation of two discrete random variables	Sheet 10 · §4.1–4.2
11	Practice problems on Markov chain, stochastic matrices, probability vectors	Sheet 11 · §4.3–4.4
12	Practice Problems on higher transition probabilities and stationary distributions	Sheet 12 · §4.5–4.6
13	Practice Problems on test of significance for large samples	Sheet 13 · §5.2–5.3
14	Practice problems on test of significance for small samples	Sheet 14 · §5.4–5.5

Suggested learning resources (as printed)

Textbooks

1. Ronald E. Walpole, Raymond H Myers, Sharon L Myers & Keying Ye, “Probability and Statistics for Engineers and Scientists”, Pearson Education, 9th edition, 2012. [Textbook-1]
2. Murray R. Spiegel, Larry J. Stephens, Statistics, Schaum's outline series, 6th edition. [Textbook-2]
3. Koshy, Thomas. Elementary number theory with applications. Academic press, 2nd Edition, 2009. [Textbook-3]
4. Seymour Lipschutz and Marc Lars Lipson, Probability, Schaum's Outline Series, Second edition, 2011. [Textbook-4]

Reference books / Manuals

1. Erwin Kreyszig, “Advanced Engineering Mathematics”, John Wiley & Sons, 11th Edition, 2006.
2. B. S. Grewal, “Higher Engineering Mathematics”, Khanna publishers, 45th Ed., 2021.

3. Irwin Miller & Marylees Miller, John E. Freund's "Mathematical Statistics with Applications", Pearson · Dorling Kindersley Pvt. Ltd. India, 8th edition, 2014.
4. P. G. Hoel, S. C. Port and C. J. Stone, "Introduction to Probability Theory", Universal Book Stall, (Reprint).
5. S C Gupta and V K Kapoor, "Fundamentals of Mathematical Statistics", S Chand and Company, 12th edition.
6. S. Ross, "A First Course in Probability", Pearson Education India, 6th Ed., 2002.
7. David M Burton: "Elementary Number Theory", Mc Graw Hill, 7th Ed., 2017.

Web links and e-resources

1. Probability and statistics: <https://nptel.ac.in/courses/111105090>
2. Introduction to Probability and Statistics for Machine Learning (Class Central / CodeSignal).
3. Probability and Statistics: Class Central course listing; <https://probabilitycourse.com/> (introduction to probability, statistics and random processes).
4. MIT OCW 18.05 Introduction to Probability and Statistics (Spring 2022).
5. VTU e-Shikshana Program.
6. Probability, Distributions and Statistics: GeeksforGeeks mathematics collection.

Teaching-learning process (innovative delivery methods)

The syllabus offers "sample strategies that educators may adopt": **Flipped Classroom Approach** (materials before class, problem-solving in class); **Problem-Based Learning (PBL)**; **ICT-Enabled Teaching** (MATLAB, Python and online simulation tools — "may be conducted in Tutorial hours"); and **Collaborative Learning** (in Tutorial hours). [The source template's sub-bullets are shared with the transform-series math course and mention Fourier/Z-transform topics; they are noted here rather than reproduced as if course-specific.]

Assessment structure (as printed)

The assessment is divided equally between CIE and SEE, 50% each. To be eligible for the SEE, the CIE score must be at least 40% of 50 (i.e., 20 marks). For the SEE, 35% of 50 (i.e., 18 marks) is required. A student passes the course when $CIE + SEE \geq 40$ of 100.

CCA + Internal Assessment = CIE [25 + 25 = 50]. A minimum of two Internal Assessment tests (25 marks total); Continuous Comprehensive Assessment worth 25 marks, with at most two learning activities:

#	Learning activity	Max marks
1	Problem-solving assignment at RBTL-3/4	15
2	Solving GATE questions related to the syllabus content	10

Rubrics (both activities): Superior (5) / Good (4) / Fair (2) / Needs Improvement (1), scored on Understanding of Concepts, Problem-Solving Ability and Mathematical Accuracy for activity 1, and Analytical Thinking (plus a second row printed as "Excellent / Good / Moderate / Weak reasoning") for activity 2. The CIE note adds: first test after two modules, second after the remaining three.

Term-work / self-learning hours (TW & SL = 50 h): LA-1 assignment (10 h), LA-2 GATE questions (10 h), "MATLAB/Python computations for the problems related to RSA algorithms, Probability, Statistics and Sampling Distribution concepts" (10 h); self-study: RSA in cryptography via GeeksforGeeks (3 h), real-life applications of continuous distributions (2 h), applications of RSA/probability/statistics/sampling (10 h), industrial applications of statistics, probability, sampling distributions and congruences (5 h).

SEE pass rules as printed: for a pass in SEE, a minimum of 35 marks out of 100 is stated; for the declaration of a pass grade, CIE (out of 50) plus SEE scaled down to 50 must be ≥ 40 ; if the total is below 40 the grade is 'F' (Fail).

FRONT MATTER

List of Contents

Page numbers in roman (i-ix) are front matter; the body restarts at arabic 1 and runs continuously through the appendices — both match the printed footers and the PDF bookmarks.

Title Page	i
Course at a Glance	ii
Preface	iii
Prescribed Syllabus — Verbatim	iv
List of Contents	vii
PART A — THEORY · MODULES 1-5	1
Module 1 — Modular Arithmetic	2
1.1 Concepts, Theorems & Rules	2
1.2 Worked Problems (Machine-Verified)	4
1.3 Computation Log	5
1.4 Module Summary	6
1.5 Checkpoint	6
1.6 Practice Set	7
Module 2 — Statistics — Curve Fitting, Correlation and Regression	8
2.1 Concepts, Theorems & Rules	8
2.2 Worked Problems (Machine-Verified)	9
2.3 Computation Log	11
2.4 Module Summary	12
2.5 Checkpoint	13
2.6 Practice Set	13
Module 3 — Probability Distributions	14
3.1 Concepts, Theorems & Rules	14
3.2 Worked Problems (Machine-Verified)	15
3.3 Computation Log	17
3.4 Module Summary	18
3.5 Checkpoint	18
3.6 Practice Set	18
Module 4 — Joint Distributions and Markov Chains	19
4.1 Concepts, Theorems & Rules	19
4.2 Worked Problems (Machine-Verified)	20
4.3 Computation Log	21
4.4 Module Summary	22
4.5 Checkpoint	22
4.6 Practice Set	22
Module 5 — Statistical Inference — Tests of Significance	24
5.1 Concepts, Theorems & Rules	24
5.2 Worked Problems (Machine-Verified)	25
5.3 Computation Log	27
5.4 Module Summary	28
5.5 Checkpoint	28
5.6 Practice Set	28
PART B — TUTORIALS · SHEETS 1-14	30
Tutorial Sheet 1 — Linear Congruences	31
1.1 Objective & Formula Review	31
1.2 Solved in Full	31
1.3 Practice Set	31
1.4 Computation Log	31

Tutorial Sheet 2 — Linear Diophantine Equations & Systems of Congruences	33
2.1 Objective & Formula Review	33
2.2 Solved in Full	33
2.3 Practice Set	33
2.4 Computation Log	34
Tutorial Sheet 3 — Euler, Wilson and Fermat	35
3.1 Objective & Formula Review	35
3.2 Solved in Full	35
3.3 Practice Set	35
3.4 Computation Log	35
Tutorial Sheet 4 — Fitting the Linear and Quadratic Forms	37
4.1 Objective & Formula Review	37
4.2 Solved in Full	37
4.3 Practice Set	38
4.4 Computation Log	38
Tutorial Sheet 5 — The Power Curve and Correlation	39
5.1 Objective & Formula Review	39
5.2 Solved in Full	39
5.3 Practice Set	40
5.4 Computation Log	40
Tutorial Sheet 6 — Lines of Regression and Rank Correlation	41
6.1 Objective & Formula Review	41
6.2 Solved in Full	41
6.3 Practice Set	42
6.4 Computation Log	42
Tutorial Sheet 7 — Mathematical Expectations	43
7.1 Objective & Formula Review	43
7.2 Solved in Full	43
7.3 Practice Set	43
7.4 Computation Log	44
Tutorial Sheet 8 — Binomial and Poisson Distributions	45
8.1 Objective & Formula Review	45
8.2 Solved in Full	45
8.3 Practice Set	46
8.4 Computation Log	46
Tutorial Sheet 9 — Normal Distributions	47
9.1 Objective & Formula Review	47
9.2 Solved in Full	47
9.3 Practice Set	47
9.4 Computation Log	47
Tutorial Sheet 10 — Covariance and Correlation of Two Discrete RVs	49
10.1 Objective & Formula Review	49
10.2 Solved in Full	49
10.3 Practice Set	49
10.4 Computation Log	50
Tutorial Sheet 11 — Markov Chains, Stochastic Matrices, Probability Vectors	51
11.1 Objective & Formula Review	51
11.2 Solved in Full	51
11.3 Practice Set	51
11.4 Computation Log	52
Tutorial Sheet 12 — Higher Transition Probabilities and Stationary Distributions	53
12.1 Objective & Formula Review	53

12.2 Solved in Full	53
12.3 Practice Set	53
12.4 Computation Log	54
Tutorial Sheet 13 — Tests of Significance for Large Samples	55
13.1 Objective & Formula Review	55
13.2 Solved in Full	55
13.3 Practice Set	55
13.4 Computation Log	56
Tutorial Sheet 14 — Tests of Significance for Small Samples	57
14.1 Objective & Formula Review	57
14.2 Solved in Full	57
14.3 Practice Set	58
14.4 Computation Log	58
PART C — APPENDICES	59
Appendix A — Formula Sheet — One Page per Module	60
Appendix B — Statistical Tables, Computed for This Book	62
Appendix C — GATE-Style Practice Set (CCA Activity 2)	66
Appendix D — Reproducing Every Number in This Book	68

P A R T



Theory — Modules 1 to 5

The five modules exactly as prescribed — modular arithmetic through RSA, least squares through rank correlation, the three named distributions, joint laws and Markov chains, and inference from confidence limits to the chi-square test. Every worked number in this Part was computed by the script reprinted under it; nothing is transcribed from a book.

IN THIS PART

- Module 1 — Modular Arithmetic
- Module 2 — Statistics — Curve Fitting, Correlation and Regression
- Module 3 — Probability Distributions
- Module 4 — Joint Distributions and Markov Chains
- Module 5 — Statistical Inference — Tests of Significance

Module 1 — Modular Arithmetic

Clock arithmetic is the first genuinely discrete tool a computer-science student meets: remainders replace sizes and “equal modulo m ” replaces “equal”. This chapter builds the algebra of congruences from the definition, solves the two equation types the syllabus names (linear congruences and linear Diophantine equations), states the three classical theorems — Euler, Fermat, Wilson — with computational use, and ends by running a real RSA key round-trip.

COURSE OUTCOME	CO1 — modular arithmetic, congruences, Diophantine equations, Euler/Fermat/Wilson, RSA
CONTACT HOURS & TUTORIALS	8 lecture hours · practised in Tutorial Sheets 1–3
PRESCRIBED READING	Koshy, Elementary Number Theory with Applications (Textbook-3): Ch. 4, 5, 6, 7, 9

AS PRESCRIBED (VERBATIM)

Introduction of modular arithmetic and its applications in Computer Science and Engineering. Introduction to Congruences, Linear Congruences, The Remainder theorem, Solving Polynomials, Linear Diophantine Equation, System of Linear Congruences, Euler’s Theorem, Wilson Theorem and Fermat’s little theorem. Applications of Congruences-RSA algorithm. [Textbook-3: Chapters 4, 5, 6, 7 and 9]

1.1 Concepts, Theorems & Rules

1.1 Congruences — the definition that makes arithmetic finite

Given an integer $m \geq 2$, we say **a is congruent to b modulo m**, written $a \equiv b \pmod{m}$, when $m \mid (a - b)$. Three equivalent views are worth keeping in mind: the same remainder upon division by m ; a difference that is a multiple of m ; and equality inside the ring of remainder classes $\mathbb{Z}_m = \{0, 1, \dots, m-1\}$.

PROPERTIES — AN EQUIVALENCE RELATION THAT PLAYS WELL WITH ARITHMETIC

(i) $a \equiv a$; (ii) $a \equiv b \Rightarrow b \equiv a$; (iii) transitivity; (iv) $a \equiv b, c \equiv d \Rightarrow a \pm c \equiv b \pm d$ and $ac \equiv bd$; (v) powers: $a \equiv b \Rightarrow a^k \equiv b^k$, all modulo m .

Cancellation warning. From $ac \equiv bc$ you may cancel c only after shrinking the modulus: with $d = \gcd(c, m)$, $ac \equiv bc \Rightarrow a \equiv b \pmod{m/d}$. Cancelling across a zero divisor is how wrong answers get manufactured in $\mathbb{Z}_6$.

1.2 The remainder theorem and solving polynomials over $\mathbb{Z}_m$

The remainder theorem: dividing $p(x)$ by $x - r$ leaves remainder $p(r)$. In modular dress, evaluating $p(r) \pmod{m}$ decides whether r is a root modulo m — the same trick powers fast polynomial hashing. Over $\mathbb{Z}_m$ with composite m , a degree- n polynomial can have more than n roots (Example 1.2.4 gives a quadratic with four roots in $\mathbb{Z}_6$); zero divisors let a product vanish without either factor vanishing.

1.3 Linear congruences $ax \equiv b \pmod{n}$

EXISTENCE AND COUNT

Let $d = \gcd(a, n)$. The congruence has a solution if and only if $d \mid b$; when solvable it has exactly d mutually incongruent solutions modulo n .

Method. (1) Compute $d = \gcd(a, n)$; stop if it fails to divide b . (2) Divide through: $a'x \equiv b' \pmod{n'}$ with $a' = a/d, b' = b/d, n' = n/d$. (3) Find $(a')^{-1} \pmod{n'}$ — extended Euclid, or Euler/Fermat when the modulus is prime. (4) $x_0 \equiv (a')^{-1}b' \pmod{n'}$; the d solutions are $x_0 + kn'$ for $k = 0, \dots, d-1$.

1.4 Linear Diophantine equations $ax + by = c$

Solvable in integers iff $g = \gcd(a, b) | c$. Run the Euclidean algorithm and read back $g = as + bt$; scaling by c/g yields a particular pair (x_0, y_0) , and the complete family is

$$x = x_0 + \frac{b}{g}t, \quad y = y_0 - \frac{a}{g}t, \quad t \in \mathbb{Z}.$$

Every step is reversible, so the family is exactly the solution set — the favourite “show that” in internal papers.

1.5 Systems of linear congruences — the Chinese Remainder route

For pairwise coprime moduli, CRT guarantees a single class modulo $N = n_1 n_2 \cdots n_k$. Construction: with $M_i = N/n_i$ and $M_i y_i \equiv 1 \pmod{n_i}$,

$$x \equiv \sum_{i=1}^k r_i M_i y_i \pmod{N}.$$

Non-coprime moduli: solvable iff every pair passes the consistency test $\gcd(n_i, n_j) | (r_j - r_i)$; solve as in 1.3 and the answer lives modulo the lcm.

1.6 Euler’s function and Euler’s theorem

$\phi(n)$ counts the classes of $\{1, \dots, n\}$ coprime to n . With $n = p_1^{e_1} \cdots p_r^{e_r}$:

$$\phi(n) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right), \quad \phi(p^e) = p^e - p^{e-1};$$

Euler’s theorem. If $\gcd(a, n) = 1$ then $a^{\phi(n)} \equiv 1 \pmod{n}$. It licenses shrinking exponents modulo $\phi(n)$ — the last-digit tricks are one application line away.

1.7 Fermat’s little theorem

For prime p : $a^p \equiv a \pmod{p}$ for every a , and $a^{p-1} \equiv 1 \pmod{p}$ whenever $p \nmid a$. Two exam habits: reduce the exponent modulo $p-1$ before raising, and remember a^{p-2} is the inverse of $a \pmod{p}$.

1.8 Wilson’s theorem

p is prime if and only if $(p-1)! \equiv -1 \pmod{p}$. The practical shape: to get $k! \pmod{p}$ for k just below p , start from $(p-1)! \equiv -1$ and divide off the missing factors through their inverses — exactly what Example 1.2.6 does.

1.9 Applications of congruences — RSA

RSA is Euler’s theorem wearing a public key:

Step	Do	Why it works
Key	pick primes p, q ; $n = pq$; $\phi = (p-1)(q-1)$	recovering ϕ without p, q is the hard way back
e	choose $\gcd(e, \phi) = 1$	so an inverse exists
d	$d = e^{-1} \pmod{\phi}$	$ed \equiv 1 \pmod{\phi}$
Encrypt	$c \equiv m^e \pmod{n}$	one-way without ϕ

Step DoDecrypt $m \equiv c^d \pmod{n}$ **Why it works** $c^d = m^{\text{ed}} \equiv m$ by Euler/Fermat**1.2 Worked Problems (Machine-Verified)****EXAMPLE 1.1****Solve** $6x \equiv 15 \pmod{21}$.

$\gcd(6, 21) = 3$ divides 15, so the congruence has 3 incongruent solutions modulo 21. Divide through by 3: $2x \equiv 5 \pmod{7}$. The inverse of 2 mod 7 is 4, hence $x \equiv 4 \cdot 5 = 20 \equiv 6 \pmod{7}$. Lifting back gives **$x = 6, 13, 20$** ; the log enumerates 0...20 and lands on the same triple.

EXAMPLE 1.2**Show that** $4x \equiv 6 \pmod{12}$ **has no solution while** $8x \equiv 4 \pmod{12}$ **has four.**

First case: $\gcd(4, 12) = 4$ does not divide 6 — the criterion fails before any algebra starts, and brute force over 0...11 finds none. Second case: $\gcd(8, 12) = 4$ divides 4, so four classes are promised; the log lists **$x = 2, 5, 8, 11$** .

EXAMPLE 1.3**Solve** $6x + 51y = 39$ **in integers — and why is** $6x + 51y = 22$ **hopeless?**

$g = \gcd(6, 51) = 3$. Since $3 \nmid 22$, the second equation has no integer solutions at all. For the first, reduce to $2x + 17y = 13$ and read $3 = 51 - 8 \cdot 6$ off the Euclid chain: $6 \cdot (-8) + 51 \cdot 1 = 3$; scale by 13 to get a particular solution $(x_0, y_0) = (-104, 13)$. The full family is $x = -104 + 17t$, $y = 13 - 2t$. Two members the log checks: $t = 6 \rightarrow (-2, 1)$; $t = 7 \rightarrow (15, -1)$.

EXAMPLE 1.4**Solve** **$x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.**

Pairwise coprime moduli, so CRT applies: $N = 105$; $M_1 = 35$, $M_2 = 21$, $M_3 = 15$ with inverses 2, 1, 1; $x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 \equiv \mathbf{23 \pmod{105}}$. Contrast the non-coprime pair $x \equiv 2 \pmod{4}$, $x \equiv 4 \pmod{6}$: $\gcd(4, 6) = 2$ does divide $4 - 2$, the system stays compatible, and resolving mod $\text{lcm} = 12$ gives **$x \equiv 10 \pmod{12}$** , verified by the 0...11 scan.

EXAMPLE 1.5**Remainder, then roots:** **$p(x) = x^3 - 4x^2 + 5x - 8$ divided by $x - 2$; solve $x^2 - 3x + 2 \equiv 0 \pmod{6}$.**

By the remainder theorem the first answer is $p(2) = 8 - 16 + 10 - 8 = \mathbf{-6}$. The quadratic factors as $(x-1)(x-2)$; over $\mathbb{Z}_6$ a zero product needs no zero factor, and the log finds roots **$x = 1, 2, 4, 5$** — four roots for a degree-2 polynomial, the composite-modulus phenomenon of §1.2.

EXAMPLE 1.6**Find the remainder of 25! on division by 29.**

29 is prime, so Wilson gives $28! \equiv -1$. Split off $28 \cdot 27 \cdot 26 = 19656 \equiv 23 \pmod{29}$; then $23 \cdot 25! \equiv -1 \equiv 28$, so $25! \equiv 28 \cdot 23^{-1} = 28 \cdot 24 = 672 \equiv \mathbf{5 \pmod{29}}$ — and the brute factorial agrees exactly.

EXAMPLE 1.7**Power shrinkage: last digit of 7^{222} ; $3^{100} \pmod{7}$; $2^{90} \pmod{43}$.**

$\varphi(10) = 4$ and $7^4 \equiv 1 \pmod{10}$; $222 = 4 \cdot 55 + 2$, so $7^{222} \equiv 7^2 = 49 \rightarrow$ last digit **9**. Fermat at $p = 7$: $100 = 6 \cdot 16 + 4$, $3^4 = 81 \equiv 4$ — the log's `pow()` says the same. At $p = 43$: $90 = 42 \cdot 2 + 6$, $2^6 = 64 \equiv 21$. Two φ values for the same recipes: $\varphi(77) = 60$, $\varphi(360) = 96$.

EXAMPLE 1.8

RSA end-to-end with $p = 7$, $q = 11$, $e = 7$, message $m = 5$.

$n = 77$, $\varphi = 60$, $d = e^{-1} \pmod{\varphi} = 43$ (check $7 \cdot 43 = 301 \equiv 1 \pmod{60}$). Encrypt: $c = 5^7 \pmod{77} = 47$. Decrypt: $47^{43} \pmod{77} = 5$ — the round trip the log asserts. The same procedure at bigger classroom scale, $p = 61$, $q = 53$, $e = 17$: $n = 3233$, $d = 2753$, and $m = 123$ encrypts to $c = 855$, decrypting back to 123.

1.3 Computation Log

The session below is the verbatim terminal output of the module's companion script — every number quoted above appears here first:

python3 src/m1_modular.py

```
=====
W1 Linear congruence 6x = 15 (mod 21)
=====
gcd(a,n) = gcd(6,21) = 3
d divides b (15)? -> True (criterion: d | b)
solutions in 0..20: [6, 13, 20] (3 incongruent solutions, step 7)
divide through by d: 2x = 5 (mod 7); inverse of 2 mod 7 = 4
x = 4 * 5 mod 7 = 6; lift: x = 6 + k*7 in [0,21) -> [6, 13, 20]

W2 Solvability in both directions
(i) 4x = 6 (mod 12): gcd(4,12) = 4 does NOT divide 6 ->
brute force over 0..11 finds [] (no solutions)
(ii) 8x = 4 (mod 12): gcd(8,12) = 4 divides 4 -> d solutions expected
brute force finds [2, 5, 8, 11] (4 = d solutions)

W3 Linear Diophantine equation 6x + 51y = 22
gcd(6,51) = 3; g | 22? False -> NO solutions when g does not divide c

Now the solvable twin 6x + 51y = 39
-----
g = 3, g | 39 -> reduced: 2x + 17y = 13
sympy parametric solution: {(17*t_0 - 104, 13 - 2*t_0)}
back-substitute: 3 = 51 - 8*6 -> 6*(-8) + 51*(1) = 3
scale by 13: 6*(-104) + 51*13 = 39 -> x0 = -104, y0 = 13
check: 6*-104 + 51*13 = 39
general solution: x = -104 + 17t, y = 13 - 2t (t any integer)
t=6: (x,y) = (-2,1) check 39
t=7: (x,y) = (15,-1) check 39

W4 7x + 11y = 13 (coprime case)
parametric: {(11*t_0 - 39, 26 - 7*t_0)}
7*(-3) + 11*2 = 1 -> multiply by 13: x0 = -39, y0 = 26
check: 7*(-39) + 11*26 = 13
general: x = -39 + 11t, y = 26 - 7t; t=4 -> (5,-2); t=3 -> (-6,5)

W5 System of linear congruences (CRT)
x = 2 (mod 3), x = 3 (mod 5), x = 2 (mod 7)
CRT gives x = 23 (mod 105) brute check -> [23]
via construction: M = [35, 21, 15], inverses = [2, 1, 1]
x = sum(r_i*M_i*y_i) mod N = 23

Non-coprime moduli: x = 2 (mod 4), x = 4 (mod 6)
gcd(4,6)=2 divides 4-2=2 -> compatible; mod lcm=12 the class is [10]

W6 Remainder theorem and solving a quadratic mod m
p(x) = x^3 - 4x^2 + 5x - 8 divided by (x-2): p(2) = -6
x^2 - 3x + 2 = 0 (mod 6) -> roots in Z_6: [1, 2, 4, 5]
four roots for a quadratic: Z_6 has zero divisors, so (x-1)(x-2) = 0 need
not force x = 1 or 2; e.g. x = 5: (4)(3) = 12 = 0 (mod 6) with neither factor 0

W7 Euler phi and Euler's theorem
phi(12) = 4
```

```

phi(77) = 60
phi(360) = 96
360 = {2: 3, 3: 2, 5: 1} -> phi(360) = 360*1/2*2/3*4/5 = 96
Euler: gcd(a,n)=1 => a^phi(n) = 1 (mod n)
check 7^phi(10)=7^4 = 2401 mod 10 = 1
check 13^60 mod 77 = 1
last digit of 7^222: 7^4 = 1 (mod 10), 222 = 4*55 + 2, 7^2 = 49 -> 9

W8 Wilson's theorem (p-1)! = -1 (mod p)
p=5: 4! mod 5 = 4 (4 == -1 mod 5: True)
p=7: 6! mod 7 = 6 (6 == -1 mod 7: True)
p=11: 10! mod 11 = 10 (10 == -1 mod 11: True)
application: 25! mod 29 (29 prime)
-----
Wilson: 28! = -1 (mod 29). 28! = 28*27*26 * 25!, and 28*27*26 mod 29 = 23
so 23 * 25! = 28 (mod 29) -> 25! = 28 * inv(23) mod 29
inv(23) = 24; 25! mod 29 = 5
brute check: math.factorial(25) % 29 = 5

W9 Fermat's little theorem
3^100 mod 7: 3^6 = 1 (mod 7), 100 = 6*16 + 4, 3^4 = 81 = 4 (mod 7)
direct pow(3,100,7) = 4
2^90 mod 43: 2^42 = 1 (mod 43), 90 = 42*2 + 6, 2^6 = 64 = 21 (mod 43)
direct pow(2,90,43) = 21
FLT also: a^p = a (mod p) for every a; check 5^11 mod 11 = 5
13^100 mod 19: 100 = 18*5 + 10, 13^10 = 6

W10 RSA key generation and a round trip (classroom-sized primes)
p = 7, q = 11 -> n = 77, phi(n) = 60
e = 7 with gcd(e,phi)=1; d = e^-1 mod phi = 43 (e*d = 301 = 1 mod 60)
message m = 5: c = m^e mod n = 47; decrypt c^d mod n = 5 round-trip ok: True
m = 23: c = 23 (note: for this m, c = m happens to hold — the map fixes that point)
bigger key: p=61, q=53 -> n=3233, phi=3120, e=17, d=2753
m=123 -> c=855 -> back 123

CK checkpoint answers (computed)
5x = 22 (mod 31) -> [23]
3x = 6 (mod 9) -> [2, 5, 8] (d=3 gives 3 solutions)
x = 4 (mod 7), x = 9 (mod 11) -> x = 53 (mod 77)
phi(55) = 40; 3^40 mod 55 = 1
Wilson: 13 prime? yes; 12! mod 13 = 12
order of 2 mod 7 = 3; of 3 mod 7 = 6

[tokens saved]

```

1.4 Module Summary

Prescribed phrase	One-line recall	Seen in
Applications in CSE	hashing, checksums, random-number streams, RSA crypto	1.1, 1.9
Congruences / remainder theorem	$a \equiv b \pmod{m}$ iff $m \mid a-b$; $p(x) \pmod{(x-r)} = p(r)$	1.1–1.2
Linear congruence $ax \equiv b$	solvable iff $\gcd(a,n) \mid b$; then exactly d solutions	1.3, Ex 1.2.1
Linear Diophantine $ax + by = c$	same gcd test; general family from Euclid back-substitution	1.4, Ex 1.2.3
System of linear congruences	CRT when coprime; pairwise gcd test otherwise	1.5, Ex 1.2.4
Euler's theorem	$a^{\varphi(n)} \equiv 1$; shrink exponents mod $\varphi(n)$	1.6, Ex 1.2.7
Wilson's theorem	$(p-1)! \equiv -1 \pmod{p}$ for primes	1.8, Ex 1.2.6
Fermat's little theorem	$a^p \equiv a$; $a^{p-1} \equiv 1$ when $p \nmid a$	1.7, Ex 1.2.7
Applications — RSA	$n = pq$, $ed \equiv 1 \pmod{\varphi}$; $c \equiv m^e$, $m \equiv c^d$	1.9, Ex 1.2.8

1.5 Checkpoint

Answer on paper first; the key follows each item in the margin.

CK1. Solve $5x \equiv 22 \pmod{31}$.

✓ Ans. unique class: $x \equiv \mathbf{23} \pmod{31}$

CK2. Solve $3x \equiv 6 \pmod{9}$.

✓ Ans. $d = 3$ divides 6 — three classes: $x = \mathbf{2, 5, 8}$

CK3. Solve $x \equiv 4 \pmod{7}$, $x \equiv 9 \pmod{11}$.

✓ Ans. $x \equiv 53 \pmod{77}$

CK4. $\phi(55)$, and what does Euler then promise for $3^{\phi(55)}$?

✓ Ans. $\phi(55) = 40$; $3^{40} \equiv 1 \pmod{55}$ — the log checks it

CK5. Compare $20! \pmod{23}$ with $12! \pmod{13}$ (least positive residues).

✓ Ans. $20! \equiv 11$ via the Wilson trick; $12! \equiv 12 \equiv -1$ by Wilson itself — the larger is $12 \pmod{13}$

CK6. Why can a quadratic have four roots in $\mathbb{Z}_6$?

✓ Ans. zero divisors: $(x-1)(x-2) \equiv 0$ can hold with neither factor $\equiv 0$ (e.g. $4 \cdot 3 = 12 \equiv 0$)

CK7. State the cancellation rule used on $6x \equiv 12 \pmod{9}$.

✓ Ans. $d = \gcd(6,9) = 3$: divide a, b, m all by 3 $\rightarrow 2x \equiv 4 \pmod{3} \rightarrow x \equiv 2, 5, 8 \pmod{9}$

CK8. Give d for RSA with $p = 5$, $q = 17$, $e = 11$.

✓ Ans. $n = 85$, $\phi = 64$, $d = 35$ ($11 \cdot 35 = 385 = 6 \cdot 64 + 1$) — see Tutorial 3, P8

1.6 Practice Set

1. Solve $14x \equiv 22 \pmod{71}$ — then compare with Tutorial Sheet 1, Solved 2.
2. Find all positive solutions of $7x + 11y = 13$ (particular pair $(-39, 26)$ from the log) before writing the general family.
3. Solve $x \equiv 3 \pmod{5}$, $x \equiv 1 \pmod{7}$ two ways — construction and successive substitution — and time yourself.
4. Explain in one line why $18x \equiv 6 \pmod{36}$ is impossible even though 6 looks harmless.
5. Compute $3^{1000} \pmod{13}$ with no calculator; justify the exponent shrinkage $\phi(13) = 12$.
6. Design an RSA exchange for $p = 3$, $q = 11$, $e = 7$, message 9: give n , ϕ , d , c and the decryption check (the answer pattern is Tutorial 3, P8).

Module 2 — Statistics — Curve Fitting, Correlation and Regression

A scatter of measurements is not a law, but the method of least squares turns it into one: of all candidate curves, keep the single one whose squared misses add up smallest. This chapter derives the normal equations for the three prescribed forms — straight line, parabola, and the power curve $y = ax^b$ — then crosses into association: the correlation coefficient, the two regression lines and the angle between them, and rank correlation for data that cannot be measured, only compared.

COURSE OUTCOME	CO2 — least squares fitting, correlation, regression, rank correlation
CONTACT HOURS & TUTORIALS	8 lecture hours · practised in Tutorial Sheets 4-6
PRESCRIBED READING	Walpole (TB-1) Ch. 11-12 · Spiegel (TB-2) Ch. 13-14

AS PRESCRIBED (VERBATIM)

Principles of least squares, Curve fitting by the method of least squares in the form: $y = a + bx$, $y = a + bx + cx^2$, and $y = ax^b$. Correlation, Coefficient of correlation, Lines of regression, Angle between regression lines, rank correlation. [Textbook-1: Chapters 11 and 12, Textbook-2: Chapters 13 and 14]

2.1 Concepts, Theorems & Rules

2.1 The principle of least squares

For data (x_i, y_i) , $i = 1 \dots n$, and a trial curve $y = f(x; \text{parameters})$, the residuals are $e_i = y_i - f(x_i)$. Least

squares picks the parameters that minimise $S = \sum_{i=1}^n e_i^2$. Squaring kills sign-cancellation, penalises wild misses, and makes S a differentiable quadratic in the parameters — so the minimiser is the unique solution of “set the partial derivatives to zero”, a square linear system called the **normal equations**.

2.2 The straight line $\Pi = s + tO$

Setting the two partial derivatives to zero gives the normal equations

$$na + b\sum x = \sum y, \quad a\sum x + b\sum x^2 = \sum xy$$

and their closed form — the pair of workhorse formulas — is

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}, \quad a = \frac{\sum y - b\sum x}{n},$$

whence the line passes through the centroid of the data: $\bar{y} = a + b\bar{x}$. Fitting by hand is a table ritual — columns x , y , x^2 , xy , one sum per column.

2.3 The quadratic $\Pi = s + tO + uO^2$

Minimising against three parameters yields three normal equations, each the previous one multiplied by another power of x and summed:

$$na + b\sum x + c\sum x^2 = \sum y$$

$$a\sum x + b\sum x^2 + c\sum x^3 = \sum xy$$

$$a\sum x^2 + b\sum x^3 + c\sum x^4 = \sum x^2y$$

So a fit asks only for five power-sums of x and three weighted sums of y — the table does the bookkeeping, a 3×3 solve does the rest (Example 2.2.2 does both with the exact rationals).

2.4 The power curve $\Pi = sO^t$

Not linear in the parameters — but log-linear: taking logs turns $y = ax^b$ into $Y = \ln a + bX$ with $X = \ln x$, $Y = \ln y$, a straight line in (X, Y) . Fit the line, then un-log: $a = e^{\bar{Y} - b\bar{X}}$.

Fitting in the log scale minimises relative error, not absolute; state that when asked why the curve does not pass exactly through the measured points.

2.5 Correlation — how tightly do the two lists move together?

The Pearson coefficient is the standardised covariance:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$-1 \leq r \leq 1$; +1 perfect positive line, -1 perfect negative, 0 no linear association. r is invariant to shifts and positive rescalings of both series — marks in °C or °F, rupees or paise: same r .

2.6 Lines of regression and the angle between them

The line of y on x predicts y from x ; the line of x on y predicts x from y ; both pass through the centroid $(\bar{x}, \bar{y})$:

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}), \quad x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}).$$

Two checks every answer should survive: $r^2 = b_{yx} \cdot b_{xy}$ (geometric mean of the slopes recovers r), and both lines hit $(\bar{x}, \bar{y})$. The θ between the two lines follows from their plain slopes $m_1 = b_{yx}$, $m_2 = 1/b_{xy}$:

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|.$$

The two lines coincide exactly when $|r| = 1$ and stand at right angles when $r = 0$ — the “angle tells you r ” family of viva questions.

2.7 Rank correlation

When only orders can be assigned (jury scores, preference rankings), Spearman’s formula uses the differences of ranks:

$$\rho = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}, \quad D = R_x - R_y.$$

Same range as r ; the computation table needs only two extra columns, D and D^2 . (With ties a correction is available, but the prescribed exercises keep ranks distinct.)

2.2 Worked Problems (Machine-Verified)

EXAMPLE 2.1

Fit a straight line to $x = 1, 2, 3, 4, 5$; $y = 2, 4, 5, 4, 5$.

Table ritual:

x	y	x^2	xy
1	2	1	2
2	4	4	8
3	5	9	15
4	4	16	16
5	5	25	25

The two normal-equation closes give $b = (5 \cdot 66 - 15 \cdot 20) / (5 \cdot 55 - 15^2) = 30/50 = \mathbf{0.6}$ and $a = (20 - 0.6 \cdot 15) / 5 = \mathbf{2.2}$. The fitted $y = \mathbf{2.2 + 0.6x}$ passes through $(3, 4)$ = the centroid, as it must.

EXAMPLE 2.2

Fit $y = a + bx + cx^2$ to $(0,1)$ $(1,4)$ $(2,9)$ $(3,16)$ $(4,26)$; estimate at $x = 5$.

The five power-sums and three weighted sums:

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	4	1	1	1	4	4
2	9	4	8	16	18	36
3	16	9	27	81	48	144
4	26	16	64	256	104	416

Solving the 3×3 normal system exactly gives $a = \mathbf{1.0857}$, $b = \mathbf{1.6286}$, $c = \mathbf{1.1429}$ — read them as $76/70$, $57/35$, $4/3.5$ -ish fractions only if you like pain; decimals are fine. $y = \mathbf{1.0857 + 1.6286x + 1.1429x^2}$ and the requested $y^{(5)} = \mathbf{37.8}$.

EXAMPLE 2.3

Fit $y = a + bx$ to $x = 2, 4, 6, 8, 10$; $y = 3, 5, 6, 8, 11$ and estimate y at $x = 12$.

x	y	x^2	xy
2	3	4	6
4	5	16	20
6	6	36	36
8	8	64	64
10	11	100	110

$b = (5 \cdot 236 - 30 \cdot 33) / (5 \cdot 220 - 30^2) = 190/200 = \mathbf{0.95}$; $a = (33 - 28.5) / 5 = \mathbf{0.9}$. Line $y = \mathbf{0.9 + 0.95x}$; at $x = 12$ the fitted value is $\mathbf{12.3}$ — an interpolation one step outside, flag it as extrapolation-adjacent.

EXAMPLE 2.4

Fit the power curve $y = ax^b$ to $(2, 3.1)$ $(4, 6.0)$ $(8, 12.9)$ $(16, 24.1)$.

Log-log table:

x	y	$X = \ln x$	$Y = \ln y$	X^2	XY
2	3.1	0.6931	1.1314	0.4805	0.7842
4	6	1.3863	1.7918	1.9218	2.4839
8	12.9	2.0794	2.5572	4.3241	5.3176
16	24.1	2.7726	3.1822	7.6872	8.8230

The straight-line fit in logs gives $b = \mathbf{0.9980}$ and $\ln a = 0.4362$, so $a = \mathbf{1.5468}$: $y = \mathbf{1.5468 \cdot x^{0.9980}}$. The data are almost a line ($b \approx 1$) — the method notices that for you; prediction at $x = 10$ is $\mathbf{15.3982}$.

EXAMPLE 2.5

Find r for marks of ten students: $x = 45, 60, 35, 40, 50, 45, 60, 65, 50, 35$; $y = 65, 60, 75, 70, 60, 65, 55, 50, 60, 70$.

x	y	x^2	y^2	xy
45	65	2025	4225	2925
60	60	3600	3600	3600
35	75	1225	5625	2625
40	70	1600	4900	2800
50	60	2500	3600	3000
45	65	2025	4225	2925
60	55	3600	3025	3300
65	50	4225	2500	3250
50	60	2500	3600	3000
35	70	1225	4900	2450

Centroid: $\bar{x} = 48.5000$, $\bar{y} = 63.0000$. Numerator $n\Sigma xy - \Sigma x \Sigma y = -6800$; denominator $\sqrt{[(5 \cdot \Sigma x^2 \text{ terms}) - (5 \cdot \Sigma y^2 \text{ terms})]} = 7150.3496$; $\mathbf{r = -0.9510}$. Strong negative — as x climbs, y falls; the log re-derives the same number through raw deviations as a cross-check.

EXAMPLE 2.6

Given $\bar{x} = 8$, $\bar{y} = 10$, $\sigma_x = 2$, $\sigma_y = 3$, $r = 2/3$: write both regression lines and their angle.

$b_{yx} = r\sigma_y/\sigma_x = 1 \rightarrow y - 10 = 1 \cdot (x - 8)$, i.e. $\mathbf{y = 2 + 1x}$. $b_{xy} = r\sigma_x/\sigma_y = \mathbf{0.4444} \rightarrow x - 8 = 0.4444(y - 10)$, i.e. $\mathbf{x = 3.5556 + 0.4444y}$. Consistency: $\sqrt{(1 \times 0.4444)} = 0.6667 = r \checkmark$. Slopes as plain lines: $m_1 = 1$, $m_2 = 1/0.4444 = 2.25$; $\tan \theta = \mathbf{0.3846} \rightarrow \theta = \mathbf{21.0375^\circ}$.

EXAMPLE 2.7

Two judges rank seven candidates ($R_x = 1...7$ as listed, $R_y = 2,1,4,3,7,5,6$). Rank correlation?

candidate	R_x	R_y	D	D^2
1	1	2	-1	1
2	2	1	1	1
3	3	4	-1	1
4	4	3	1	1
5	5	7	-2	4
6	6	5	1	1
7	7	6	1	1

$\Sigma D^2 = \mathbf{10}$, so $\rho = 1 - 6 \cdot 10 / (7 \cdot 48) = 1 - 0.1786 = \mathbf{0.8214}$: fairly close agreement between the judges, with candidate 5 the biggest disagreement ($D = -2$).

2.3 Computation Log

The session below is the verbatim terminal output of the module's companion script — every number quoted above appears here first:

```
python3 src/m2_stats.py
```

```
=====
W1 Fit the straight line y = a + bx to  x=1..5, y=2,4,5,4,5
=====
```

```

n=5 Sx=15 Sy=20 Sx^2=55 Sxy=66
b = (n*Sxy - Sx*Sy)/(n*Sx^2 - (Sx)^2) = (330 - 300)/(275 - 225) = 0.6
a = (Sy - b*Sx)/n = (20 - 0.6*15)/5 = 2.2
FITTED LINE: y = 2.2 + 0.6 x

```

W2 Fit $y = a + bx$ to $x=2,4,6,8,10$ $y=3,5,6,8,11$ and estimate at $x=12$
 $Sx=30$ $Sy=33$ $Sx^2=220$ $Sxy=236$
 $b = (1180-990)/(1100-900) = 190/200 = 0.95$
 $a = (33 - 0.95*30)/5 = 0.9$
FITTED LINE: $y = 0.9 + 0.95x$; $y(12) = 0.9 + 0.95*12 = 12.3$

W3 Fit $y = a + bx + cx^2$ to $(0,1)$ $(1,4)$ $(2,9)$ $(3,16)$ $(4,26)$
 $Sx=10$ $Sx^2=30$ $Sx^3=100$ $Sx^4=354$
 $Sy=56$ $Sxy=174$ $Sx^2y=600$
normal equations (multiply sums by 1, x , x^2 and add):
 $5a + 10b + 30c = 56$
 $10a + 30b + 100c = 174$
 $30a + 100b + 354c = 600$
solve: $a=38/35=1.0857$ $b=57/35=1.6286$ $c=8/7=1.1429$
FITTED PARABOLA: $y = 1.0857 + 1.6286x + 1.1429x^2$
estimate at $x=5$: $y(5) = 37.8$

W4 Fit $y = a*x^b$ to $(2,3.1)$ $(4,6.0)$ $(8,12.9)$ $(16,24.1)$ [log-log straight line]
take logs: $Y = \ln y$, $X = \ln x$; $Y = \ln a + b X$
sums: $SX=6.9315$ $SY=8.6626$ $SXX=14.4136$ $SXY=17.4087$
 $b = 0.9980$; $\ln a = 0.4362$ $\rightarrow a = \exp(\ln a) = 1.5468$
FITTED CURVE: $y = 1.5468 * x^{0.9980}$
estimate at $x=10$: $y(10) = 15.3982$

W5 Coefficient of correlation for marks (x) and (y) of 10 students
 $Sx=485$ $Sy=630$ $Sx^2=24525$ $Sy^2=40200$ $Sxy=29875$
 $\bar{x} = 48.5000$ $\bar{y} = 63.0000$
 $r = (nSxy - SxSy)/\sqrt{(nSx^2 - (Sx)^2)(nSy^2 - (Sy)^2)}$
 $= (298750 - 305550) / \sqrt{(10025 * 5100)} = -6800 / 7150.3496$
 $r = -0.9510$
cross-check via deviations: $r = -0.9510$ match: True

W6 Regression lines and angle between them (summary statistics given)
 $\bar{x}=8.0$ $\bar{y}=10.0$ $\sigma_x=2.0$ $\sigma_y=3.0$ $r=2/3$ (0.6667)
 $b(y \text{ on } x) = r \sigma_y/\sigma_x = 0.6667*3.0/2.0 = 1$
 $\rightarrow y - 10 = 1(x - 8)$ i.e. $y = 2 + 1x$
 $b(x \text{ on } y) = r \sigma_x/\sigma_y = 0.6667*2.0/3.0 = 0.4444$
 $\rightarrow x - 8 = 0.4444(y - 10)$ i.e. $x = 3.5556 + 0.4444y$
consistency: $\sqrt{b(yx)*b(xy)} = \sqrt{0.4444} = 0.6667 == r$ (0.6667)
angle: slopes $m_1=1$ (y on x) $m_2=1/b(x \text{ on } y)=2.25$
 $\tan(\theta) = |(m_1-m_2)/(1+m_1m_2)| = 0.3846 \rightarrow \theta = 21.0375 \text{ deg}$

W7 Rank correlation between two judges
 $D = R_x - R_y$: $[-1, 1, -1, 1, -2, 1, 1]$ $S D^2 = 10$
 $\rho = 1 - 6*SD^2/(n(n^2-1)) = 1 - 60/(7*48) = 1 - 0.1786$
 $\rho = 0.8214$

CK checkpoint answers (computed)
CK1 $x=1.6$ $y=35.10$: $y = 40 + -5x$ ($y(8) = 0$)
CK2 $r = 0.9700$ (strong positive, consistent with data)
CK3 $SD^2=4$ $\rho = 0.8000$

[tokens saved]

2.4 Module Summary

Prescribed phrase	Formula to quote	Seen in
Principles of least squares	minimise $\sum \text{residual}^2$; normal equations = $\partial/\partial \text{param} = 0$	2.1
$y = a + bx$	2 normal equations; $b = (n\sum xy - \sum x \sum y)/(n\sum x^2 - (\sum x)^2)$	2.2, Ex 1/4
$y = a + bx + cx^2$	3 normal equations, power-sums $\sum x \dots \sum x^4$, $\sum y$, $\sum xy$, $\sum x^2y$	2.3, Ex 2
$y = ax^b$	log-log linearise; fit $Y = \ln a + bX$	2.4, Ex 4
Coefficient of correlation	$r = (n\sum xy - \sum x \sum y)/\sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$	2.5, Ex 5
Lines of regression	$y - \bar{y} = r(\sigma_y/\sigma_x)(x - \bar{x})$; $x - \bar{x} = r(\sigma_x/\sigma_y)(y - \bar{y})$; $r^2 = b_{yx}b_{xy}$	2.6, Ex 6 2.6, Ex 6

Prescribed phrase	Formula to quote	Seen in
Angle between regression lines	$\tan \theta = (m_1 - m_2)/(1 + m_1 m_2) $ with $m_2 = 1/b_{xy}$	
Rank correlation	$\rho = 1 - 6\sum D^2/n(n^2 - 1)$	2.7, Ex 7

2.5 Checkpoint

Answer on paper first; the key follows each item in the margin.

CK1. Fit a line to $x = 1 \dots 6$, $y = 35, 30, 25, 20, 15, 10$ and read off $y(8)$.

✓ Ans. The points are collinear: $y = 40 + (-5)x$, so $y(8) = 0$ — exact fit, residuals all zero

CK2. Why is the centroid $(\bar{x}, \bar{y})$ on every least-squares line?

✓ Ans. the first normal equation $n a + b \sum x = \sum y$ is exactly $a + b \bar{x} = \bar{y}$ after dividing by n

CK3. $\bar{x} = 10$, $\bar{y} = 8$, $\sigma_x = 2$, $\sigma_y = 4$, $r = 0.5$: slope of y on x ?

✓ Ans. $r \sigma_y / \sigma_x = 0.5 \cdot 4 / 2 = 1$; and of x on $y = 0.25$; product $= 0.25 = r^2$ ✓

CK4. $x = 2, 4, 5, 4, 3, 1, 2$ with $y = 1, 3, 4, 3, 2, 1, 1$: find r .

✓ Ans. **0.9700** — strongly positive, and the sign agrees with the visible trend

CK5. When do the two regression lines become perpendicular?

✓ Ans. $r = 0$: slopes $m_1 = 0$, $m_2 = \infty$ in the limit formulation; $\tan \theta \rightarrow \infty$ means $\theta = 90^\circ$

CK6. $\rho = 1 - 6\sum D^2/n(n^2 - 1)$ for $n = 5$, $\sum D^2 = 4$?

✓ Ans. $1 - 24/(5 \cdot 24) = \mathbf{0.8000}$

CK7. What does ‘fitted in the log scale minimises relative error’ buy and cost?

✓ Ans. buy: linear machinery and a clean b (elasticity); cost: not the same criterion as absolute least squares — points with small y get louder voices

CK8. Why must $|r| \leq 1$?

✓ Ans. Cauchy-Schwarz on the deviation vectors: $(\sum uv)^2 \leq \sum u^2 \sum v^2$ — divide by the norms

2.6 Practice Set

- Fit $y = a + bx$ to $x = 0, 1, 2, 3, 4$; $y = 1, 3, 2, 4, 5$. Give the fitted values and the residual list; verify $\sum e = 0$.
- Fit a parabola to $x = 1 \dots 5$ with $y = 0, 1, 4, 9, 16$; discover (as the log did for Tutorial 4) that $c = 1$, $a = b = 0$.
- The number of new businesses in a district (thousands) vs years since 2020 gave $y = 1.5468 x^{0.998}$ in class; re-derive it for $x = 2, 4, 8, 16$ with $y = 3.1, 6.0, 12.9, 24.1$.
- r for the pairs $(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)$ without a table — name the shape first.
- $\bar{x} = 40$, $\bar{y} = 50$, $\sigma_x = 5$, $\sigma_y = 10$, $r = 0.8$: write both lines, find the angle between them.
- Two critics rank 9 films; $\sum D^2 = 12$. Compute ρ and say in one sentence whether they agree.

Module 3 — Probability Distributions

A random variable is a function with a probability attached at every point, and a distribution is its whole story told compactly. The chapter reviews the probability kit (conditioning, Bayes), formalises expectation and variance for both discrete and continuous cases — the syllabus’s “mathematical expectation” pair — and then lives in three famous families: Binomial (with the prescribed derivations of mean and standard deviation), Poisson (rare events, and fitting λ to counts), and Normal (the table, the 68–95–99.7 rule, and the approximation that quietly connects the two worlds).

COURSE OUTCOME	CO3 — discrete and continuous random variables; Binomial, Poisson, Normal
CONTACT HOURS & TUTORIALS	9 lecture hours · practised in Tutorial Sheets 7–9
PRESCRIBED READING	Walpole (TB-1) Ch. 3–6 · Spiegel (TB-2) Ch. 6–7

AS PRESCRIBED (VERBATIM)

Review of basic probability theory. Random variables (discrete and continuous), probability mass and density functions. Mathematical expectation: mean and variance. Binomial, Poisson and normal distributions-problems (derivations for mean and standard deviation for Binomial and Poisson distributions only). [Textbook-1: Chapter-3, 4, 5, 6, Textbook-2: Chapters 6 and 7]

3.1 Concepts, Theorems & Rules

3.1 Probability review, sized for this course

Three axioms (non-negativity, totality, additivity) generate everything the module needs: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; complement $P(A^c) = 1 - P(A)$; and conditioning

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A|B) P(B).$$

Bayes’ theorem inverts a conditional when causes compete:

$$P(M_i|D) = \frac{P(M_i) P(D|M_i)}{\sum_j P(M_j) P(D|M_j)}.$$

Example 3.2.1 runs all of this on a two-dice roll and a three-machine defect line.

3.2 Random variables and their laws

Discrete: X takes countably many values with **probability mass function** $p(x) = P(X = x)$, requiring $p(x) \geq 0$, $\sum_x p(x) = 1$. Continuous: a **density** $f(x)$ with $f(x) \geq 0$, $\int f = 1$, and probabilities are areas:

$P(a \leq X \leq b) = \int_a^b f$. The distribution function $F(x) = P(X \leq x)$ rises monotonically from 0 to 1 in either world.

3.3 Mathematical expectation: mean and variance

$$E(X) = \mu = \sum_x x p(x) \quad \left[\int x f(x) dx \right], \quad E[g(X)] = \sum_x g(x) p(x) \quad \left[\int g(x) f(x) dx \right]$$

$$V(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - [E(X)]^2, \quad \sigma = \sqrt{V(X)}.$$

Linearity of E is the whole toolbox of shortcuts: $E(aX + b) = a\mu + b$; variance ignores shifts but squares scales: $V(aX + b) = a^2 V(X)$. The workhorse for hand computations is the second (computational) form of $V(X)$: get $E(X)$, get $E(X^2)$, subtract the square.

3.4 Binomial distribution

n independent trials, success probability p at every trial, X = number of successes. The law counts arrangements and attaches the right product of probabilities:

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \dots, n, \quad q = 1 - p.$$

PREScribed DERIVATION — MEAN (AND THE TWIN FOR SD)

$E(X) = \sum x \cdot C(n, x) p^x q^{n-x}$. Use $x \binom{n}{x} = n \binom{n-1}{x-1}$ and shift the index: the sum becomes $np \sum_{k=0}^{n-1} \binom{n-1}{k} p^k q^{n-1-k} = np \cdot (q+p)^{n-1} = np$.

For the variance, start from $x(x-1)$: $E[X(X-1)] = n(n-1)p^2$ by the same trick twice, then $V(X) = E[X(X-1)] + E(X) - (np)^2 = npq$ and $\sigma = \sqrt{npq}$. The log checks both closed forms symbolically for $n = 4, 8, 12$ before the numeric instance — that is the derivation the syllabus demands, verified.

3.5 Poisson distribution

Rare events over an interval: rate λ per unit, events independent in disjoint pieces —

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

It is the limit of $b(x; n, p)$ with $\lambda = np$ held fixed as n grows (why: the binomial product squeezes into $e^{-\lambda} \lambda^x / x!$ via $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$). Prescribed derivations, in the same index-shift style: $E(X) = \sum x \cdot e^{-\lambda} \lambda^x / x!$; x cancels $(x-1)!$ leaving $\lambda \sum e^{-\lambda} \lambda^{x-1} / (x-1)! = \lambda$; with $E[X(X-1)] = \lambda^2$ we get $V(X) = \lambda + \lambda^2 - \lambda^2 = \lambda$, so mean and variance are equal — the model's built-in diagnostic — and $\sigma = \sqrt{\lambda}$.

Fitting: given a frequency table of counts, $\hat{\lambda} = \sum fx / \sum f$ and the expected frequencies are $N \cdot p(x)$. Example 3.2.3 fits a real table and then asks the model how good it is.

3.6 Normal distribution

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}, \quad Z = \frac{X-\mu}{\sigma} \sim N(0, 1).$$

The area under the bell is tabled only in standard form (Appendix B reproduces the table — computed, to four places, exactly as printed). The 68-95-99.7 rule is a table shortcut the log evaluates from raw integrals: within 1σ area **0.6827**, within 2σ **0.9545**, within 3σ **0.9973**.

Normal approximation to the binomial. For large n , $X \sim N(np, npq)$; with continuity correction

$$P(X \geq k) \approx 1 - \Phi\left(\frac{k - 0.5 - np}{\sqrt{npq}}\right). \text{ Example 3.2.6 compares the approximation to the exact binomial tail.}$$

3.2 Worked Problems (Machine-Verified)

EXAMPLE 3.1

Two fair dice. P(sum = 7)? Given the sum is 7, is the first die equally likely even or odd? Three machines make 50/30/20% of output with defect rates 2/3/5%; a defective item is sampled — cause probabilities?

$P(\text{sum} = 7) = 6/36 = \mathbf{0.1667}$; conditioning: $P(A|B)$ with $A = \text{"first die even"}$, $B = \text{"sum 7"}$: $(3/36)/(6/36) = 1/2$ — symmetry says no bias, the arithmetic agrees. Bayes over the machines: $P(D) = 1/2 \cdot 0.02 + 3/10 \cdot 0.03 + 1/5 \cdot 0.05 = \mathbf{29/1000}$ and the posteriors $P(M_1|D) = 10/29 = \mathbf{0.3448}$, $P(M_2|D) = 9/29 = \mathbf{0.3103}$, $P(M_3|D) = 10/29 = \mathbf{0.3448}$: the low-volume, high-defect machine M_3 ties with the high-volume M_1 as most likely culprit — the entire lesson of Bayes in one row of numbers.

EXAMPLE 3.2

X has pmf 1/16, 4/16, 6/16, 4/16, 1/16 on 0,1,2,3,4. Find $E(X)$, $V(X)$, $sd(X)$, and $E(2X-1)$, $V(2X-1)$.

x	P(X=x)	xP	x ² P
0	1/16	0	0
1	1/4	0.25	0.25
2	3/8	0.75	1.5
3	1/4	0.75	2.25
4	1/16	0.25	1

The xP-column sums to **2.0000**, the x²P-column to **5.0000**; hence $V = 5.0000 - 2.0000^2 = \mathbf{1.0000}$, $\sigma = \mathbf{1.0000}$. Linearity: $E(2X-1) = 2 \cdot 2 - 1 = 3$; $V(2X-1) = 4 \cdot 1 = 4$. (The pmf is $\text{Bin}(4, \frac{1}{2})$ — $np = 2$ and $npq = 1$, and the direct sums match the closed forms, as they must.)

EXAMPLE 3.3

Binomial $n = 10$, $p = 0.4$: $P(X = 5)$, $P(X \leq 2)$, $P(X \geq 7)$; mean, variance, sd.

$P(X=5) = C(10,5) \cdot 0.4^5 \cdot 0.6^5 = \mathbf{0.2007}$; $P(X \leq 2) = \mathbf{0.1673}$; $P(X \geq 7) = \mathbf{0.0548}$. Closed forms: $\mu = np = \mathbf{4.0000}$, $\sigma^2 = npq = \mathbf{2.4000}$, $\sigma = \mathbf{1.5492}$. The log additionally runs the symbolic check behind these two formulas (the $x \cdot C(n,x)$ identity) for $n = 4, 8, 12$ — this is the derivation section written out in full in §3.4.

EXAMPLE 3.4

Defect counts per lot, 200 lots: $x = 0,1,2,3,4$ with frequencies 102, 67, 24, 6, 1. Fit a Poisson; P(at least one defect)?

$\hat{\lambda} = (67 + 48 + 18 + 4)/200 = \mathbf{0.6850}$. Expected frequencies

x	observed f	P(X=x)	expected N·P
0	102	0.5041	100.818
1	67	0.3453	69.0604
2	24	0.1183	23.6532
3	6	0.0270	5.4008
4	1	0.0046	0.9249

The fit is close — the observed and expected columns hug — and the goodness-of-fit χ^2 the log prints is tiny. $P(\text{at least one}) = 1 - P(0) = 1 - e^{-0.685} = 1 - \mathbf{0.5041} = \mathbf{0.4959}$. And the model's own signature: mean = variance = λ , so $sd = \sqrt{\lambda} = \mathbf{0.8276}$.

EXAMPLE 3.5

$X \sim N(100, \sigma = 10)$. Find $P(X > 115)$, $P(X < 85)$, $P(90 < X < 115)$.

Standardise: $z = (115-100)/10 = 1.5 \rightarrow P = 1 - \Phi(1.5) = \mathbf{0.0668}$. By symmetry $P(X < 85) = \Phi(-1.5) = \mathbf{0.0668}$. The band: $\Phi(1.5) - \Phi(-1) = \mathbf{0.7745}$. Every figure here is the table value to four places; Appendix B prints the whole table from the same integral, so the book never disagrees with itself.

EXAMPLE 3.6

Top 5% cut-off of $N(70, \sigma = 5)$.

Read the table backwards: z with $\Phi(z) = 0.95$ is **1.6449** (precisely 1.645 — the one-tail 5% z from Appendix B); $x = 70 + 1.6449 \cdot 5 = \mathbf{78.2243}$.

EXAMPLE 3.7

Fair coin tossed 100 times. Approximate $P(\geq 60 \text{ heads})$ and compare with the exact value.

$\mu = np = 50$, $\sigma = \sqrt{npq} = \mathbf{5.0000}$. Continuity correction: $z = (59.5 - 50)/5 = 1.9$, $P \approx 1 - \Phi(1.9) = \mathbf{0.0287}$; exact binomial tail **0.0284** — the correction buys an error of three in the fourth decimal, which is why the exam answers it with the 0.5 shift.

3.3 Computation Log

The session below is the verbatim terminal output of the module's companion script — every number quoted above appears here first:

python3 src/m3_dist.py

```
=====
W1 Classical probability: two dice
=====
|S| = 36    P(sum=7) = 6/36 = 1/6 = 0.1667
P(double)   = 6/36 = 1/6
P(sum=7 AND double) = 3/36 -> 0 : events mutually exclusive? False
P(double | sum=7) = 0/6 (exclusive). Conditional demo, P(A|B) = P(AB)/P(B):
  A = first die even, B = sum = 7: P(A) = 1/2, P(B) = 1/6, P(AB) = 3/36
  P(A|B) = (3/36)/(6/36) = 1/2 = 0.5000

Bayes: machines M1,M2,M3 make 50%,30%,20%; defect rates .02,.03,.05
P(D) = 0.0290 = 29/1000 (exact)
P(M1|D) = 10/29 = 0.3448
P(M2|D) = 9/29 = 0.3103
P(M3|D) = 10/29 = 0.3448

W2 Mathematical expectation of a discrete RV
pmf:  x      0      1      2      3      4
      P(x) 1/16 4/16 6/16 4/16 1/16 (binomial-shape; sums to 1)
E(X) = sum x p = 2 = 2.0000
E(X^2) = sum x^2 p = 5 = 5.0000
V(X) = E(X^2) - [E(X)]^2 = 1 = 1.0000; sd = 1.0000
E(2X-1) = 2E(X)-1 = 3      V(2X-1) = 4 V(X) = 4

W3 Binomial distribution n=10, p=0.4
P(X= 0) = C(10,0) 0.4^0 0.6^10 = 0.0060
P(X= 2) = C(10,2) 0.4^2 0.6^8 = 0.1209
P(X= 5) = C(10,5) 0.4^5 0.6^5 = 0.2007
P(X=10) = C(10,10) 0.4^10 0.6^0 = 0.0001
P(X <= 2) = 0.1673      P(X >= 7) = 0.0548
mean = np = 4.0000    var = npq = 2.4000    sd = 1.5492

Derivation check with sympy (mean via k*C(n,k) = n*C(n-1,k-1)):
n= 4: E = 4*p    V = -4*p**2 + 4*p (expect 4*p and 4*p*q)
n= 8: E = 8*p    V = -8*p**2 + 8*p (expect 8*p and 8*p*q)
n=12: E = 12*p   V = -12*p**2 + 12*p (expect 12*p and 12*p*q)
-> for every n tested the closed forms E = np, V = npq hold symbolically;
    the same algebra on sum k*C(n,k) = n*sum C(n-1,k-1) is the textbook derivation.
numeric check n=10 p=0.4: E2 = 18.4    V = 2.40

W4 Poisson: fitting lambda to a frequency table (defective items in 200 lots)
total lots = 200; lambda = sum f*k / sum f = 137/200 = 0.6850
P(X=x) = e^-lam lam^x / x!, e^-0.6850 = 0.5041
P(0) = 0.5041    expected freq = 200 * 0.5041 = 100.8
P(1) = 0.3453    expected freq = 200 * 0.3453 = 69.1
P(2) = 0.1183    expected freq = 200 * 0.1183 = 23.7
P(3) = 0.0270    expected freq = 200 * 0.0270 = 5.4
P(4) = 0.0046    expected freq = 200 * 0.0046 = 0.9
goodness-of-fit chi-square vs Poisson model = 0.1530 (small -> good fit)
P(at least 1 defective) = 1 - P(0) = 1 - 0.5041 = 0.4959
mean = var = lambda = 0.6850; sd = sqrt(lambda) = 0.8276

W5 Normal distribution areas (X ~ N(100, sigma=10))
P(X > 115): z = (115-100)/10 = 1.5000; P = 1 - Phi(1.5) = 0.0668
P(X < 85): z = -1.5000; P = Phi(-1.5) = 0.0668
P(90<X<115): Phi(1.5) - Phi(-1.0) = 0.7745
within one sd: P(90<X<110) = 0.6827
within two sd: P(80<X<120) = 0.9545
within three sd: = 0.9973

W6 Finding a cut-off: top 5% of N(70, sd=5)
z such that Phi(z)=0.95: z = 1.6449; x = mu + z*sd = 70 + 1.6449*5 = 78.2243
```

```

W7 Normal approximation to Binomial(100, 0.5): P(X >= 60), continuity correction
mu = 50, sigma = sqrt(25) = 5.0000
with correction: z = (59.5 - 50)/5 = 1.9000; P = 1 - Phi(1.9000) = 0.0287
exact binomial: P(X >= 60) = 0.0284 (close)
no correction: z = 2 -> P = 0.0228

CK checkpoint answers (computed)
CK1 P(double | even sum) = 6/18 = 1/3
CK2 c=1/10 pmf E(X) = ... see tutorial; P(X=4 | n=8,p=.5) = C(8,4)/256 = 35/128 = 0.2734
CK4 mean 6 var 3: q = 3/6 = 1/2 -> p = 1/2 -> n = 6/p = 12
CK7 N(500,40): P(X<460) = 0.1587; top-10% z = 1.2816

[tokens saved]

```

3.4 Module Summary

Prescribed phrase	Formula to quote	Seen in
Random variables, pmf/pdf	$p(x) \geq 0, \sum p = 1; f \geq 0, \int f = 1; F = P(X \leq x)$	3.2
Mathematical expectation	$E(X) = \sum xp; V = E(X^2) - \mu^2; E[g(X)]$ with g inside the sum	3.3, Ex 2
Binomial	$C(n,x)p^xq^{n-x}; \mu = np; \sigma = \sqrt{npq}$ (derived 3.4)	3.4, Ex 3
Poisson	$e^{-\lambda}\lambda^x/x!; \mu = V = \lambda; \text{fit } \lambda = \sum fx/\sum f$	3.5, Ex 4
Normal	$Z = (X-\mu)/\sigma; 68-95-99.7; \text{table} = \Phi$	3.6, Ex 5-6
Approximation	binomial $\rightarrow N(np, npq)$ with ± 0.5 correction	3.6, Ex 7

3.5 Checkpoint

Answer on paper first; the key follows each item in the margin.

CK1. Roll two dice: $P(\text{double} \mid \text{sum even})$?

✓ Ans. all 6 doubles sit among the 18 even-sum outcomes: $1/3$ — every double is even, so conditioning changes nothing

CK2. Why does a pmf-table's x^2P column, not the xP column, first appear when computing V ?

✓ Ans. V needs $E(X^2) = \sum x^2p$; the xP column only feeds μ — the shortcut $V = E(X^2) - \mu^2$ uses both

CK3. $n = 8, p = 1/2$: $P(\text{exactly 4 successes})$?

✓ Ans. $C(8,4)/2^8 = 70/256 = \mathbf{0.2734}$

CK4. A binomial has mean 6 and variance 3. Give n and p .

✓ Ans. $np = 6, npq = 3 \rightarrow q = 1/2, p = 1/2, n = \mathbf{n = 12, p = 1/2}$ (i.e. $n = 12$)

CK5. Poisson: why must mean = variance, and how do you use that on real data?

✓ Ans. both equal λ (derived in 3.5); on data, compare sample var with sample mean — a gap says the model is wrong, as the χ^2 row in Ex 4 quantifies

CK6. $\Phi(1.645) = 0.95$ — use it to name the top-10% z .

✓ Ans. top 10% is $z = \mathbf{1.2816}$ (1.2816); 1.645 is the top-5% point

CK7. Which single step turns binomial coefficients into $e^{-\lambda}$?

✓ Ans. the limit $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$ together with $n(n-1)\dots(n-x+1) \approx n^x$

CK8. $X \sim N(500, 40)$: $P(X < 460)$?

✓ Ans. $z = -1 \rightarrow \Phi(-1) = \mathbf{0.1587}$

3.6 Practice Set

- Two urns: U_1 has 3 white/2 black, U_2 has 1 white/4 black; a fair coin picks the urn, a white ball is drawn. Which urn is more likely? (Bayes, as in Ex 3.2.1.)
- Given $P(X=x) = c \cdot x$ for $x = 1 \dots 4$, find c , $E(X)$, $V(X)$, and $E((X-2)^2)$ by two routes.
- A machine produces 2% defective. In a sample of 50: expected defectives, sd, and $P(\text{more than 2})$ via binomial; compare with a Poisson shortcut.
- Fit Poisson to counts of typos per page on 200 pages: 84, 71, 30, 10, 5 for $x = 0 \dots 4$; report $\hat{\lambda}$ and the first two expected frequencies.
- Heights $N(65 \text{ in}, 3 \text{ in})$: find the 10th-90th percentile band (two table lookups).
- Verify the Poisson mean derivation for $\lambda = 2$ numerically: $\sum x \cdot p(x)$ over $x = 0 \dots 12$, then argue the tail.

Module 4 — Joint Distributions and Markov Chains

Two variables measured on the same trials tell more than two separate laws: their joint table carries the co-movement — which is exactly what expectation, covariance and correlation extract. The chapter's second half jumps disciplines: a probability vector moves through a stochastic matrix, powers of the matrix are multi-step forecasts, and regular chains forget their starting line and settle onto a stationary distribution. Every number in both halves — from $\sum xy \cdot p$ to $\pi P = \pi$ — was solved exactly with sympy rationals before being rounded into this text.

COURSE OUTCOME	CO4 — joint laws, covariance/correlation; transition probabilities, stationary distributions, absorbing states
CONTACT HOURS & TUTORIALS	8 lecture hours · practised in Tutorial Sheets 10-12
PRESCRIBED READING	Walpole (TB-1) Ch. 3-6 · Lipschutz & Lipson (TB-4) Ch. 8

AS PRESCRIBED (VERBATIM)

Joint Probability distribution for two discrete random variables, expectation, covariance and correlation. Markov Chain: Introduction to Stochastic Process, Probability Vectors, Stochastic matrices, Regular stochastic matrices, Markov chains, Higher transition probabilities, Stationary distribution of Regular Markov chains. [Textbook-1: Chapter-3, 4, 5, 6, Textbook-4: Chapter-8]

4.1 Concepts, Theorems & Rules

4.1 The joint law of two discrete random variables

$f(x, y) = P(X = x, Y = y)$ on a grid of value-pairs, with all entries non-negative and summing to 1. From the joint table everything else is marginalisation:

$$f_1(x) = \sum_y f(x, y), \quad f_2(y) = \sum_x f(x, y), \quad P(Y = y | X = x) = \frac{f(x, y)}{f_1(x)}.$$

X and Y are **independent** iff the table is a product: $f(x, y) = f_1(x)f_2(y)$ for every cell.

4.2 Expectation, covariance, correlation

$$E(XY) = \sum_{x,y} xy f(x, y), \quad \text{Cov}(X, Y) = E(XY) - E(X)E(Y),$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1], \quad V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y).$$

Independence forces $\text{Cov} = 0$ (hence $\rho = 0$); the converse fails — the log's 2×2 product table is the standard counterexample family with uncorrelated-but-dependent pairs to hand in §4.1. Exam habit: write $E(XY)$ through the table, then Cov , then ρ — three lines, no algebraic detours.

4.3 Stochastic processes and Markov chains

A **stochastic process** is a family of random variables indexed by time; the chain in this syllabus has finitely many states and the **Markov property**: the future depends on the past only through the present.

The one-step data are the numbers $p_{ij} = P(X_{t+1} = j | X_t = i)$, assembled into a matrix P whose i -th row is the outlook from state i . A **probability vector** is a nonnegative row summing to 1; a **stochastic matrix** has every row a probability vector; and the chain evolves by left-multiplication: $x^{(t+1)} = x^{(t)}P$, so after t steps $x^{(t)} = x^{(0)}P^t$.

4.4 Regular stochastic matrices

P is **regular** when some power P^k has all entries strictly positive. Regularity is what makes the limit theorems clean: it rules out absorbing walls and periodic cycles: the swap matrix with ones off the diagonal never becomes positive — its powers just toggle between P and the identity).

4.5 Higher transition probabilities

The (i, j) entry of P^k is exactly the k -step probability of going from state i to state j : multiplying once more adds one step of all possible intermediate histories, and that is the whole proof, by induction, of the Chapman-Kolmogorov bookkeeping. Example 4.2.3 computes P^2 and P^3 for the two-state weather chain, then hands them to a forecast.

4.6 Stationary distribution of a regular Markov chain

A stationary (equilibrium) vector satisfies

$$\pi P = \pi, \quad \sum_j \pi_j = 1.$$

For a regular chain this system has a unique solution, and every starting vector converges to it:

$\lim_{t \rightarrow \infty} x^{(0)} P^t = \pi$. Two ways to finish a 2-state instance: solve the 3-equation linear system (2 balance + 1 normalisation), or read the ratio directly from the balance equation — the log solves both, in exact rationals, for the 2- and 3-state chains.

ABSORBING STATES (CO4'S EXTENSION)

A state with $p = 1$ on itself absorbs: once entered, never left. The hitting probabilities $a_i = P(\text{reach the absorbing set} \mid \text{start at } i)$ solve the same fixed-point equation restricted to the transient states; §4.2.4 works the two-state absorption line and confirms it by marching $P^{20} = 0.4000$ to four places.

4.2 Worked Problems (Machine-Verified)

EXAMPLE 4.1

Joint law from weights: $P(x,y) \propto w$ with $W = [[1,2],[2,3],[1,1]]$ over $x \in \{1,2,3\}$, $y \in \{1,2\}$; total weight 10. Marginals, means, Cov, ρ .

$P(x,y)$	$y=1$	$y=2$	$f(x)$
$x=1$	1/10	1/5	3/10
$x=2$	1/5	3/10	1/2
$x=3$	1/10	1/10	1/5
$g(y)$	2/5	3/5	1

Marginals $f(x) = (3, 5, 2)/10$, $g(y) = (4, 6)/10$. $E(X) = 19/10 = \mathbf{1.9}$, $E(Y) = \mathbf{1.6}$, and $E(XY) = 30/10 = \mathbf{3}$. $\text{Cov} = 3 - 1.9 \cdot 1.6 = \mathbf{-0.04}$; with $V(X) = 0.49$, $V(Y) = 0.24$ (log-checked), $\rho = -0.2/(\sqrt{0.49} \cdot \sqrt{0.24}) = \mathbf{-0.1166}$ — a mild negative link: bigger x tends to ride with smaller y . The log closes by enumerating $X+Y$ directly and confirming $V(X+Y) = 0.65$ matches $VX + VY + 2\text{Cov}$.

EXAMPLE 4.2

For the same pair, compute $V(2X + 3Y)$.

$4 \cdot V(X) + 9 \cdot V(Y) + 2 \cdot 2 \cdot 3 \cdot \text{Cov} = 4 \cdot 0.49 + 9 \cdot 0.24 + 12 \cdot (-0.04) = \mathbf{4.6}$. The $2ab \cdot \text{Cov}$ term is where sign matters: a negative covariance makes the sum less variable than the parts — the diversification effect one line long.

EXAMPLE 4.3

Weather chain S/R with $P = [[0.8, 0.2], [0.4, 0.6]]$: start from $x_0 = (1/2, 1/2)$. Predict days 1-3, write P^2 , name the stationary forecast.

Successive multiplication in the log: $x_1 = (0.6, 0.4)$, $x_2 = (0.64, 0.36)$, $x_3 = (0.656, 0.344)$ — the sequence already leans to the equilibrium. P^2 has all-positive entries, so the chain is regular, and $\pi P = \pi$ gives $\pi = (2/3, 1/3) = (0.6667, 0.3333)$. $x_0 P^{20} = (0.6665, 0.3335)$ in the log: three decades later, still chasing $2/3$ — the convergence is the theorem, visible.

EXAMPLE 4.4

3-state chain with $P = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{bmatrix}$: is it regular? Find π .

P^2 already has all positive entries (the log lists them), hence regular. Solving the five equations (three balance + normalisation) in exact rationals: $\pi = (0.2500, 0.5000, 0.2500)$ — the middle state carries half the long-run mass because its row is flat; symmetry across the outer states is the quick sanity check.

EXAMPLE 4.5

Absorption: states 0 and 2 are absorbing; from state 1 the outlook is (0.3, 0.5, 0.2). Probability of being absorbed at 2 starting from 1?

$a_1 = 0.5 \cdot a_1 + 0.2 \cdot 1 \rightarrow 0.5a_1 = 0.2 \rightarrow a_1 = 0.4$. Marching P^{20} the other way, the (1,2) entry reaches **0.4000** — same number, two methods, one line of difference below 10^{-15} .

4.3 Computation Log

The session below is the verbatim terminal output of the module's companion script — every number quoted above appears here first:

python3 src/m4_joint_markov.py

```
=====
W1 Joint pmf on weights (values / 10); X in {1,2,3}, Y in {1,2}
=====
weight table (divide by 10):
  x=1: y=1 -> 1/10   y=2 -> 2/10
  x=2: y=1 -> 2/10   y=2 -> 3/10
  x=3: y=1 -> 1/10   y=2 -> 1/10
marginals: f(x) = ['3/10', '5/10', '2/10']   g(y) = ['4/10', '6/10']
E(X) = 1.9   E(Y) = 1.6
E(XY) = 3 (sum xy f)
V(X) = 0.49   V(Y) = 0.24   sd(X) = 0.7000   sd(Y) = 0.4899
Cov(X,Y) = E(XY) - E(X)E(Y) = -0.04
rho = Cov/(sd*sd) = -0.1166
check V(X+Y) = VX + VY + 2Cov = 0.65
direct enumeration V(X+Y) = 0.65 == formula: True

W2 Properties: V(aX+bY), independent vs uncorrelated
with V(X)=?, V(Y)=?, Cov=? from W1: V(2X-3Y) = 4V(X)+9V(Y)-12Cov
    = 4*0.49 + 9*0.24 - 12*(-0.04) = 4.6

W3 Two-state weather chain P = [[0.8,0.2],[0.4,0.6]] (rows: S, R)
=====
row 0 sums to 1 (stochastic matrix: yes)
row 1 sums to 1 (stochastic matrix: yes)
x0 = [1/2, 1/2]
x1 = x0 P = [0.6, 0.4]
x2 = x0 P^2 = [0.64, 0.36]
x3 = x0 P^3 = [0.656, 0.344]
P^2 = [[18/25, 7/25], [14/25, 11/25]] all entries > 0 -> regular
P^3 = [[86/125, 39/125], [78/125, 47/125]]
stationary: pi P = pi, A+B=1 -> pi = [0.6667, 0.3333]
check pi P = [[2/3, 1/3]]
convergence: P^10[0] = ['0.6667', '0.3333'] -> rows -> pi (0.6667, 0.3333)

W4 Three-state chain (textbook-style): rows sum to 1
P = [[1/2, 1/2, 0], [1/4, 1/2, 1/4], [0, 1/2, 1/2]]
P^2 = [[3/8, 1/2, 1/8], [1/4, 1/2, 1/4], [1/8, 1/2, 3/8]]
stationary pi = [0.25, 0.5, 0.25]
decimals: (0.2500, 0.5000, 0.2500)
P^2 strictly positive? True -> regular (P^3 too: True)
```

```

W5 Absorbing chain: states 0,2 absorbing; row 1 = [3/10, 1/2, 1/5]
hitting probability a1 of state 2: a1 = 1/2*a1 + 1/5*1 -> a1 = 0.4
check by limit: (P^n)[1,2] as n->20 = 0.4000 -> 0.4
expected visits before absorption from state 1: 1/(1-1/2) = 2

CK checkpoint answers
CK-a: if f(x,y) = f(x)g(y) then Cov = 0: example P = [[1/4,1/4],[1/4,1/4]]
E(X)=1.5 E(Y)=1.5 E(XY)=2.25 Cov=0
CK-b: x = (1/3, 2/3), Q = [[2/3, 1/3], [1/4, 3/4]]; x Q^2 = (0.412, 0.588)
stationary of Q: (0.4286, 0.5714)

```

[tokens saved]

4.4 Module Summary

Prescribed phrase	Formula to quote	Seen in
Joint distribution (discrete)	$f(x,y)$; marginals by summing rows/columns; independence = product table	4.1, Ex 1
Expectation of XY	$E(XY) = \sum xy f(x,y)$	4.2, Ex 1
Covariance	$\text{Cov} = E(XY) - E(X)E(Y)$	4.2, Ex 1
Correlation	$\rho = \text{Cov}/(\sigma_x \sigma_y)$; $V(aX+bY)$ with $2ab \cdot \text{Cov}$	4.2, Ex 2
Stochastic process / chain	Markov property; $p_{ij} = P(\text{next state } j \mid \text{now } i)$	4.3
Probability vector	nonnegative, sums to 1; $x' = xP$	4.3
Stochastic matrix	rows sum to 1	4.3, Ex 3
Regular stochastic matrix	some power all positive	4.4, Ex 4
Higher transition probabilities	$(P^k)_{ij} = k\text{-step probability}$	4.5, Ex 3
Stationary distribution	$\pi P = \pi$, $\sum \pi = 1$; unique limit for regular chains	4.6, Ex 3-4

4.5 Checkpoint

Answer on paper first; the key follows each item in the margin.

CK1. A joint table has all zero covariance but fails independence — possible?

✓ Ans. yes — ρ only forbids linear association; the $\frac{1}{4}$ -equal 2×2 pattern swaps signs off-diagonal and is the smallest example

CK2. Which row of a stochastic matrix is a probability vector? all, or just one?

✓ Ans. every row, by definition — a stochastic matrix is rows of probability vectors (4×4 matrix with one bad row is not stochastic)

CK3. For the weather chain, $x_0 = (1, 0)$ — which day's forecast is exactly $(0.656, 0.344)$?

✓ Ans. day 3: $x_0 P^3 = (0.656, 0.344) = \mathbf{(0.656, 0.344)}$ from the same power the log prints

CK4. Does $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ have a stationary vector? Is it regular?

✓ Ans. $\pi = (\frac{1}{2}, \frac{1}{2})$ is stationary, yet no power is positive — stationary does not imply regular, and this chain oscillates forever

CK5. What changes in the π computation when a state is absorbing?

✓ Ans. absorption probabilities solve a reduced system on transient states only; π itself (with mass 1 on the absorbing state) is the long-run distribution

CK6. $V(3X - Y)$ given $V(X) = 4$, $V(Y) = 1$, $\text{Cov} = 0.5$?

✓ Ans. $9 \cdot 4 + 1 - 2 \cdot 3 \cdot 1 \cdot 0.5 = 36 + 1 - 3 = \mathbf{34}$

CK7. $E(XY) = E(X)E(Y)$ — what did you just prove without naming it?

✓ Ans. zero covariance ($\rho = 0$ when both variances are positive)

CK8. State the 2-state balance shortcut.

✓ Ans. $\pi_1/\pi_2 = p_{21}/p_{12}$; normalise — e.g. $0.4/0.2 = 2 \rightarrow \pi = (2/3, 1/3)$ ✓

4.6 Practice Set

- Build a 2×2 joint table with independent rows where $E(X) = 1.2$: give $f(x)$, $g(y)$, and the table.
- For $W = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} / 6$ over $x, y \in \{1, 2\}$: compute Cov and ρ ; then $V(X+Y)$ two ways (formula and enumeration).

3. A shopper flips a fair coin to pick brand A or B each week; A retains 70%, B retains 60%. Write P , start from $(\frac{1}{2}, \frac{1}{2})$, and give the two-week and long-run shares (pattern: Ex 4.2.3 and Tutorial 12).
4. For $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix}$: show P^2 , verify regularity, solve $\pi P = \pi$ exactly.
5. Which of $(0.4, 0.7, -0.1)$, $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $(1, 0, 0)$ are probability vectors? Where does each belong?
6. Absorption twin: from state 1 the outlook is $(0.4, 0.4, 0.2)$ with 0 and 2 absorbing — probability of landing at 0?

Module 5 — Statistical Inference — Tests of Significance

Inference answers the question a sample always raises: how much of what I see is real? The machinery is one sentence long — pretend H_0 is true, compute how far the sample strayed from it in standard-error units, and compare that distance to a threshold set by the level of significance. This chapter walks the sampling distribution and the central limit theorem (with a tiny population enumerated by hand), then takes each prescribed test — large-sample mean and proportion, their two-sample comparisons, Student's t for small samples, and the chi-square goodness-of-fit — and computes every z , t , χ^2 and p -value from data, not from a remembered answer.

COURSE OUTCOME	CO5 — hypothesis testing, confidence intervals, large- and small-sample conclusions
CONTACT HOURS & TUTORIALS	9 lecture hours · practised in Tutorial Sheets 13-14
PRESCRIBED READING	Walpole (TB-1) Ch. 8, 10 · Spiegel (TB-2) Ch. 8-10, 12

AS PRESCRIBED (VERBATIM)

Sampling distribution, standard error. Levels of significance. Testing of hypothesis. Test of significances. Confidence limits, simple sampling of attributes. Sampling variables, central limit theorem and confidence limit for unknown mean.

AS PRESCRIBED (VERBATIM)

Test of significance for large samples (mean, proportion). Comparison of large samples (mean, proportion).

AS PRESCRIBED (VERBATIM)

Test of Significance for means of two small samples, students-' t ' distribution, and Chi-square distribution as a test of goodness of fit. [Textbook-1: Chapter 8 and 10, Textbook-2: Chapters-8, 9, 10 and 12]

5.1 Concepts, Theorems & Rules

5.1 Sampling distribution and standard error

Draw samples of size n from a population and the statistic you compute — say the mean — has its own distribution, the **sampling distribution**. Its sd is the **standard error**:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \left[\text{finite pop. } N: \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right].$$

Example 5.2.1 enumerates the sampling distribution of the mean for $n = 2$ from the four-number population $\{2, 4, 6, 8\}$ and watches the formula hold exactly.

Central limit theorem. Whatever the parent distribution (finite variance), the standardised mean $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ tends to $N(0, 1)$; $n \geq 30$ is the working rule. Consequence — the confidence limit for an unknown mean uses s in place of σ for large n .

5.2 Hypotheses, significance level, confidence limits

H_0 is the no-effect claim; H_1 the alternative; the **level of significance** α (5% and 1% are the customs) caps the probability of rejecting a true H_0 . Confidence limits are the same geometry inverted:

$$\bar{X} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}} \quad (95\% : Z_{\alpha/2} = 1.96).$$

Attributes (proportions) of a large sample obey the same law: $\hat{p} \pm z\sqrt{\hat{p}q/n}$ — “simple sampling of attributes”.

5.3 Tests for large samples

Question	Statistic	H ₀ says
One mean (σ or large- n s)	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	$z \sim N(0,1)$
One proportion	$z = (\hat{p} - p_0)/\sqrt{(p_0 q_0/n)}$	$z \sim N(0,1)$
Two means, large	$z = (\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$	$z \sim N(0,1)$
Two proportions	$z = (\hat{p}_1 - \hat{p}_2)/\sqrt{[\bar{p}\bar{q}(1/n_1 + 1/n_2)]}$, $\bar{p}$ pooled	$z \sim N(0,1)$

Verdict language is fixed: two-tail $p = 2[1 - \Phi(|z|)]$; $p < \alpha \rightarrow$ “the difference is significant”, otherwise “no significant difference at level α ”. z beyond 1.96 $\Rightarrow p$ below 0.05; 2.576 $\Rightarrow$ 0.01.

5.4 Small samples — Student’s t

With $n < 30$ and normal parent, σ unknown, the standardised mean follows Student’s t with $n - 1$ degrees of freedom — heavier tails than the normal, converging to it. Two-sample version for means:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}, \quad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad df = n_1 + n_2 - 2.$$

Compare $|t|$ with $t(\alpha/2, df)$ from Appendix B; that table was generated for this book from the t-distribution definition itself, so the critical values are the honest ones.

5.5 Chi-square as a test of goodness of fit

Bucket the data, compute what the model expected in each bucket, and measure the disagreement:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}, \quad df = k - 1 - (\text{parameters estimated}).$$

Small χ^2 (relative to the table cut at df) $\rightarrow$ the model fits; large $\rightarrow$ the model is rejected. The dice example in 5.2.4 and its digits twin both come straight out of the computation log, components and all.

5.2 Worked Problems (Machine-Verified)

EXAMPLE 5.1

Enumerate the sampling distribution of $\bar{x}$ for $n = 2$ from $\{2, 4, 6, 8\}$ (all ordered samples, with replacement) and confirm the standard-error formula.

16 ordered samples give means 2, 3, 3, 4, 4, 4, 5, ..., 8 — the full table

sample mean	2	3	4	5	6	7	8
frequency /16	1	2	3	4	3	2	1
probability	1/16	2/16	3/16	4/16	3/16	2/16	1/16
$x \cdot p$	0.1250	0.3750	0.7500	1.2500	1.1250	0.8750	0.5000

Weighted by their 16-fold frequencies, $E(\bar{x}) = 5$ = the population mean (unbiasedness, visible), and $sd(\bar{x}) = 1.5811 = \sigma/\sqrt{2} = 2.2361/1.4142$. The parent is flat-uniform; the $n = 2$ mean is already triangular — the log’s last two lines note the climb toward normality as n grows (CLT in one picture).

EXAMPLE 5.2

A factory claims mean lifetime $\mu_0 = 1500$ h, known $\sigma = 250$ h. A sample of $n = 100$ gives $\bar{x} = 1450$ h. Test at 5% and at 1%.

$SE = 250/\sqrt{100} = 25$. $z = (1450 - 1500)/25 = -2.0000$; two-tail $p = 2(1 - \Phi(2)) = 0.0455$. At 5%: $|z| = 2 > 1.96 \rightarrow$ reject — the claim misses significance. At 1%: $2 < 2.576 \rightarrow$ not rejected. One dataset, two answers, because the level is a choice, not a law. This is exactly the pattern of §5.3 row 1.

EXAMPLE 5.3

A party claims 75% support; your poll of 400 finds 320 supporters. Significant at 5%? one-tailed?

$\hat{p} = 0.80$, $SE_0 = \sqrt{(0.75 \cdot 0.25/400)} = 0.02165$, $z = \mathbf{2.3094}$; two-tail $p = \mathbf{0.0209} \rightarrow$ significant at 5%. One-tail (the “support grew” reading): $p = \mathbf{0.0105}$ — clears 0.05 but just misses 0.01, so the direction clause matters: H_1 decides which p you are allowed to quote.

EXAMPLE 5.4

Two wire types: A $n = 120$, $\bar{x} = 420$, $\sigma = 24$; B $n = 100$, $\bar{x} = 411$, $\sigma = 22$. Compare the means.

$SE = \sqrt{(24^2/120 + 22^2/100)} = \mathbf{3.1048}$; $z = 9/SE = \mathbf{2.8987}$; $p = \mathbf{0.0037} < 0.05 \rightarrow$ the 9-unit advantage is real at 5%; $z = 2.8987 > 2.576$ too, so it survives the 1% line as well. The log prints both comparisons verbatim.

EXAMPLE 5.5

Town voters: 60% of 500 support a policy; in the neighbouring town 52% of 400. Difference of proportions?

Pooled $\bar{p} = (300 + 208)/900 = \mathbf{0.5644}$; $SE = \sqrt{[\bar{p}\bar{q}(1/500 + 1/400)]} = 0.0333$; $z = \mathbf{2.4052}$, $p = \mathbf{0.0162} < 0.05 \rightarrow$ significant. Note the pooled standard error — under H_0 both towns share one proportion, so one estimate feeds the SE.

EXAMPLE 5.6

$n = 100$ mean-burning times, $\bar{x} = 2.1$ h, $s = 0.6$ h: give the 95% and 99% confidence limits for μ ; then do the attributes version for 320/400.

$SE = 0.6/10 = 0.06$. 95%: $2.1 \pm 1.96(0.06) = (\mathbf{1.9824}, \mathbf{2.2176})$; 99%: $\pm 2.576(0.06) = (\mathbf{1.9454}, \mathbf{2.2546})$ — the price of a second 9 is a wider interval. Attributes: $\hat{p} = 0.8$, $SE = \sqrt{(0.8 \cdot 0.2/400)} = \mathbf{0.0200}$, 95% CL (0.7608, 0.8392) in the log.

EXAMPLE 5.7

Two repair crews, 9 and 7 jobs: strengths A = (45, 42, 51, 44, 46, 41, 49, 47, 44), B = (38, 47, 41, 44, 36, 42, 48). Test equality of means with t at 5%.

$\bar{x}_A = 45.4444$, $\bar{x}_B = 42.2857$; $s^2_A = 10.2778$, $s^2_B = 19.5714$ (the log's per-sample lines); pooled $s^2_p = \mathbf{14.2608}$, so $t = \mathbf{1.6598}$ with $df = 14$. The log pulls $t_{0.025}(14) = 2.1448$ from the distribution's own definition; $|1.66| < 2.14 \rightarrow$ **no significant difference** ($p = 0.1192$). The small-sample test is honest about small data: it declines to conclude.

EXAMPLE 5.8

Battery claim $\mu_0 = 165$ days, $n = 10$ observed: (164, 160, 158, 161, 163, 159, 162, 160, 165, 158).

$\bar{x} = 161$, $s = 2.4495$, $t = \mathbf{-5.1640}$, $df = 9$, $p = 0.0006$; $|t| = 5.16 > t_{0.025}(9) = 2.2622 \rightarrow$ the claim is rejected at 5% (and at 1%). One detail the exam loves: here s replaces σ because the mean must be estimated from the same ten cells.

EXAMPLE 5.9

A die is rolled 600 times: faces 93, 107, 98, 110, 88, 104. Fair? Then the same question for 1000 random digits.

Expected 100 per face. Component table

face	observed	expected	(O-E)	(O-E) ² /E
1	93	100	-7	0.49
2	107	100	7	0.49
3	98	100	-2	0.04
4	110	100	10	1
5	88	100	-12	1.44

face	observed	expected	(O-E)	(O-E)^2/E
6	104	100	4	0.16

$\chi^2 = 3.6200$ against $df = 5$: the 5% cut-off is 11.0705 and $p = 0.6053 \rightarrow$ the die passes for fair. Digits: $\chi^2 = 5.0600$ vs cut-off 16.9190, $p = 0.8290$ — also uniform. Both verdicts come from the same two lines: compute, compare, conclude.

5.3 Computation Log

The session below is the verbatim terminal output of the module's companion script — every number quoted above appears here first:

python3 src/m5_inference.py

```
=====
W1 Sampling distribution of the mean, population {2,4,6,8}, n=2 (with replacement)
=====
population mean = 5    variance = 5    sd = 2.2361
all 16 ordered samples -> distinct means: [2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0]
mean 2.0: frequency 1    probability 1/16
mean 3.0: frequency 2    probability 2/16
mean 4.0: frequency 3    probability 3/16
mean 5.0: frequency 4    probability 4/16
mean 6.0: frequency 3    probability 3/16
mean 7.0: frequency 2    probability 2/16
mean 8.0: frequency 1    probability 1/16
E(mean) = 5 (== mu)    V(mean) = 2.5 (== var/n = 5/2)
sd(mean) = 1.5811 = sigma/sqrt(n) = 1.5811

CLT in numbers: sampling distribution above is not normal (it is a
triangular 16-point pile-up) but its sd already equals sigma/sqrt(n);
n=10 gives mean 5, sd 2.2361/sqrt(10) = 0.7071 with an approximately
normal shape — this is the central limit theorem at work.

W2 One mean, large sample: bulbs claim mu0=1500h, sigma=250, n=100, xbar=1450
SE = sigma/sqrt(n) = 250/10 = 25
z = (1450-1500)/25 = -2.0000    two-tail p = 2*(1-Phi(2)) = 0.0455
|z| = 2 > 1.96 -> reject H0 at 5%; p=0.0455 < 0.05 (also > 2.576? False -> not significant at 1%)

W3 One proportion: claim p0 = 0.75; sample n=400, favourable 320
phat = 320/400 = 0.8; SE0 = sqrt(.75*.25/400) = 0.0217
z = (0.8-0.75)/0.0217 = 2.3094    two-tail p = 0.0209
one-tail p = 0.0105 -> significant at 5%: True

W4 Two large samples (wires): A n=120 xbar 420 sigma 24; B n=100 xbar 411 sigma 22
SE = sqrt(24^2/120 + 22^2/100) = 3.1048
z = (420-411)/3.1048 = 2.8987; p = 0.0037
compare with 1.96 -> no significant difference at 5%: False

W5 Two towns supporting a policy: 60% of 500 vs 52% of 400
pooled p = (0.6*500 + 0.52*400)/900 = 0.5644
SE = sqrt(p q (1/n1+1/n2)) = 0.0333
z = 0.08/0.0333 = 2.4052    two-tail p = 0.0162
|z| > 1.96? True -> significant at 5%: True

W6 Confidence limits (large n): n=100, xbar=2.1h, s=0.6h
95% CL: 2.1 +/- 1.96*0.06 = (1.9824, 2.2176)
99% CL: 2.1 +/- 2.576*0.06 = (1.9454, 2.2546)
attributes version: 320/400 -> p=0.8, SE=sqrt(pq/n)=0.02, 95% CL = (0.7608, 0.8392)
recomputed: SE = 0.0200; CL = (0.7608, 0.8392)

W7 Student's t, two small samples
sample A: n=9, mean = 45.4444, s^2 = 10.2778
sample B: n=7, mean = 42.2857, s^2 = 19.5714
pooled s^2 = [8*10.2778 + 6*19.5714]/14 = 14.2608
t = (45.4444-42.2857)/sqrt(14.2608*(1/9+1/7)) = 1.6598    df = 14
two-tail p = 0.1192; t_0.025(14) = 2.1448 -> significant at 5%: False

W8 One-sample t: battery life n=10: 164 160 158 161 163 159 162 160 165 158, mu0=165
n=10, mean = 161, s^2 = 6.0000, s = 2.4495
t = (161-165)/sqrt(6.0000/10) = -5.1640    df=9
two-tail p = 0.0006; t_0.025(9) = 2.2622
reject mu0=165 at 5%? True
```

```

W9 Chi-square goodness of fit: die rolled 600 times
expected 100 per face; components (O-E)^2/E:
  face 1: (93-100)^2/100 = 0.490
  face 2: (107-100)^2/100 = 0.490
  face 3: (98-100)^2/100 = 0.040
  face 4: (110-100)^2/100 = 1.000
  face 5: (88-100)^2/100 = 1.440
  face 6: (104-100)^2/100 = 0.160
chi2 = 3.6200  df = 5  p = 0.6053  critical(5%) = 11.0705
chi2 < critical -> the die is fair (no evidence against): True

digits example: 1000 random digits, expected 100 each
chi2 = 5.0600,  df = 9,  critical(5%) = 16.9190,  p = 0.8290

CK checkpoint answers
CK1 n=400 xbar=6.1 s=0.9 vs mu=6: SE=0.0450 z=2.2222 p=0.0263 -> significant: True
CK2 proportions 55% n=200 vs 48% n=250: pooled 0.5111, z = 1.4761, p = 0.1399

[tokens saved]

```

5.4 Module Summary

Prescribed phrase	Formula to quote	Seen in
Sampling distribution, standard error	$\sigma/\sqrt{n}$ (finite population: $\times\sqrt{[(N-n)/(N-1)]}$)	5.1, Ex 1
Levels of significance	$\alpha = P(\text{reject true } H_0)$; 5% / 1% customs	5.2, Ex 2
Testing hypothesis	H_0, H_1 ; two- vs one-tail p; “significant” language	5.2–5.3
Confidence limits	$\bar{x} \pm z \cdot s/\sqrt{n}$; $\hat{p} \pm z\sqrt{(\hat{p}\hat{q}/n)}$	5.2, Ex 6
CLT / unknown mean	$(\bar{x}-\mu)/(\sigma/\sqrt{n}) \rightarrow N(0,1)$; s stands in for σ when n large	5.1, Ex 1
Large-sample tests (mean, proportion) & comparisons	z-formulas table in 5.3	5.3, Ex 2–5
Two small samples (Student’s t)	pooled s^2_p ; t with df n_1+n_2-2	5.4, Ex 7
Chi-square goodness of fit	$\Sigma(O-E)^2/E$ with df by the –1 rule	5.5, Ex 9

5.5 Checkpoint

Answer on paper first; the key follows each item in the margin.

CK1. Why does the standard error carry a $\sqrt{n}$, not an n ?

✓ Ans. independence: variances add, so sd of the sum is $\sqrt{n} \cdot \sigma$ and dividing by n leaves $\sigma/\sqrt{n}$

CK2. State the two errors a test can make and which one α controls.

✓ Ans. type I (reject true H_0) — controlled by α ; type II (accept false H_0) — lowered by bigger n

CK3. A z of -2.1 at 5% two-tail: verdict? one-tail?

✓ Ans. reject both ways ($|z| > 1.96$; one-tail even more so)

CK4. For $n = 400$, $\bar{x} = 6.1$, $s = 0.9$ against $\mu_0 = 6$: z and the log’s p-decision?

✓ Ans. $z = 2.2222$ — exceeds 1.96: significant at 5%, visible at 2.22 vs 1.96

CK5. 55% of 200 vs 48% of 250: difference of proportions?

✓ Ans. pooled 0.5111, $z = 1.4761$, $p = 0.1399 \rightarrow$ not significant at 5% — a 7-point gap with small samples is noise-sized

CK6. Which df for: χ^2 on 5 categories? two-sample t with $n_1 = 10$, $n_2 = 8$?

✓ Ans. df = 4; df = 16 — the –1 and the $n_1 + n_2 - 2$ rules

CK7. t’s table entries exceed z’s at the same α — why?

✓ Ans. estimating σ from s adds uncertainty; heavier tails, shrinking as df grows

CK8. Confidence limits for the unknown mean when $n = 64$, $\bar{x} = 2.1$, $s = 0.6$?

✓ Ans. $2.1 \pm 1.96 \cdot 0.075 = (1.953, 2.247)$ — one half-row of Ex 5.2.6 with n changed; the log method is identical

5.6 Practice Set

1. Enumerate the sampling distribution of $\bar{x}$ for samples of size 2 (no replacement) from $\{1, 3, 5\}$: list means, find $E(\bar{x})$, and check the finite-population correction factor.

2. A courier claims mean delivery 32 min; your 36-delivery sample gives 34.2 min, $s = 4.5$. Test at 5%, then re-test as a one-tailed claim (“no faster”).
3. Day-care survey: 66 of 150 parents prefer outdoor play. Test the 50/50 claim and give 95% CL for the proportion.
4. Two bottling lines: $\bar{x}_1 = 503$, $s_1 = 4$, $n_1 = 50$; $\bar{x}_2 = 500$, $s_2 = 5$, $n_2 = 45$. Compare at 5%.
5. Samples (21, 18, 22, 25, 19) and (17, 20, 16, 19, 18) of etch depths: two-sample t at 5%; state df and the pooled s^2 .
6. Observed counts 30, 22, 24, 24 (uniform model): χ^2 , df , verdict at 5%; then at 1%.
7. Fit-or-falsify: rolls of a die 120 times, faces 15, 22, 18, 19, 24, 22 — χ^2 and verdict; compare with the 600-roll answer in Ex 5.2.9.

P A R T

B

Tutorials — Sheets 1 to 14

The fourteen prescribed practice problems in full: each sheet opens with its verbatim objective, gathers the formulas it drills, works three or four problems end-to-end with machine-verified answers, lists the remaining practice items with results, and closes with the computation log that produced them. Two hours each, as the scheme requires.

IN THIS PART

- Sheet 1 — Linear Congruences
- Sheet 2 — Linear Diophantine Equations & Systems of Congruences
- Sheet 3 — Euler, Wilson and Fermat
- Sheet 4 — Fitting the Linear and Quadratic Forms
- Sheet 5 — The Power Curve and Correlation
- Sheet 6 — Lines of Regression and Rank Correlation
- Sheet 7 — Mathematical Expectations
- Sheet 8 — Binomial and Poisson Distributions
- Sheet 9 — Normal Distributions
- Sheet 10 — Covariance and Correlation of Two Discrete RVs
- Sheet 11 — Markov Chains, Stochastic Matrices, Probability Vectors
- Sheet 12 — Higher Transition Probabilities and Stationary Distributions
- Sheet 13 — Tests of Significance for Large Samples
- Sheet 14 — Tests of Significance for Small Samples

Tutorial Sheet 1 — Linear Congruences

OBJECTIVE (VERBATIM)	Practice problems on linear congruences
DRILLS	Module 1, §1.3 · criterion + reduction + inverse + lift
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on linear congruences

1.1 Objective & Formula Review

$ax \equiv b \pmod{n}$

$d = \gcd(a, n)$

d solutions mod n

inverse of a'

Recall

solvability iff $d \mid b$

reduce to $a'x \equiv b' \pmod{n'}$, $x_0 = (a')^{-1}b'$, lift by n'

extended Euclid, or $a'^{\varphi(n')-1}$ when n' prime (Fermat: $a'^{n'-2}$)

1.2 Solved in Full

SOLVED 1

Solve $7x \equiv 3 \pmod{11}$.

$\gcd(7, 11) = 1$ divides 3 — one class. $7^{-1} \pmod{11} = 8$ ($7 \cdot 8 = 56 \equiv 1$). $x \equiv 8 \cdot 3 = 24 \equiv \mathbf{2 \pmod{11}}$.

SOLVED 2

Solve $14x \equiv 22 \pmod{71}$.

71 is prime, so one class. Euclid: $71 = 5 \cdot 14 + 1 \rightarrow 14^{-1} \equiv -5 \equiv 66 \pmod{71}$. $x = 66 \cdot 22 = 1452 \equiv \mathbf{32 \pmod{71}}$ — $\text{pow}(14, -1, 71)$ in the log, same 66, same 32.

SOLVED 3

Does $18x \equiv 6 \pmod{36}$ have a solution?

$\gcd(18, 36) = 18$, and $18 \nmid 6$ — **no solution**. The criterion settles it; no trial needed.

SOLVED 4

Solve $4x \equiv 8 \pmod{12}$.

$d = 4 \mid 8 \rightarrow$ four classes. $4x \equiv 8 \pmod{12} \div 4$ gives $x \equiv 2 \pmod{3}$; lift: $\mathbf{x = 2, 5, 8, 11}$.

1.3 Practice Set

P1	$9x \equiv 12 \pmod{30}$.	$x = \mathbf{8, 18, 28}$ ($d = 3$)
P2	$5x \equiv 7 \pmod{13}$.	$x \equiv \mathbf{4}$
P3	$3x \equiv 1 \pmod{12}$.	no solution ($3 \nmid 1$)
P4	$12x \equiv 9 \pmod{15}$.	$x = \mathbf{2, 7, 12}$

1.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 1 linear congruences
=====
SOLVED 1.
```



```

solve 7x = 3 (mod 11): d = gcd(7,11) = 1
d | 3 -> reduce: 7x = 3 (mod 11)
inverse of 7 mod 11 = 8; x = 2 (mod 11)
solutions mod 11: [2]

```

SOLVED 2.

```

solve 14x = 22 (mod 71): d = gcd(14,71) = 1
d | 22 -> reduce: 14x = 22 (mod 71)
inverse of 14 mod 71 = 66; x = 32 (mod 71)
solutions mod 71: [32]

```

SOLVED 3.

```

solve 18x = 6 (mod 36): d = gcd(18,36) = 18
d does NOT divide 6 -> NO solution

```

SOLVED 4.

```

solve 4x = 8 (mod 12): d = gcd(4,12) = 4
d | 8 -> reduce: 1x = 2 (mod 3)
inverse of 1 mod 3 = 1; x = 2 (mod 3)
solutions mod 12: [2, 5, 8, 11]

```

PRACTICE (answers at the end of the run)

P5: $9x = 12 \pmod{30}$

```

solve 9x = 12 (mod 30): d = gcd(9,30) = 3
d | 12 -> reduce: 3x = 4 (mod 10)
inverse of 3 mod 10 = 7; x = 8 (mod 10)
solutions mod 30: [8, 18, 28]

```

P6: $5x = 7 \pmod{13}$

```

solve 5x = 7 (mod 13): d = gcd(5,13) = 1
d | 7 -> reduce: 5x = 7 (mod 13)
inverse of 5 mod 13 = 8; x = 4 (mod 13)
solutions mod 13: [4]

```

P7: $3x = 1 \pmod{12}$

```

solve 3x = 1 (mod 12): d = gcd(3,12) = 3
d does NOT divide 1 -> NO solution

```

P8: $12x = 9 \pmod{15}$

```

solve 12x = 9 (mod 15): d = gcd(12,15) = 3
d | 9 -> reduce: 4x = 3 (mod 5)
inverse of 4 mod 5 = 4; x = 2 (mod 5)
solutions mod 15: [2, 7, 12]

```

[tokens saved]

Tutorial Sheet 2 — Linear Diophantine Equations & Systems of Congruences

OBJECTIVE (VERBATIM)	Practice Problems on Linear Diophantine Equation, System of Linear Congruences
DRILLS	Module 1, §1.4–1.5
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice Problems on Linear Diophantine Equation, System of Linear Congruences

2.1 Objective & Formula Review

$ax + by = c$

solvability

particular pair

general family

system

Recall

$g = \gcd(a, b)$ must divide c

Euclid chain read back: $g = as + bt$; scale by c/g

$x = x_0 + (b/g)t$, $y = y_0 - (a/g)t$

pairwise consistency $g_{ij} \mid (r_j - r_i)$; CRT when coprime

2.2 Solved in Full

SOLVED 1

Solve $9x + 5y = 12$.

$\gcd(9, 5) = 1$, so solutions exist for every right side. The smallest particular pair the log hunts down is $(x_0, y_0) = (3, -3)$: $27 - 15 = 12$. The whole family, as the log prints it, is **$x = 3 + 5t$, $y = -3 - 9t$ (e.g. $(3, -3)$)** — $t = -1$ gives $(-2, 6)$, $t = 1$ gives $(8, -12)$.

SOLVED 2

Solve $15x + 12y = 17$.

$\gcd(15, 12) = 3$ and $3 \nmid 17 \rightarrow$ **no solution**. Say it in one line in the exam: the left side is always a multiple of 3; 17 is not.

SOLVED 3

Solve $7x + 5y = 1$.

Euclid: $7 = 1 \cdot 5 + 2$, $5 = 2 \cdot 2 + 1 \rightarrow 1 = 5 - 2(7-5) = 3 \cdot 5 - 2 \cdot 7 \rightarrow (x_0, y_0) = (-2, 3)$. Family **$x = -2 + 5t$, $y = 3 - 7t$ (e.g. $(-2, 3)$)** — the log's minimal scan lands on exactly the pair Euclid produced.

SOLVED 4

Solve $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{5}$, $x \equiv 5 \pmod{6}$.

Moduli are not pairwise coprime, so check pairs: $\gcd(4, 6) = 2 \mid (5-1) \checkmark$, others are 1. Compatible $\rightarrow$ solve over $\text{lcm}(4, 5, 6) = 60$: **$x = 17 \pmod{60}$** (brute force over the full cycle agrees).

2.3 Practice Set

P1	$12x + 20y = 8$.	$\gcd 4 \mid 8$; family $x = -1 + 5t$, $y = 1 - 3t$ (e.g. $(-1, 1)$)
P2	$x \equiv 2 \pmod{4}$, $x \equiv 5 \pmod{6}$.	$\gcd(4, 6) = 2$ does not divide 3 $\rightarrow$ no solution
P3	$6x + 10y = 3$.	$\gcd(6, 10) = 2 \nmid 3 \rightarrow$ no solution
P4	$x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.	CRT: $x = 23 \pmod{105}$

2.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 2 Diophantine equations and systems
=====
SOLVED 1.
9x + 5y = 12: g = gcd(9,5) = 1
  reduced 9x + 5y = 12; sympy: [(5*t_0 - 12, 24 - 9*t_0)]
  particular: (x0,y0) = (3,-3) check 12 = 12

SOLVED 2.
15x + 12y = 17: g = gcd(15,12) = 3
  g does not divide 17 -> no integer solutions

SOLVED 3.
7x + 5y = 1: g = gcd(7,5) = 1
  reduced 7x + 5y = 1; sympy: [(5*t_0 - 2, 3 - 7*t_0)]
  particular: (x0,y0) = (-2,3) check 1 = 1
  Euclid walk: 7 = 1*5+2, 5 = 2*2+1 -> 1 = 5-2*2 = 5-2*(7-5) = 3*5-2*7
  hence 7*(-2) + 5*3 = 1: (x0,y0)=(-2,3); general x = -2+5t, y = 3-7t

SOLVED 4. System x = 1 (mod 4), x = 2 (mod 5), x = 5 (mod 6)
  moduli not pairwise coprime -> check consistency: x=1 (mod 4) and x=5 (mod 6)
  need gcd(4,6)=2 | (5-1)=4 OK; then solve the reduced pair and lift.
  brute over one full lcm(4,5,6)=60 cycle: x = 17 (mod 60)

PRACTICE
P5: 12x+20y = 8
  12x + 20y = 8: g = gcd(12,20) = 4
  reduced 3x + 5y = 2; sympy: [(5*t_0 + 4, -3*t_0 - 2)]
  particular: (x0,y0) = (-1,1) check 8 = 8
P6: x = 2 (mod 4), x = 5 (mod 6)
  gcd(4,6)=2 must divide 5-2=3; it does not -> search confirms []
  -> NO solution (incompatible pair)
P7: 6x + 10y = 3
  gcd = 2 does not divide 3 -> no solutions
P8: x = 2 (mod 3), x = 3 (mod 5), x = 2 (mod 7)
  CRT: x = 23 (mod 105)

[tokens saved]
```

Tutorial Sheet 3 — Euler, Wilson and Fermat

OBJECTIVE (VERBATIM)	Practice problems on Euler's Theorem, Wilson Theorem and Fermat's little theorem.
DRILLS	Module 1, §1.6-1.9
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on Euler's Theorem, Wilson Theorem and Fermat's little theorem.

3.1 Objective & Formula Review

Theorem	Work use
$\varphi(n) = n \prod (1 - 1/p)$	count of units in $\mathbb{Z}_n$
Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ ($\gcd(a, n) = 1$)	shrink exponents mod $\varphi(n)$
Fermat: $a^{p-1} \equiv 1 \pmod{p}$ (p prime)	shrink exponents mod $p-1$; inverse a^{p-2}
Wilson: $(p-1)! \equiv -1 \pmod{p}$	factorials just below p ; primality signature
RSA	$n = pq$, $\varphi = (p-1)(q-1)$, $ed \equiv 1 \pmod{\varphi}$, $c \equiv m^e$

3.2 Solved in Full

SOLVED 1

Compute $\varphi(315)$ and check Euler for base 2.

$315 = 3^2 \cdot 5 \cdot 7 \rightarrow \varphi = 315 \cdot (2/3)(4/5)(6/7) = \mathbf{144}$. Then $2^{144} \equiv 1 \pmod{315}$ — the log prints `pow(2, 144, 315) = 1`.

SOLVED 2

What is the order of 3 mod 7, and how does it compare with Fermat's $p-1$?

3, 2, 6, 4, 5, 1: the order is **6** — here it equals $p-1 = 6$, i.e. 3 is a primitive root mod 7. Fermat promises the cycle divides 6; order says exactly 6.

SOLVED 3

$6^{100} \pmod{7}$.

$6 \equiv -1 \rightarrow (-1)^{100} = \mathbf{1}$. The log's pow agrees; FLT shrinking would give the same with more writing.

SOLVED 4

$20! \pmod{23}$.

Wilson: $22! \equiv -1$; peel $22 \cdot 21 = (-1)(-2) = 2$: $2 \cdot 20! \equiv -1 \rightarrow 20! \equiv -1 \cdot 2^{-1} = -12 \equiv \mathbf{11}$. Brute factorial confirms.

3.3 Practice Set

P1	$\varphi(1001)$.	$1001 = 7 \cdot 11 \cdot 13 \rightarrow \mathbf{720}$
P2	Last two digits of 3^{100} .	$\varphi(100) = 40$: $3^{100} = (3^{40})^2 \dots \rightarrow \mathbf{01}$
P3	Wilson check at $p = 13$.	$12! \equiv \mathbf{12} \pmod{13} (= -1 \pmod{13})$
P4	RSA $p = 5$, $q = 17$, $e = 11$, $m = 9$.	$d = 35$, $c = 49$ ($n = 85$, $\varphi = 64$)

3.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 3 Euler / Wilson / Fermat
=====
```

SOLVED 1. $\phi(315)$ and Euler's theorem check

$$315 = \{3: 2, 5: 1, 7: 1\} = 3^2 * 5 * 7$$

$$\phi(315) = 315 * (1 - 1/3) * (1 - 1/5) * (1 - 1/7) = 144$$

$$\gcd(2, 315) = 1 \rightarrow 2^{144} \equiv 1 \pmod{315}; \quad \text{pow}(2, \phi(315)) = 1$$

SOLVED 2. order of 3 modulo 7 (why FLT exponents shrink)

$$3^1 \pmod{7} = 3$$

$$3^2 \pmod{7} = 2$$

$$3^3 \pmod{7} = 6$$

$$3^4 \pmod{7} = 4$$

$$3^5 \pmod{7} = 5$$

$$3^6 \pmod{7} = 1$$

$$3^6 \equiv 1 \pmod{7} \text{ matches Fermat } (p-1 = 6); \text{ the least such exponent is } \text{ord}_7(3) = 6$$

SOLVED 3. $6^{100} \pmod{7}$

$$6 \equiv -1 \pmod{7}; \quad 6^{100} \equiv (-1)^{100} \equiv 1; \quad \text{direct pow}(6, 100, 7) = 1$$

SOLVED 4. Wilson: remainder of $20!$ divided by 23

$$23 \text{ prime} \rightarrow 22! \equiv -1 \pmod{23} \quad (\text{Wilson})$$

$$22! = 22 * 21 * 20!; \quad 22 \equiv -1, 21 \equiv -2 \pmod{23} \rightarrow (-1)(-2) * 20! \equiv -1$$

$$2 * 20! \equiv -1 \rightarrow 20! \equiv -1 * \text{inv}(2) \equiv -12 \pmod{23} \equiv 11$$

$$\text{computed: } 11; \quad \text{brute: } 20! \pmod{23} = 11$$

PRACTICE

P5: $\phi(1001)$ ($1001 = 7 * 11 * 13$)

$$\phi(1001) = 1001 * (6/7) * (10/11) * (12/13) = 720$$

P6: last two digits of 3^{100} (i.e. $\pmod{100}$)

$$\phi(100) = 40, \gcd(3, 100) = 1 \rightarrow 3^{40} \equiv 1 \pmod{100} \rightarrow 3^{100} = (3^{40})^2 * 3^{20}$$

$$3^{20} \pmod{100} = 1; \quad 3^{100} \pmod{100} = 1 \rightarrow \text{last two digits: } 01$$

P7: Wilson check for $p = 13$

$$12! \pmod{13} = 12 \quad (12 \equiv -1 \pmod{13}: \text{True})$$

P8: RSA with $p=5, q=17, e=11$

$$n = 85, \phi = 64, d = 35; \quad \text{encrypt } m=9: c = 49; \text{ decrypt: } 9$$

[tokens saved]

Tutorial Sheet 4 — Fitting the Linear and Quadratic Forms

OBJECTIVE (VERBATIM)	Practice problems on fitting linear form and quadratic form
DRILLS	Module 2, §2.2-2.3
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on fitting linear form and quadratic form

4.1 Objective & Formula Review

Form

$$y = a + bx$$

$$y = a + bx + cx^2$$

table ritual

Normal equations

$$\Sigma y = na + b\Sigma x; \Sigma xy = a\Sigma x + b\Sigma x^2$$

three rows: multiply by 1, x, x^2 and sum ($\Sigma x^2 y$ appears)

columns x, y, x^2 , x^3 , x^4 , xy, $x^2 y \rightarrow$ one Σ row $\rightarrow$ solve

4.2 Solved in Full

SOLVED 1

Fit $y = a + bx$ to (0,1) (1,2) (2,4) (3,4); report fitted values and residuals.

The sums give $a = 1.1$, $b = 1.1$ — the line $y = 1.1 + 1.1x$.

x	y (given)	y (fitted)	residual
0	1	1.1000	-0.1000
1	2	2.2000	-0.2000
2	4	3.3000	0.7000
3	4	4.4000	-0.4000

Residuals add to zero (first normal equation in the flesh), and the fitted value at $x = 4$ is **5.5**.

SOLVED 2

Fit a line to $x = 1 \dots 5$, $y = 2, 4, 5, 4, 5$ and predict at $x = 6$.

x	y	x^2	xy
1	2	1	2
2	4	4	8
3	5	9	15
4	4	16	16
5	5	25	25

$a = 2.2$, $b = 0.6 \rightarrow y = 2.2 + 0.6x$; $y(6) = 5.8$. (Same data as Module 2 Example 2.2.1 — fitting is deterministic.)

SOLVED 3

Fit $y = a + bx + cx^2$ to (1,2) (2,3) (3,6) (4,11).

Power-sum table:

x	y	x^2	x^3	x^4	xy	$x^2 y$
1	2	1	1	1	2	2
2	3	4	8	16	6	12
3	6	9	27	81	18	54

x	y	x ²	x ³	x ⁴	xy	x ² y
4	11	16	64	256	44	176

Solving the 3×3 gives $a = 3$, $b = -2$, $c = 1$: the data sit on $y = x^2 - 2x + 3$ exactly, so the residuals vanish — an exact fit is allowed and worth naming. Prediction: $y(5) = 18$.

SOLVED 4

Can a straight line be the least-squares answer for $x = 1 \dots 4$, $y = 0, 1, 4, 9$?

Only in the least-squares sense: $a = -4$, $b = 3 \rightarrow y = -4 + 3x$ passes near the points but cannot follow curvature — the residual pattern is systematic (high, low, low, high), the classic sign to reach for the quadratic form.

4.3 Practice Set

P1	Quadratic through $x = 0 \dots 4$, $y = 0, 1, 4, 9, 16$.	a=0, b=0, c=1 — recovers $y = x^2$ exactly
P2	Predict $x = 4$ on the SOLVED-1 line.	$1.1 + 1.1 \cdot 4 = 5.5$
P3	Line for $x = 2, 4, 5, 4, 3, 1, 2$; $y = 1, 3, 4, 3, 2, 1, 1$.	y = -0.3571 + 0.8333x

4.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 4  linear and quadratic fits
=====
SOLVED 1.  Fit y = a + bx to  (0,1) (1,2) (2,4) (3,4)
n=4  Sx=6  Sy=11  Sx^2=14  Sxy=22
b = 1.1  a = 1.1  ->  y = 1.1 + 1.1x
fitted values and residuals:
x=0:  y_hat=1.1000  residual=-0.1000
x=1:  y_hat=2.2000  residual=-0.2000
x=2:  y_hat=3.3000  residual=+0.7000
x=3:  y_hat=4.4000  residual=-0.4000
check: sum of residuals = 0 as required (normal equation): True
estimate at x=4: y = 5.5

SOLVED 2.  Fit y = a + bx to  x=1..5, y=2,4,5,4,5
n=5  Sx=15  Sy=20  Sx^2=55  Sxy=66
b = 0.6  a = 2.2  ->  y = 2.2 + 0.6x
y = 2.2 + 0.6x ;  y(6) = 5.8

SOLVED 3.  Quadratic fit to (1,2) (2,3) (3,6) (4,11)
normal equations:
4a + 10b + 30c = 22
10a + 30b + 100c = 70
30a + 100b + 354c = 244
solution:  a = 3,  b = -2,  c = 1
FITTED:  y = 3 + (-2)x + 1x^2  (exact here: y = x^2 - 2x + 3)
predicted y(5) = 18

PRACTICE
P4: fit line to x=1..6, y=0,1,4,9,16,25 restricted to the first 4 points?
n=4  Sx=10  Sy=14  Sx^2=30  Sxy=50
b = 3  a = -4  ->  y = -4 + 3x
y = -4 + 3x (a straight line cannot follow a parabola closely)
P5: quadratic through x=0..4, y=0,1,4,9,16  (should recover y = x^2)
a=0, b=0, c=1  ->  y = x^2 exactly as expected
P6: predict at x=4 from solved 1
y(4) = a+4b = 1.1 + 4.4 = 5.5 for the line y = 1.1 + 1.1x
P7: fit a line to x=2,4,5,4,3,1,2  y=1,3,4,3,2,1,1
b = 0.8333,  a = -0.3571  ->  y = -0.3571 + 0.8333x

[tokens saved]
```

Tutorial Sheet 5 — The Power Curve and Correlation

OBJECTIVE (VERBATIM)	Practice problems on $y = ax^b$ and correlation
DRILLS	Module 2, §2.4–2.5
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on $y = ax^b$ and correlation

5.1 Objective & Formula Review

Tool

$$y = ax^b$$

r

interpretation

Recall

log-log: $Y = \ln a + bX$; fit line; un-log the intercept

$$(n\sum xy - \sum x \sum y) / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$$

$|r| \geq 0.95$ very strong; ≥ 0.8 strong; ≤ 0.3 negligible

5.2 Solved in Full

SOLVED 1

Fit $y = ax^b$ to (1,2.0) (2,4.1) (3,6.3) (4,8.0) (5,10.2).

x	y	$X=\ln x$	$Y=\ln y$	X^2	XY
1	2.0	0.0000	0.6931	0.0000	0.0000
2	4.1	0.6931	1.4110	0.4805	0.9780
3	6.3	1.0986	1.8405	1.2069	2.0221
4	8.0	1.3863	2.0794	1.9218	2.8827
5	10.2	1.6094	2.3224	2.5903	3.7377

In logs: $b = 1.0082$, $\ln a = 0.7039 \rightarrow a = 2.0217$. $y = 2.0217 \cdot x^{1.0082}$ — the fit itself tells you the data are essentially proportional. Estimate $y(6) = 12.3100$.

SOLVED 2

Find r : $x = 10, 12, 13, 15, 17, 18, 20, 22$; $y = 78, 80, 83, 85, 88, 92, 95, 97$.

x	y	x^2	y^2	xy
10	78	100	6084	780
12	80	144	6400	960
13	83	169	6889	1079
15	85	225	7225	1275
17	88	289	7744	1496
18	92	324	8464	1656
20	95	400	9025	1900
22	97	484	9409	2134

The five column sums feed $r = (8 \cdot 11280 - 127 \cdot 698) / \sqrt{(951 \cdot 2716)} = 1594 / 1607.14 = 0.9918$: a near-perfect positive, almost exactly linear, pair of series.

SOLVED 3

$x = 1, 2, 3, 4$ with $y = 2, 4, 6, 8$: r ?

Collinear data: the formula collapses to **1**. Worth remembering: $r = +1$ is the algebraic signature of exact proportionality.

SOLVED 4

Interpret: $r = 0.9918$ for 8 pairs — “99% of y is explained by x ”?

No — r is the linear correlation; “explained” is $r^2 = 0.9837 \rightarrow$ about 98% of the variance of y on this scale. Name r^2 correctly and the examiner stops worrying.

5.3 Practice Set

P1	r for $x = 1 \dots 5$, $y = 5, 4, 3, 2, 1$.	-1 (exact negative line)
P2	Power law through (2,3) (4,6) (8,12).	$a = 1.5$, $b = 1$
P3	$n = 10$, $\Sigma x = 50$, $\Sigma y = 60$, $\Sigma x^2 = 300$, $\Sigma y^2 = 390$, $\Sigma xy = 330$: r ?	$300/\sqrt{(500 \cdot 300)} = \mathbf{0.7746}$
P4	Fit $y = ax^b$ through (1,1) (4,2) (9,3) (16,4).	$a = 1$, $b = 0.5$ — the square-root law
P5	If every x -value is identical, what is r ?	undefined (zero variance in x) — Σx^2 term is 0

5.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 5  power curve and correlation
=====
SOLVED 1.  Fit  $y = a \cdot x^b$  to (1,2.0) (2,4.1) (3,6.3) (4,8.0) (5,10.2)
  log:  $Y = \ln a + bX$ ;    $SX=4.7875$   $SY=8.3465$   $SXX=6.1995$   $SXY=9.6205$ 
   $b = 1.0082$ ;  $\ln a = 0.7039$    $\rightarrow a = 2.0217$ 
  FITTED:  $y = 2.0217 x^{1.0082}$   (nearly proportional:  $b \sim 1$ )
  estimate  $y(6) = 12.3100$ 

SOLVED 2.  correlation for  $x=10,12,13,15,17,18,20,22$   $y=78,80,83,85,88,92,95,97$ 
   $Sx=127$   $Sy=698$   $Sx^2=2135$   $Sy^2=61240$   $Sxy=11280$ 
  numerator =  $nSxy - SxSy = 1594$ 
  denom =  $\sqrt{(951 \cdot 2716)} = 1607.1453$ 
   $r = 0.9918$   (very strong positive association)

SOLVED 3.  perfect line:  $x=1..4$   $y=2,4,6,8$ 
   $r = 1$  exactly 1: the formula gives 100% fit for collinear points

PRACTICE
P4:  $x=1..5$   $y=5,4,3,2,1$    $\rightarrow r = -1$  (check)
   $r = -1.0000$ 
P5: power law through (2,3) (4,6) (8,12): exactly  $y = 1.5x$ 
   $b = 1$ ,  $a = 1.5$  on log-log (any 3 collinear-through-origin points give this)
P6: given  $n=10$   $Sx=50$   $Sy=60$   $Sx^2=300$   $Sy^2=390$   $Sxy=330$    $\rightarrow r = ?$ 
   $r = 300/\sqrt{(500 \cdot 300)} = 0.7746$ 
P7: fit  $y = a x^b$  exactly to (1,1) (4,2) (9,3) (16,4):  $y = \sqrt{x}$ 
  log-log gives  $b = 1/2$ ,  $\ln a = 0$    $\rightarrow a = 1$   (three or more exact points pin it)
P8: if all  $x$  equal,  $r$  is: 0 / undefined / 1 ?
   $Sx^2$  term vanishes:  $nSx^2 - (Sx)^2 = 0$ , so  $r$  is undefined (no spread  $\rightarrow$  no association)

[tokens saved]
```

Tutorial Sheet 6 — Lines of Regression and Rank Correlation

OBJECTIVE (VERBATIM)	Practice problems on lines of regression and rank correlation
DRILLS	Module 2, §2.6–2.7
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on lines of regression and rank correlation

6.1 Objective & Formula Review

Tool

y on x

x on y

checks

angle

rank

Recall

$$y - \bar{y} = r(\sigma_y/\sigma_x)(x - \bar{x})$$

$$x - \bar{x} = r(\sigma_x/\sigma_y)(y - \bar{y})$$

$$r^2 = b_{yx}b_{xy}; \text{ lines meet at } (\bar{x}, \bar{y})$$

$$\tan \theta = |(m_1 - m_2)/(1 + m_1m_2)| \text{ with } m_1 = b_{yx}, m_2 = 1/b_{xy}$$

$$\rho = 1 - 6\Sigma D^2/n(n^2 - 1)$$

6.2 Solved in Full

SOLVED 1

$\bar{x} = 36, \bar{y} = 85, \sigma_x = 11, \sigma_y = 8, r = 0.66$: write both lines; predict y at $x = 45$.

$b_{yx} = 0.66 \cdot 8/11 = 0.4800 \rightarrow y - 85 = 0.48(x - 36)$, i.e. $y = 67.7200 + 0.48x$. $b_{xy} = 0.66 \cdot 11/8 = 0.9075 \rightarrow x = -41.1375 + 0.9075y$. Product $0.48 \cdot 0.9075 = 0.4356 = r^2 \checkmark$. Prediction: $y(45) = 89.3200$.

SOLVED 2

Regression of y on x for $x = 1 \dots 5, y = 2, 4, 5, 4, 5$ — what is it, exactly?

The line $y = 2.2 + 0.6x$ — identical to the least-squares line of the data, because least squares = regression of y on x. Say that sentence in Part B papers.

SOLVED 3

Angle between $y = 0.8x + 1$ and $x = 0.5y + 2$.

Both as $y = mx + c$: $m_1 = 0.8, m_2 = 1/0.5 = 2$. $\tan \theta = 0.4615$, so $\theta = 24.7751^\circ$.

SOLVED 4

Judges rank 10 candidates: $R_x = 1 \dots 10, R_y = 3, 1, 2, 6, 4, 5, 9, 7, 10, 8$.

candidate	R_x	R_y	D	D^2
1	1	3	-2	4
2	2	1	1	1
3	3	2	1	1
4	4	6	-2	4
5	5	4	1	1
6	6	5	1	1
7	7	9	-2	4
8	8	7	1	1

candidate	R _x	R _y	D	D ²
9	9	10	-1	1
10	10	8	2	4

$\Sigma D^2 = 22 \rightarrow \rho = 1 - 6 \cdot 22 / 990 = \mathbf{0.8667}$: close but not complete agreement; three candidates differ by two places ($D = \pm 2$ for candidates 1, 4 and 7).

6.3 Practice Set

P1	Lines $y = 1.2x + 3$ and $x = 0.5y + 1$: recover r .	$r^2 = 0.6 \rightarrow r = \mathbf{0.7746}$
P2	$n = 6$, $\Sigma D^2 = 20$: ρ ?	0.4286
P3	Slopes 0.5 and 2: what is r ?	$r = 1$ — the lines coincide
P4	On SOLVED 1's x -on- y line, estimate x at $y = 90$.	40.5375

6.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 6 regression and rank correlation
=====
SOLVED 1. Regression lines from summaries: xbar=36, ybar=85, sx=11, sy=8, r=0.66
  b(y on x) = 0.4800 -> y = 85.0 + 0.4800 (x - 36.0) = 67.7200 + 0.4800x
  b(x on y) = 0.9075 -> x = 36.0 + 0.9075 (y - 85.0) = -41.1375 + 0.9075y
  check r = sqrt(product of slopes) = 0.6600
  estimate: at x=45, y = 89.3200

SOLVED 2. Regression of y on x for x=1..5, y=2,4,5,4,5 (shared table with M2)
  slope b = 0.6, intercept a = 2.2 (least squares line = y on x line)
  line: y = 2.2 + 0.6x

SOLVED 3. angle between y = 0.8x + 1 and x = 0.5y + 2
  slopes 0.8 and 1/0.5 = 2.0
  tan(th) = |(0.8-2.0)/(1+1.6000)| = 0.4615 -> th = 24.7751 deg

SOLVED 4. rank correlation of 10 candidates
  D: [-2, 1, 1, -2, 1, 1, -2, 1, -1, 2]
  SD^2 = 22; rho = 1 - 6*22/(10*99) = 1 - 0.1333 = 0.8667

PRACTICE
P5: two lines y=1.2x+3 and x=0.5y+1 -> r^2 = b1*b2 = 0.6, r = sqrt(0.6)
  r = 0.7746
P6: n=6, SD^2=20 -> rho = 1 - 120/(6*35) = 0.4286
P7: if the two regression slopes are 0.5 and 2, what is r?
  r^2 = 0.5*2 = 1 -> r = 1 (perfect correlation, the two lines coincide)
P8: using the SOLVED 1 line x - 36 = 0.9075 (y - 85), estimate x at y = 90
  x = 36 + 0.9075*(90-85) = 40.5375
```

[tokens saved]

Tutorial Sheet 7 — Mathematical Expectations

OBJECTIVE (VERBATIM)	Practice problems on mathematical expectations
DRILLS	Module 3, §3.2-3.3
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on mathematical expectations

7.1 Objective & Formula Review

Fact

$E(X)$

$V(X)$

linearity

functions

Recall

$\sum x \cdot p(x)$ — or $\int x f(x) dx$

$E(X^2) - [E(X)]^2$; $\sigma = \sqrt{V}$

$E(aX + b) = aE(X) + b$; $V(aX + b) = a^2V(X)$

$E[g(X)] = \sum g(x)p(x)$ — feed g before summing

7.2 Solved in Full

SOLVED 1

pmf $P(0) = 1/4 \cdot (1/2)$, $3/8$, $3/8$, $1/8$ on $0 \dots 3$: E , V , and $E(3X+2)$.

x	$P(x)$	xP	x^2P
0	1/8	0	0
1	3/8	0.375	0.375
2	3/8	0.75	1.5
3	1/8	0.375	1.125

$E(X) = 1.5$, $V(X) = 0.75$; $E(3X+2) = 3 \cdot 1.5 + 2 = 6.5$.

SOLVED 2

Continuous: $f(x) = (2x+1)/2$ on $(0,1)$. Find $E(X)$, $V(X)$.

Integrate: $E = 7/12 = 0.5833$; $E(X^2) = \int_0^1 x^2(x+1/2)dx = 5/12$, so $V = 5/12 - (7/12)^2 = 11/144 = 0.0764$.
Check the density first: $\int_0^1 f = 1$ ✓ (the log prints that line before any expectation).

SOLVED 3

A die pays $\pounds(\text{face})^2$; entry costs $\pounds 10$. Fair?

$E(\text{payoff}) = (1+4+9+16+25+36)/6 = 91/6 = 15.1667 > 10$ — the game profits the player by **5.1667** per turn. House edge is negative: wrong side of the table for the organiser.

SOLVED 4

$P(x) = k \cdot x^2$ on $x = 1 \dots 4$: find k , then $E(X)$.

$1 = k(1+4+9+16) = 30k \rightarrow k = 1/30$; $E = (1+8+27+64)/30 = 3.3333$.

7.3 Practice Set

P1	$P(x) = x/10$ on $1 \dots 4$: E and V ?	$E = 3$, $V = 1$
P2	$f(x) = 3x^2$ on $(0,1)$: E and V ?	$E = 0.75$, $V = 0.0375$
P3	$E(X) = 5$, $V(X) = 4$: $E(2X-1)$, $\text{sd}(2X-1)$?	$E = 9$, $\text{sd} = 4$
P4	Two fair dice, $X = \text{sum}$: E and V ?	$E = 7$, $V = 5.8333$

7.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 7 expectations
=====
SOLVED 1. pmf P(0)=1/8 P(1)=3/8 P(2)=3/8 P(3)=1/8
E(X) = 1.5 E(X^2) = 3
V(X) = E(X^2) - E(X)^2 = 0.75 = 0.7500; sd = 0.8660
E(3X+2) = 3*1.5+2 = 6.5; V(3X+2) = 9 V(X) = 6.75

SOLVED 2. continuous: f(x) = (2x+1)/2 on 0<x<1
check integral of f = 1 (valid density)
E(X) = int t f = 7/12 = 0.5833
E(X^2) = 5/12 = 0.4167
V(X) = 11/144 = 0.0764

SOLVED 3. fair die pays x^2 rupees for face x; cost of playing = 10
E(payoff) = (1+4+9+16+25+36)/6 = 91/6 = 15.1667
net expectation = 15.1667 - 10 = 5.1667 -> gain per play

SOLVED 4. find k, then E(X): P(x) = k x^2 for x = 1,2,3,4
1 = k(1+4+9+16) = 30k -> k = 0.0333
E(X) = k(1+8+27+64) = 96/30 = 3.3333

PRACTICE
P5: P(x) = x/10 for x=1..4 -> E(X) = ? V(X) = ?
E = 3, E2 = 10, V = 1
P6: f(x)=3x^2 on (0,1): E = 3/4, V = ?
E = 3/4 = 0.7500; V = 3/5 - 9/16 = 3/80 = 0.0375
P7: E(X)=5, V(X)=4 -> E(2X-1) and sd(2X-1)?
E = 2*5-1 = 9; V = 4*4 = 16, sd = 4
P8: two dice, X = sum. E(X) = ? V(X) = ?
E = 7 V = 5.8333 (mean 7 as symmetry predicts)

[tokens saved]
```

Tutorial Sheet 8 — Binomial and Poisson Distributions

OBJECTIVE (VERBATIM)	Practice problems on Binomial and Poisson distributions
DRILLS	Module 3, §3.4–3.5
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on Binomial and Poisson distributions

8.1 Objective & Formula Review

Family

binomial

poisson

fit

Recall

$C(n,x)p^xq^{n-x}$; $\mu = np$; $\sigma^2 = npq$; $\sigma = \sqrt{npq}$

$e^{-\lambda}\lambda^x/x!$; $\mu = \sigma^2 = \lambda$

$\hat{\lambda} = \Sigma fx/\Sigma f$; expected = $N \cdot P(x)$; tail via $1 - \Sigma$ or $-\Sigma$

8.2 Solved in Full

SOLVED 1

Fair coin six times: P(exactly 4 heads), P(at least 5).

$P(4) = C(6,4)/2^6 = 15/64 = \mathbf{0.2344}$; $P(\geq 5) = (6+1)/64 = \mathbf{0.1094}$. $\mu = 3$, $\sigma = \sqrt{1.5} = 1.2247$ — the mean-derivation of §3.4 in one line.

SOLVED 2

10% defectives, sample 15: P(≤ 2) and σ .

$P(0)+P(1)+P(2) = \mathbf{0.8159}$; $\sigma = \sqrt{1.35} = \mathbf{1.1619}$. The log evaluates each pmf term too, so partial credit is checkable.

SOLVED 3

Defects per 100 m over 200 pieces: 77, 75, 33, 11, 3, 1 for $x = 0 \dots 5$. Fit a Poisson.

$\hat{\lambda} = 191/200 = \mathbf{0.9550}$.

x	observed	P(X=x)	expected
0	77	0.3848	76.9624
1	75	0.3675	73.4991
2	33	0.1755	35.0958
3	11	0.0559	11.1722
4	3	0.0133	2.6674
5	1	0.0025	0.5095

The fitted P-column reproduces the observed shape; χ^2 between the two = **0.6723** — comfortable. $P(\text{at most 1 defect}) = e^{-\lambda}(1+\lambda) = \mathbf{0.7523}$.

SOLVED 4

A binomial has mean 4 and variance 3. Recover n, p, q.

$np = 4$, $npq = 3 \rightarrow q = 3/4 \rightarrow \mathbf{n = 16, p = 1/4, q = 3/4}$. The pair (μ, σ^2) determines a binomial — two equations, two unknowns, no guessing.

8.3 Practice Set

P1	10 fair coins once: P(more heads than tails)?	0.3770
P2	Poisson $\lambda = 3$: P(4 calls); P(2 or fewer)?	P(4) = 0.1680, P(≤ 2) = 0.4232
P3	$\lambda = 1$ per day complaints: P(at least 1 today)?	0.6321
P4	$n = 20$, $p = 0.05$: P(no defectives) and σ ?	P(0) = 0.3585, $\sigma = \sqrt{0.95} \approx 0.9747$

8.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 8  binomial and Poisson
=====
SOLVED 1.  fair coin, n=6:  P(exactly 4 heads), P(at least 5 heads)
  P(X=4) = C(6,4)(1/2)^6 = 15/64 = 0.2344
  P(X=5) = C(6,5)(1/2)^6 = 6/64 = 0.0938
  P(X=6) = C(6,6)(1/2)^6 = 1/64 = 0.0156
  P(X >= 5) = (C(6,5)+C(6,6))/64 = 7/64 = 0.1094
  mean = np = 3,  sd = sqrt(npq) = sqrt(1.5) = 1.2247

SOLVED 2.  10% of items defective, sample n=15
  P(X=0) = 0.2059
  P(X=1) = 0.3432
  P(X=2) = 0.2669
  P(X <= 2) = 0.8159    P(X >= 3) = 0.1841
  mean = 1.5,  sd = sqrt(15*0.1*0.9) = 1.1619

SOLVED 3.  Poisson fit:  defects per 100 m of cloth, 200 pieces
  lambda = (0*77+1*75+2*33+3*11+4*3+5*1)/200 = 191/200 = 0.9550
  P(0) = 0.3848;  expected = 76.9624
  P(1) = 0.3675;  expected = 73.4991
  P(2) = 0.1755;  expected = 35.0958
  P(3) = 0.0559;  expected = 11.1722
  P(4) = 0.0133;  expected = 2.6674
  P(5) = 0.0025;  expected = 0.5095
  fit check: chi-square = 0.6723 -> the Poisson model is acceptable
  P(no more than 1 defect) = 0.7523

SOLVED 4.  given binomial mean 4 and variance 3: find n and p
  np = 4, npq = 3  ->  q = 3/4,  p = 1/4,  n = 16
  check: np = 4.0,  npq = 3.0

PRACTICE
P5: 10 coins tossed once: P(more heads than tails)?
  = (210+120+45+10+1)/1024 = 0.3770
P6: calls per hour lambda=3: P(exactly 4) and P(2 or fewer)
  P(4) = 0.1680;  P(<=2) = 0.4232
P7: P(at least one defect) with lambda=1?
  1 - e^-1 = 0.6321
P8: n=20 p=0.05: mean, sd, P(X=0)?
  mean = 1,  sd = sqrt(0.95) = 0.9747,  P(0) = 0.95^20 = 0.3585

[tokens saved]
```

Tutorial Sheet 9 — Normal Distributions

OBJECTIVE (VERBATIM)	Practice Problems on Normal distributions
DRILLS	Module 3, §3.6 · table = Appendix B
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice Problems on Normal distributions

9.1 Objective & Formula Review

Recipe

area left of x

band

cut-off

68-95-99.7

Recall

$z = (x - \mu)/\sigma$, read $\Phi(z)$

$\Phi(z_2) - \Phi(z_1)$

read table backwards: $x = \mu + z\sigma$

0.6827, 0.9545, 0.9973 of the mass

9.2 Solved in Full

SOLVED 1

Heights $N(64.8 \text{ in}, \sigma = 2.5)$: $P(X > 70)$?

$z = 5.2/2.5 = 2.0800$; $P = 1 - \Phi(2.08) = \mathbf{0.0188}$ — about 1 in 53.

SOLVED 2

Same law: $P(60 < X < 68)$?

$z = -1.92$ and $+1.28$: $\Phi(1.28) - \Phi(-1.92) = \mathbf{0.8723}$: 87% of the population sits in that 8-inch band.

SOLVED 3

$X \sim N(500, 40)$: the value exceeded by the top 1%?

$\Phi(z) = 0.99 \rightarrow z = \mathbf{2.3263}$ (2.326); $x = 500 + 2.3263 \cdot 40 = \mathbf{593.0539}$.

SOLVED 4

Name the three sigma-coverage probabilities and use them to sanity-check a claim of “68% within $\pm 1.5\sigma$ ”.

$\pm 1\sigma$: **0.6827**; $\pm 2\sigma$: **0.9545**; $\pm 3\sigma$: **0.9973**. $\pm 1.5\sigma$ covers about 86.6%, so 68% at 1.5σ is a misquote — the numbers themselves refuse the claim.

9.3 Practice Set

P1	$P(-1.96 < Z < 1.96)$?	0.9500 — the 95% of inference chapter 5
P2	$X \sim N(500, 40)$: $P(X > 460)$?	0.8413
P3	Marks $N(72, 15)$: top-10% cutoff?	91.2233
P4	Coin $n = 200$, $P(\text{more than 110 heads})$ by normal approximation?	0.0896

9.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 9 normal distribution
```



```
=====
SOLVED 1. heights  $X \sim N(64.8 \text{ in}, \text{sd } 2.5)$ :  $P(X > 70)$ 
   $z = (70 - 64.8) / 2.5 = 2.0800$ 
   $P = 1 - \Phi(2.0800) = 1 - 0.9812 = 0.0188$ 

SOLVED 2. same distribution:  $P(60 < X < 68)$ 
   $z1 = -1.9200, z2 = 1.2800$ 
   $P = \Phi(1.2800) - \Phi(-1.9200) = 0.8997 - 0.0274 = 0.8723$ 

SOLVED 3. find the cut-off for the top 1% of  $N(500, \text{sd } 40)$ 
   $\Phi(z) = 0.99 \rightarrow z = 2.3263; x = 500 + 2.3263 * 40 = 593.0539$ 

SOLVED 4. sigma levels
   $P(\mu - 1 \text{ sd} < X < \mu + 1 \text{ sd}) = 0.6827$ 
   $P(\mu - 2 \text{ sd} < X < \mu + 2 \text{ sd}) = 0.9545$ 
   $P(\mu - 3 \text{ sd} < X < \mu + 3 \text{ sd}) = 0.9973$ 

PRACTICE
P5:  $X \sim N(0, 1)$ :  $P(-1.96 < Z < 1.96) = ?$ 
   $= 0.9500$ 
P6:  $X \sim N(500, 40)$ :  $P(X > 460) = ?$ 
   $z = -1.0000; P = \Phi(1.0000) = 0.8413$ 
P7: marks  $X \sim N(72, 15)$ : top 10% cut-off?
   $z_{0.90} = 1.2816; x = 72 + 15 * 1.2816 = 91.2233$ 
P8: normal approximation,  $n=200$   $p=0.5$ ,  $P(X > 110)$  with continuity correction
   $\mu=100, \text{sd} = \sqrt{50} = 7.0711; z = (109.5 - 100) / 7.0711 = 1.3435$ 
   $P = 1 - \Phi(1.3435) = 0.0896$ 

[tokens saved]
```

Tutorial Sheet 10 — Covariance and Correlation of Two Discrete RVs

OBJECTIVE (VERBATIM)	Practice problems on covariance, correlation of two discrete random variables
DRILLS	Module 4, §4.1–4.2
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on covariance, correlation of two discrete random variables

10.1 Objective & Formula Review

Fact	Recall
$E(XY)$	$\sum xy f(x,y)$ over the grid
Cov	$E(XY) - E(X)E(Y)$
ρ	$\text{Cov}/(\sigma_X \sigma_Y)$, always in $[-1, 1]$
$V(aX+bY)$	$a^2VX + b^2VY + 2ab \text{Cov}$

10.2 Solved in Full

SOLVED 1

Joint weights $[[1,2,1],[1,2,3]]/10$ on $x \in \{1,2\}$, $y \in \{1,2,3\}$: $E(X)$, $E(Y)$, Cov , ρ .

$P(x,y)$	$y=1$	$y=2$	$y=3$	$f(x)$
$x=1$	1/10	1/5	1/10	2/5
$x=2$	1/10	1/5	3/10	3/5
$g(y)$	1/5	2/5	2/5	1

Row sums $(4, 6)/10$, columns $(2, 4, 4)/10$. $E(X) = 1.6$, $E(Y) = 2.2$; $E(XY) = (1+4+3 + 2+8+18)/10 = 3.6$, so $\text{Cov} = 3.6 - 1.6 \cdot 2.2 = 0.0800$, $\rho = 0.2182$ — mild positive linkage.

SOLVED 2

For the same pair: $V(X+Y)$ and $V(X-Y)$.

$V(X) = 0.24$, $V(Y) = 0.56$ (log), so $V(X+Y) = 0.24 + 0.56 + 2(0.08) = 0.9600$ and $V(X-Y) = 0.24 + 0.56 - 0.16 = 0.6400$. Covariance decides whether the sum is more or less spread than the parts.

SOLVED 3

Is the table independent?

Cell (1,1) is 1/10 but $f(1) \cdot g(1) = 0.4 \cdot 0.2 = 0.08 \neq 0.1 \rightarrow$ **not independent**, and the non-zero ρ already said so.

SOLVED 4

$V(2X+3Y)$ for the same data?

$4(0.24) + 9(0.76) + 12(0.08) = 6.9600$.

10.3 Practice Set

P1	$X \perp Y$, $VX = 2$, $VY = 5$: $\text{Cov}(3X, 2Y)$ and $V(3X - 2Y)$?	Cov = 0, V = 38
P2	$\text{Cov} = 6$, $\sigma_X = 3$, $\sigma_Y = 4$: ρ ?	0.5
P3	Product table $[[2,2],[2,2]]/8$: covariance?	

Cov = 0.0 (independence again visible)**P4** $VX = VY = 1, \rho = 0.5: V(X+Y)?$ **3**

10.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 10 joint laws, covariance, correlation
=====
SOLVED 1. joint pmf with integer weights over 10:
rows x = 1,2 ; columns y = 1,2,3
x=1: y=1: 1/10 y=2: 2/10 y=3: 1/10
x=2: y=1: 1/10 y=2: 2/10 y=3: 3/10
marginals: f(x) = ['4/10', '6/10'], g(y) = ['2/10', '4/10', '4/10']
E(X) = 1.6 E(Y) = 2.2
E(XY) = 3.6 Cov = E(XY) - E(X)E(Y) = 0.0800
V(X) = 0.2400 V(Y) = 0.5600 rho = 0.2182

SOLVED 2. V(X+Y), V(X-Y) from SOLVED 1
V(X+Y) = VX + VY + 2Cov = 0.9600
V(X-Y) = VX + VY - 2Cov = 0.6400

SOLVED 3. independence check on the same table:
test f(x,y) = f(x)g(y) entrywise: e.g. (1,1): P = 1/10, f*g = 4*4/100 = 0.1600
1/10 = 0.1000 vs 16/100 = 0.1600 -> NOT independent (consistent with rho != 0)

SOLVED 4. linear transform: with Cov(X,Y) as above, V(2X+3Y)?
V(2X+3Y) = 4VX + 9VY + 12Cov = 6.9600

PRACTICE
P5: if X, Y independent with V(X)=2, V(Y)=5: Cov(3X,2Y) and V(3X-2Y)?
Cov = 0; V = 9*2 + 4*5 = 38
P6: rho = Cov/(sx sy) with Cov=6, sx=3, sy=4 -> rho = ?
rho = 6/12 = 0.5
P7: joint weights [[2,2],[2,2]]/8: covariance?
E(X)=E(Y)=1.5, E(XY)=2.25, Cov=0.0 -> product law, zero covariance
P8: if V(X)=V(Y)=1 and rho=0.5: V(X+Y)?
Cov = rho*1*1 = 0.5; V(X+Y) = 1+1+2*0.5 = 3

[tokens saved]
```

Tutorial Sheet 11 — Markov Chains, Stochastic Matrices, Probability Vectors

OBJECTIVE (VERBATIM)	Practice problems on Markov chain, stochastic matrices, probability vectors
DRILLS	Module 4, §4.3–4.4
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on Markov chain, stochastic matrices, probability vectors

11.1 Objective & Formula Review

Definition

probability vector
stochastic matrix
regular
chain step

Test to run

all entries ≥ 0 and $\Sigma = 1$
every row a probability vector (rows $\Sigma = 1$)
some power P^k all positive
 $x' = xP$

11.2 Solved in Full

SOLVED 1

Which of $v_1 = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$, $v_2 = (\frac{3}{4}, \frac{1}{4})$, $v_3 = (\frac{2}{3}, \frac{1}{2})$ are probability vectors?

v1, v2 yes; v3 no (sum = 7/6) — v_1 and v_2 sum to 1 with nonnegative parts; v_3 sums to 7/6.

SOLVED 2

Which of $A = [[\frac{1}{3}, \frac{2}{3}], [\frac{1}{4}, \frac{3}{4}]]$, $B = [[\frac{1}{2}, \frac{2}{3}], [\frac{1}{2}, \frac{1}{3}]]$, $C = [[0, 1], [1, 0]]$ are stochastic?

Row sums: $A = 1, 1$; $B = 7/6, 5/6$; $C = 1, 1$. **A and C stochastic; B no (row sums 7/6, 5/6).**

SOLVED 3

Are they regular?

C not regular (periodic, swaps states); A regular: C's powers only ever alternate I and C — periodic, never positive; A already has all positive entries.

SOLVED 4

Today a customer is with brand 1 with prob 0.7; transition matrix $P = [[0.8, 0.2], [0.5, 0.5]]$. Forecasts for the next two days?

$x_0 P =$ and $x_0 P^2$ from the log: **$x_1 = (0.71, 0.29)$, $x_2 = (0.713, 0.287)$** — the chain starts at 0.71 and creeps toward its equilibrium $\pi = (0.7143, 0.2857)$, which is what the next sheet computes properly.

11.3 Practice Set

P1	Is $(0.4, -0.1, 0.7)$ a probability vector?	no (negative component)
P2	Make $[[0.5, 0.2], [?, 1]]$ row-stochastic: the missing row entry?	each row to 1: $[[0.5, 0.5], [?, 1]]$ — missing piece pattern 0.3 per row
P3	$x_0 = (1, 0)$, $P = [[0.6, 0.4], [0.2, 0.8]]$: x_1 and x_2 ?	$x_1 = (0.6, 0.4)$, $x_2 = (0.44, 0.56)$
P4	$[[1, 0], [0.3, 0.7]]$: stochastic? regular?	stochastic, not regular

11.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 11 probability vectors and stochastic matrices
=====
SOLVED 1. which are probability vectors?
  v1 = ['1/2', '1/3', '1/6']: sum = 1, nonneg = True -> vector? True
  v2 = ['3/4', '1/4']: sum = 1, nonneg = True -> vector? True
  v3 = ['2/3', '1/2']: sum = 1.1667, nonneg = True -> vector? False

SOLVED 2. which matrices are stochastic (rows sum to 1)?
  A: row sums ['1', '1'] -> stochastic: True
  B: row sums ['1.1667', '0.8333'] -> stochastic: False
  C: row sums ['1', '1'] -> stochastic: True

SOLVED 3. regularity check
  C = [[0,1],[1,0]]: C^2 = I -> every even power has zeros on 0-entries,
  C^2 = [['1', '0'], ['0', '1']], C^3 = [['0', '1'], ['1', '0']]
  no power of C is strictly positive -> C is stochastic but NOT regular
  A has all entries > 0 already -> A is regular

SOLVED 4. one step: x0 = (0.7, 0.3), P = [[0.8,0.2],[0.5,0.5]]
  x1 = x0 P = [0.71, 0.29]
  x2 = x0 P^2 = [0.713, 0.287]

PRACTICE
P5: is v = (0.4, -0.1, 0.7) a probability vector?
  sums to 1 but has a negative entry -> NO
P6: 3x3 row [0.5, 0.2, ?] stochastic -> missing entry = 0.3
P7: x0=(1,0), P=[[0.6,0.4],[0.2,0.8]]: x1 and x2?
  x1 = [0.6, 0.4], x2 = [0.44, 0.56]
P8: square matrix with row [1, 0] and [0.3, 0.7]: stochastic? regular?
  stochastic yes; not regular: entry (0,1) is 0 in every power (absorbing state)

[tokens saved]
```

Tutorial Sheet 12 — Higher Transition Probabilities and Stationary Distributions

OBJECTIVE (VERBATIM)	Practice Problems on higher transition probabilities and stationary distributions
DRILLS	Module 4, §4.5–4.6
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice Problems on higher transition probabilities and stationary distributions

12.1 Objective & Formula Review

Task	Move
k-step prob	entry (i,j) of P^k
t-step forecast	$x_0 P^t$
stationary	solve $\pi P = \pi$ with $\sum \pi = 1$
2-state shortcut	$\pi_1/\pi_2 = p_{21}/p_{12}$

12.2 Solved in Full

SOLVED 1

P = [[0.9,0.1],[0.2,0.8]]: write P^2 , the two-step sunny→sunny probability, the three-step sunny→rainy probability, and π .

P^2 rows: **(0.83, 0.17) / (0.34, 0.66)**; two-step S→S = **0.83**; three-step S→R = **0.219**. Balance $0.1\pi_1 = 0.2\pi_2$ + normalisation: $\pi = \textbf{(0.6667, 0.3333)}$; and $x_0 P^{20} = \textbf{(0.6665, 0.3335)}$ — the forecast line up to rounding.

SOLVED 2

Brand war: A retains 70%, B retains 60%, today 50-50. Shares in a month, two months, forever?

$x_1 = \textbf{(0.55, 0.45)}$, $x_2 = \textbf{(0.565, 0.435)}$, and the stationary solution $\pi = \textbf{(0.5714, 0.4286)} = (4/7, 3/7)$ — read off the balance equation $0.3A = 0.4B$ in one line, then normalise.

SOLVED 3

3-state chain (rows $[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}]$, $[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$, $[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}]$): stationary vector?

Four equations (three balance, one normalisation), solved in the log: $\pi = \textbf{(0.3721, 0.3488, 0.2791)}$ — state 1 holds a slight edge; the flat middle row spreads its own mass evenly, so the outer rows decide the tilt.

SOLVED 4

Show convergence numerically for SOLVED 1's P.

row 0 of $P^{12} = \textbf{((0.6713, 0.3287))}$, a hair above $\pi = (0.6667, 0.3333)$ — regular chains exponentially approach π , the visible proof of the theorem.

12.3 Practice Set

P1	$P = [[\frac{1}{2}, \frac{1}{2}], [0, 1]]$: closed form of P^n ?	$P^n = [[\frac{1}{2}^n, 1 - \frac{1}{2}^n], [0, 1]]$
P2	π for $[[0.75, 0.25], [0.6, 0.4]]$?	(0.7059, 0.2941)
P3	$\pi = (0.2, 0.5, 0.3)$: long-run share of state 2?	0.5

12.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 12  higher powers and stationary distributions
=====
SOLVED 1.  P = [[0.9,0.1],[0.2,0.8]]: P^2, P^3, and two-step probabilities
  P^2 = [['0.83', '0.17'], ['0.34', '0.66']]
  P^3 = [['0.781', '0.219'], ['0.438', '0.562']]
  2-step S->S prob = 0.83;  3-step S->R = 0.219
  stationary:  pi = (0.6667, 0.3333) = (0.6667, 0.3333)
  from x0 = (1/2,1/2):  x1 = ('0.55', '0.45'),  x5 = ('0.6387', '0.3613')
  x20 = ('0.6665', '0.3335') -> converging to pi

SOLVED 2.  market shares:  A/B brand retention 0.7/0.6; today (0.5, 0.5)
  after 1 month:  (0.55, 0.45);  after 2 months: (0.565, 0.435)
  long run:  (0.5714, 0.4286) = (0.5714, 0.4286)

SOLVED 3.  3-state chain, solve pi P = pi by hand
  pi = (0.3721, 0.3488, 0.2791)
  check pi P = pi : [[16/43, 15/43, 12/43]] == [[16/43, 15/43, 12/43]]

SOLVED 4.  converge check P^12 vs pi
  row 0 of P^12 = ['0.6713', '0.3287']  (stationary was 0.6667/0.3333 for P=[[.9,.1],[.2,.8]]? here it is
  ('0.6667', '0.3333'))

PRACTICE
P5: P = [[0.5,0.5],[0,1]]: P^n?
  P^2 = [['0.25', '0.75'], ['0', '1']],  P^4 = [['0.0625', '0.9375'], ['0', '1']]  (upper-left halves ->
  0)
P6: stationary of [[0.75,0.25],[0.6,0.4]]?
  pi = (0.7059, 0.2941) = (0.7059, 0.2941)
P7: if a chain is regular with pi = (0.2,0.5,0.3), long-run share of state 2?
  0.5 in the long run, from any starting vector

[tokens saved]
```

Tutorial Sheet 13 — Tests of Significance for Large Samples

OBJECTIVE (VERBATIM)	Practice Problems on test of significance for large samples
DRILLS	Module 5, §5.1–5.3
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice Problems on test of significance for large samples

13.1 Objective & Formula Review

Question	z
one mean	$(\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
one proportion	$(\hat{p} - p_0)/\sqrt{(p_0q_0/n)}$
two means	$(\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
two proportions	with pooled $\bar{p}$ in the SE
verdict	$ z $ vs 1.96 (5%) or 2.576 (1%); $p = 2(1 - \Phi(z))$

13.2 Solved in Full

SOLVED 1

$\mu_0 = 50$ kg, $\sigma = 10$, $n = 400$, $\bar{x} = 50.8$: test at 5%.

SE = 0.5, $z = 1.6000$, $p = 0.1096 > 0.05 \rightarrow$ not significant: an 0.8 kg wobble at $n = 400$ with $\sigma = 10$ is still noise.

SOLVED 2

Claim $p_0 = 0.5$; 480 of 1000: verdict?

$z = -1.2649$, $p = 0.2059 \rightarrow$ do not reject; 20 short of half is inside the $\pm 1.96\sqrt{(250)} = \pm 31$ band.

SOLVED 3

Two teaching methods: (285, σ 12, n 36) vs (279, σ 10, n 49): significant at 5%?

SE = $\sqrt{(144/36 + 100/49)} = \sqrt{6.0408} = 2.4578$; $z = 2.4412$, $p = 0.0146 \rightarrow$ significant at 5%, just barely; not at 1%.

SOLVED 4

City 320/500 vs village 240/400 support: compare.

0.64 vs 0.60, pooled 0.6222 , $z = 1.2299$, $p = 0.2187 \rightarrow$ no significant difference: the SE of the pooled proportions swallows a 4-point gap.

13.3 Practice Set

P1	95% CL, $n = 225$, $\bar{x} = 72$, $s = 4.5$.	(71.412, 72.588)
P2	$\bar{x} = 1540$, $\sigma = 200$, $n = 100$ vs $\mu_0 = 1500$, at 1%?	$z = 2.0000$, not significant at 1%
P3	112 of 400 (28%) vs $p_0 = 25\%$: z ?	$z = 1.3856$ — not significant ($p \approx 0.17$)
P4	Which z is the two-tailed 1% cut-off?	2.576

13.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 13  large-sample tests (z)
=====
SOLVED 1.  one mean:  mu0 = 50 kg, sd = 10, n = 400, xbar = 50.8
  SE = 10/sqrt(400) = 0.5
  z = (50.8-50)/0.5 = 1.6000    two-tail p = 0.1096
  at 5% (|z| > 1.96):  not significant

SOLVED 2.  one proportion:  claimed p0 = 0.5, observed 480 of 1000
  phat = 0.48;  SE0 = sqrt(0.25/1000) = 0.0158
  z = -1.2649;  p = 0.2059  ->  do not reject H0 at 5%

SOLVED 3.  two means (large):  (285,12,36) vs (279,10,49)
  SE = sqrt(144/36 + 100/49) = sqrt(4.0000 + 2.0408) = 2.4578
  z = 6/2.4578 = 2.4412;  p = 0.0146  ->  significant at 5%

SOLVED 4.  two proportions:  320/500 vs 240/400
  p1 = 0.64,  p2 = 0.6,  pooled p = 0.6222
  SE = 0.0325;  z = 1.2299;  p = 0.2187 -> not significant at 5%

PRACTICE
P5: 95% confidence limits for mean, n=225, xbar=72, s=4.5
  SE = 0.3;  CL = 72 +/- 1.96*0.3 = (71.412, 72.588)
P6: xbar=1540, sd=200, n=100, mu0=1500: z and decision at 1%?
  z = 2.0000;  2.576 threshold -> not significant at 1%
P7: sample 112/400 (28%) vs p0 = 0.25: z?
  z = 1.3856;  p = 0.1659 -> not reject at 5%
P8: which is the two-tailed 1% critical z? 1.645 / 1.96 / 2.576
  2.576 (1.96 is 5% two-tailed; 1.645 is 5% one-tailed)

[tokens saved]
```

Tutorial Sheet 14 — Tests of Significance for Small Samples

OBJECTIVE (VERBATIM)	Practice problems on test of significance for small samples
DRILLS	Module 5, §5.4–5.5
FORMAT	2 contact hours · problems worked at the bench, log reprinted below

AS PRESCRIBED (VERBATIM)

Practice problems on test of significance for small samples

14.1 Objective & Formula Review

Tool

one-sample t
two-sample t
 χ^2 GOF
tables

Recall

$(\bar{x} - \mu_0)/(s/\sqrt{n})$, $df = n-1$
pooled s^2_p ; $df = n_1 + n_2 - 2$
 $\Sigma(O-E)^2/E$; $df = k-1$ ($-m$ for m fitted parameters)
Appendix B has t and χ^2 critical values for every df 1–30

14.2 Solved in Full

SOLVED 1

Breaking strengths (kg): A = (14.1, 13.6, 14.8, 12.9, 13.7, 13.5, 14.6, 12.5, 14.4, 13.9), B = (13.2, 12.8, 14.0, 13.1, 12.6, 13.5, 13.0, 12.9). Equality of means at 5%?

$\bar{x}_A = 13.8000$ ($n = 10$), $\bar{x}_B = 13.1375$ ($n = 8$); the pooled variance comes to 1.9353, $t = 2.2622$ with $df = 16$, and the log draws the 2.5% two-tail critical value $t_{0.025}(16) = 2.1199$ from the distribution definition. $|t| = 2.2622 > 2.1199 \rightarrow$ significant at 5%, $p = 0.0380$; at 1% the verdict flips. Two decimal places of margin — quote the p and the level together, as here.

SOLVED 2

9 repair times (28,33,29,31,27,30,32,26,34) vs claimed $\mu = 30$: t ?

$\bar{x} = 30$ — dead on the claim, so $t = 0.0000$, $p = 1.0000$: the data neither support nor refute anything; a null sample for a null hypothesis.

SOLVED 3

Weekday accidents 32, 25, 27, 30, 26 — uniform over the week?

Expected 28 each.

day	Mon	Tue	Wed	Thu	Fri
observed	32	25	27	30	26
expected	28	28	28	28	28
$(O-E)^2/E$	0.5714	0.3214	0.0357	0.1429	0.1429

$\chi^2 = 1.2143$, $df = 4$, critical = **9.4877**, $p = 0.8757$: no evidence of weekday bias — Friday's 26 against an expected 28 is the largest single-day wobble and still noise.

SOLVED 4

Die rolled 120 times: 23,18,21,25,15,18 — fair?

$\chi^2 = 3.4000$ against $df = 5$ cut-off 11.07; $p = 0.6386 \rightarrow$ keep playing with this die (no evidence of bias at any standard level).

14.3 Practice Set

P1	$t_{0.025}$ for $df = 14$?	2.1448
P2	$n = 8$, $\bar{x} = 21$, $s = 2.4$, $\mu_0 = 24$: t ?	-3.5355 — significant ($ t > 2.365$)
P3	$\chi^2 = 12.5$, $df = 4$: verdict?	reject ($12.5 > 9.4877$)
P4	Why $df = n - 1$ in the one-sample t ?	$df = n-1$: one estimated parameter

14.4 Computation Log

Every solved answer above is a token pulled from this log:

```
=====
TUTORIAL 14  small-sample tests
=====
SOLVED 1.  two small samples (t, pooled): breaking strengths
  A: n=10, mean=13.8000, s^2=0.5267; B: n=8, mean=13.1375, s^2=0.1941
  pooled s^2 = 0.3812
  t = 2.2622   df = 16   p = 0.0380
  t_0.025(16) = 2.1199  ->  significant at 5%

SOLVED 2.  one-sample t:  9 repairs, claimed mean 30 min
  mean = 30,   s^2 = 7.5000,   s = 2.7386
  t = (30-30)/sqrt(7.5000/9) = 0.0000   df = 8
  p = 1.0000  ->  no evidence against the claim at 5%

SOLVED 3.  chi-square goodness of fit:  accidents by weekday
  total = 140,   expected = 28 per day
  day 1:  (32-28)^2/28 = 0.5714
  day 2:  (25-28)^2/28 = 0.3214
  day 3:  (27-28)^2/28 = 0.0357
  day 4:  (30-28)^2/28 = 0.1429
  day 5:  (26-28)^2/28 = 0.1429
  chi2 = 1.2143   df = 4   p = 0.8757   critical(5%) = 9.4877
  ->  no deviation from uniform: True

SOLVED 4.  chi-square for a 6-sided die, 120 rolls
  expected 20 per face
  chi2 = 3.4000,   df = 5,   critical(5%) = 11.0705,   p = 0.6386
  ->  fair (no evidence otherwise)

PRACTICE
P5: t critical value at 5% two-tail for df = 14?
  2.1448
P6: n=8 sample, mean 21, s = 2.4, H0 mu = 24: t?
  t = -3/(2.4/2.8284) = -3.5355;  |t| vs t_0.025(7) = 2.3646: significant: True
P7: chi2 = 12.5, df = 4:  verdict at 5%?
  critical = 9.4877 -> reject goodness of fit: True
P8: why df = n-1 for one-sample t?
  one parameter (the mean) is estimated from the data; 7 independent deviations remain for n=8 style
  counts

[tokens saved]
```

P A R T

C

Appendices

The reference apparatus of the course: a formula sheet per module, statistical tables computed rather than copied, the GATE-style set priced by the CCA rubric, and the commands that reproduce every number in this book.

IN THIS PART

- Appendix A — Formula Sheet — One Page per Module
- Appendix B — Statistical Tables, Computed for This Book
- Appendix C — GATE-Style Practice Set (CCA Activity 2)
- Appendix D — Reproducing Every Number in This Book

Appendix A — Formula Sheet — One Page per Module

The five sheets below are the complete tool list of the course: nothing in any worked example of this book uses a formula that is not on its module's sheet. Each entry is exactly the form used in the computations of Parts A and B, so the logs reproduce line-for-line.

A.1 · Module 1 — modular arithmetic

Item	Statement
Congruence	$a \equiv b \pmod{n} \Leftrightarrow n \mid (a - b)$; multiply-compatible, and cancellable only after dividing the modulus by $\gcd(c, n)$
d-criterion	$ax \equiv b \pmod{n}$ is solvable iff $d = \gcd(a, n)$ divides b ; then exactly d solutions $\pmod{n}$
Reduction	$a'x \equiv b' \pmod{n'}$ with $a = da'$, $b = db'$, $n = dn'$; one solution is $x_0 = (a')^{-1}b'$ and the rest are $x_0 + kn'$, $k = 1, \dots, d-1$
Diophantine	$ax + by = c$ is solvable iff $d = \gcd(a, b) \mid c$; from one solution (x_0, y_0) : $x = x_0 + (b/d)t$, $y = y_0 - (a/d)t$
Chinese Remainder	$x \equiv r_i \pmod{n_i}$ with the n_i pairwise coprime has exactly one class $\pmod{N = \prod n_i}$
Euler	$\gcd(a, n) = 1 \Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$; $\varphi(pq) = (p-1)(q-1)$ for distinct primes
Fermat	p prime, $p \nmid a \Rightarrow a^{p-1} \equiv 1 \pmod{p}$; inverse $a^{-1} \equiv a^{p-2}$
Wilson	p prime $\Leftrightarrow (p-1)! \equiv -1 \pmod{p}$
RSA	$n = pq$, $ed \equiv 1 \pmod{\varphi(n)}$; $C \equiv M^e \pmod{n}$, $M \equiv C^d \pmod{n}$

A.2 · Module 2 — least squares, correlation, regression

Item	Formula
Line $y = a + bx$	$na + b\sum x = \sum y$ and $a\sum x + b\sum x^2 = \sum xy$ — solve the pair
Parabola $y = a + bx + cx^2$	three normal equations: the sums $\sum y$, $\sum xy$, $\sum x^2y$ taken with weights 1, x , x^2 — the full system is worked in §2.2
Curve $y = ax^b$	$\log y = \log a + b \log x$ — fit a straight line in $(\log x, \log y)$, then un-log the intercept
Correlation	$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2} \sqrt{n\sum y^2 - (\sum y)^2}}$
Regression lines	$y - \bar{y} = b_{yx}(x - \bar{x})$ and $x - \bar{x} = b_{xy}(y - \bar{y})$; both pass through $(\bar{x}, \bar{y})$; $r^2 = b_{yx}b_{xy}$
Angle between them	$\tan \theta = \frac{1 - b_{yx}b_{xy}}{b_{yx} + b_{xy}}$ (the acute angle)
Rank correlation	$\rho = 1 - \frac{6\sum D^2}{n(n^2 - 1)}$, D = difference of ranks

A.3 · Module 3 — distributions

Model	Density / mass, mean, variance
pmf / pdf	$\sum p(x) = 1, p(x) \geq 0; \int f dx = 1; E[g(X)] = \sum g p = \int g f dx$
Variance	$V(X) = E(X^2) - [E(X)]^2; V(aX + b) = a^2 V(X)$
Binomial(n, p)	$P(X = x) = \binom{n}{x} p^x q^{n-x}; \text{mean } np, \text{variance } npq, \text{SD } \sqrt{npq} \text{ — derivations in §3.4}$
Poisson(λ)	$P(X = x) = e^{-\lambda} \lambda^x / x!; \text{mean} = \text{variance} = \lambda$
Normal $N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}; \text{standardise } Z = (X - \mu)/\sigma; 68\text{-}95\text{-}99.7 \text{ rule}$
Approximations	Binomial $\rightarrow$ Poisson for large n, small p ($\lambda = np$); Binomial $\rightarrow$ Normal for np and nq both ≥ 5

A.4 · Module 4 — joint distributions and Markov chains

Item	Formula
Marginals	$f(x) = \sum_y h(x, y), g(y) = \sum_x h(x, y); \text{independence} \Leftrightarrow h = f \cdot g \text{ in every cell}$
Covariance	$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$
Correlation	$\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y), -1 \leq \rho \leq 1; \text{independent} \Rightarrow \rho = 0 \text{ (not conversely)}$
Variance of a sum	$V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y)$
Chain step	$x_{n+1} = x_n P$ (row vector $\times$ stochastic matrix); after n steps, $x_0 P^n$
Regular chain	P^k all-positive for some k $\Rightarrow x_n \rightarrow \pi$ from every start
Equilibrium	$\pi P = \pi, \sum \pi_i = 1$ — solve the homogeneous system plus the normalisation

A.5 · Module 5 — inference

Item	Formula
Standard error	$\sigma_{\bar{x}} = \sigma/\sqrt{n}$ when σ is known; estimated by $s/\sqrt{n}$ otherwise
Confidence limits	$\bar{x} \pm z_{\alpha/2} \sigma_{\bar{x}}$ — large sample; substitute s for σ (CLT)
One-sample z	$z = (\bar{x} - \mu_0) / (s/\sqrt{n}); \text{reject } H_0 \text{ at level } \alpha \text{ when } z > z_{\alpha/2} \text{ (two-tail)}$
One proportion	$z = (p - p_0) / \sqrt{p_0 q_0 / n}$
Two proportions	$z = (p_1 - p_2) / \sqrt{\bar{p} \bar{q} (1/n_1 + 1/n_2)}$ with pooled $\bar{p}$
Two means, small	$t = (\bar{x}_1 - \bar{x}_2) / (s_p \sqrt{1/n_1 + 1/n_2})$ with df $n_1 + n_2 - 2$; $s_p^2 =$ pooled sample variance
Chi-square GOF	$\chi^2 = \sum (O - E)^2 / E$; df = cells - 1 - parameters estimated
p-value	P(test statistic at least as extreme as observed H_0 true); reject $\Leftrightarrow p < \alpha$

Symbols: Σ sums run over the data of the problem at hand; $z_{\alpha/2}$, $t_{\alpha}(\text{df})$ and $\chi^2_{\alpha}(\text{df})$ critical values are in Appendix B; “cell” means one entry of a joint or contingency table.

Appendix B — Statistical Tables, Computed for This Book

Every value in this appendix was produced by `src/gen_tables.py` using `scipy` on this machine — the book never transcribes a printed table; any disagreement with a printed appendix is pure rounding. Distributions: `norm`, `t`, `chi2` from `scipy.stats`; binomial coefficients from `math.comb`.

B.1 · Standard normal cumulative probabilities $\Phi(z) = P(Z \leq z)$

z	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

B.2 · The five z-values you will actually be asked for

tail prob (two-tail)	tail prob (one-tail)	z
0.20	0.10	1.2816
0.10	0.05	1.6449

tail prob (two-tail)	tail prob (one-tail)	z
0.05	0.03	1.9600
0.02	0.01	2.3263
0.01	0.01	2.5758

B.3 · Student's t — two-tail critical values $t_{\alpha}(df)$

df	80%	90%	95%	98%	99%
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554
9	1.3830	1.8331	2.2622	2.8214	3.2498
10	1.3722	1.8125	2.2281	2.7638	3.1693
11	1.3634	1.7959	2.2010	2.7181	3.1058
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123
14	1.3450	1.7613	2.1448	2.6245	2.9768
15	1.3406	1.7531	2.1314	2.6025	2.9467
16	1.3368	1.7459	2.1199	2.5835	2.9208
17	1.3334	1.7396	2.1098	2.5669	2.8982
18	1.3304	1.7341	2.1009	2.5524	2.8784
19	1.3277	1.7291	2.0930	2.5395	2.8609
20	1.3253	1.7247	2.0860	2.5280	2.8453
21	1.3232	1.7207	2.0796	2.5176	2.8314
22	1.3212	1.7171	2.0739	2.5083	2.8188
23	1.3195	1.7139	2.0687	2.4999	2.8073
24	1.3178	1.7109	2.0639	2.4922	2.7969
25	1.3163	1.7081	2.0595	2.4851	2.7874
26	1.3150	1.7056	2.0555	2.4786	2.7787
27	1.3137	1.7033	2.0518	2.4727	2.7707
28	1.3125	1.7011	2.0484	2.4671	2.7633
29	1.3114	1.6991	2.0452	2.4620	2.7564
30	1.3104	1.6973	2.0423	2.4573	2.7500
40	1.3031	1.6839	2.0211	2.4233	2.7045
60	1.2958	1.6706	2.0003	2.3901	2.6603
120	1.2886	1.6577	1.9799	2.3578	2.6174

two-tail critical values $t_{\alpha}(df)$

B.4 · Chi-square — upper-tail points

df	.995	.99	.975	.95	.05	.025	.01	.005
1	7.8794	6.6349	5.0239	3.8415	0.0039	0.0010	0.0002	0.0000
2	10.5966	9.2103	7.3778	5.9915	0.1026	0.0506	0.0201	0.0100
3	12.8382	11.3449	9.3484	7.8147	0.3518	0.2158	0.1148	0.0717
4	14.8603	13.2767	11.1433	9.4877	0.7107	0.4844	0.2971	0.2070

df	.995	.99	.975	.95	.05	.025	.01	.005
5	16.7496	15.0863	12.8325	11.0705	1.1455	0.8312	0.5543	0.4117
6	18.5476	16.8119	14.4494	12.5916	1.6354	1.2373	0.8721	0.6757
7	20.2777	18.4753	16.0128	14.0671	2.1673	1.6899	1.2390	0.9893
8	21.9550	20.0902	17.5345	15.5073	2.7326	2.1797	1.6465	1.3444
9	23.5894	21.6660	19.0228	16.9190	3.3251	2.7004	2.0879	1.7349
10	25.1882	23.2093	20.4832	18.3070	3.9403	3.2470	2.5582	2.1559
11	26.7568	24.7250	21.9200	19.6751	4.5748	3.8157	3.0535	2.6032
12	28.2995	26.2170	23.3367	21.0261	5.2260	4.4038	3.5706	3.0738
13	29.8195	27.6882	24.7356	22.3620	5.8919	5.0088	4.1069	3.5650
14	31.3193	29.1412	26.1189	23.6848	6.5706	5.6287	4.6604	4.0747
15	32.8013	30.5779	27.4884	24.9958	7.2609	6.2621	5.2293	4.6009
16	34.2672	31.9999	28.8454	26.2962	7.9616	6.9077	5.8122	5.1422
17	35.7185	33.4087	30.1910	27.5871	8.6718	7.5642	6.4078	5.6972
18	37.1565	34.8053	31.5264	28.8693	9.3905	8.2307	7.0149	6.2648
19	38.5823	36.1909	32.8523	30.1435	10.1170	8.9065	7.6327	6.8440
20	39.9968	37.5662	34.1696	31.4104	10.8508	9.5908	8.2604	7.4338
21	41.4011	38.9322	35.4789	32.6706	11.5913	10.2829	8.8972	8.0337
22	42.7957	40.2894	36.7807	33.9244	12.3380	10.9823	9.5425	8.6427
23	44.1813	41.6384	38.0756	35.1725	13.0905	11.6886	10.1957	9.2604
24	45.5585	42.9798	39.3641	36.4150	13.8484	12.4012	10.8564	9.8862
25	46.9279	44.3141	40.6465	37.6525	14.6114	13.1197	11.5240	10.5197
26	48.2899	45.6417	41.9232	38.8851	15.3792	13.8439	12.1981	11.1602
27	49.6449	46.9629	43.1945	40.1133	16.1514	14.5734	12.8785	11.8076
28	50.9934	48.2782	44.4608	41.3371	16.9279	15.3079	13.5647	12.4613
29	52.3356	49.5879	45.7223	42.5570	17.7084	16.0471	14.2565	13.1211
30	53.6720	50.8922	46.9792	43.7730	18.4927	16.7908	14.9535	13.7867
40	66.7660	63.6907	59.3417	55.7585	26.5093	24.4330	22.1643	20.7065
50	79.4900	76.1539	71.4202	67.5048	34.7643	32.3574	29.7067	27.9907

upper-tail points: $P(X^2 > \text{value}) = 1 - q$

Read as: the entry under heading .025 in row 9 satisfies $P(\chi^2(9) > \text{value}) = 0.025$. The goodness-of-fit test (Module 5, §5.5) uses the .05 column.

B.5 · Binomial coefficients up to $n = 11$

$n \setminus k$	0	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1											
2	1	2	1										
3	1	3	3	1									
4	1	4	6	4	1								
5	1	5	10	10	5	1							
6	1	6	15	20	15	6	1						
7	1	7	21	35	35	21	7	1					
8	1	8	28	56	70	56	28	8	1				
9	1	9	36	84	126	126	84	36	9	1			
10	1	10	45	120	210	252	210	120	45	10	1		
11	1	11	55	165	330	462	462	330	165	55	11	1	
12	1	12	66	220	495	792	924	792	495	220	66	12	1

For larger n the book computes $\text{math.comb}(n, x) \cdot p^x \cdot q^{(n-x)}$ directly (see the logs of Sheet 8); never round a coefficient by hand when a table is out of range.

B.6 · Generation log (verbatim)

The run that produced every cell above:

```
computing statistical tables with scipy 1.17.1 and math.comb
z table:      35 rows x 10 cols  (Phi(1.96) = 0.9750)
t table:      33 df rows x 5 alpha cols
chi-square:   32 df rows x 8 tail cols
binom coeffs: n = 1..12
key quantile: z_(one-tail 0.100) = 1.2816
key quantile: z_(one-tail 0.050) = 1.6449
key quantile: z_(one-tail 0.025) = 1.9600
key quantile: z_(one-tail 0.010) = 2.3263
wrote out/tables.json
```

Appendix C — GATE-Style Practice Set (CCA Activity 2)

Twelve single-correct questions in the GATE pattern, pitched at the syllabus as printed; the CCA rubric prices ten of them at 1 mark each. The options, and the key, are machine-generated with the set itself ([src/gate.py](#)) — the numbers below are the values that script computed, not the values a printed key would quote. Attempt all twelve closed-book, against the clock: twelve minutes.

Question

- 1 The remainder when 7^{100} is divided by 13 is
(a) 3 (b) 9 (c) 11 (d) 12
- 2 The solution of $3x \equiv 5 \pmod{11}$ is $x \equiv$
(a) 4 (b) 9 (c) 7 (d) 10
- 3 The number of solutions of $x^2 \equiv 1 \pmod{105}$ is
(a) 2 (b) 4 (c) 6 (d) 8
- 4 RSA with $p = 5$, $q = 11$, $e = 3$: the ciphertext $C = 12$ decrypts to the plaintext
(a) 7 (b) 23 (c) 27 (d) 12
- 5 The number of solutions of $3x + 5y = 100$ in positive integers is
(a) 5 (b) 6 (c) 12 (d) 20
- 6 By Wilson's theorem, $12! \pmod{13}$ equals
(a) 0 (b) 1 (c) 11 (d) 12
- 7 $X \sim \text{Binomial}(100, 0.02)$. The exact $P(X \leq 1)$ is (Poisson(2) approximation shown for contrast)
(a) 0.1326 (b) 0.2706 (c) 0.4060 (d) 0.4033
- 8 $P(|Z| < 1.96)$ for a standard normal variable equals
(a) 0.9000 (b) 0.9500 (c) 0.9750 (d) 0.9900
- 9 For data lying exactly on the line $y = 3 - 2x$, the correlation coefficient r is
(a) -2 (b) -1 (c) 0 (d) 1
- 10 Two regression lines have slopes $b_{yx} = 0.8$ and $b_{xy} = 0.2$. The angle between them is
(a) 21.80° (b) 33.69° (c) 40.03° (d) 49.97°
- For the two-state chain with $p_{11} = 0.9$ and $p_{21} = 0.2$ (rows summing to 1), the long-run probability
- 11 of state 1 is
(a) 0.1111 (b) 0.2000 (c) 0.3333 (d) 0.6667
- A goodness-of-fit test with 5 cells and no estimated parameters is carried out at how many degrees
- 12 of freedom?
(a) 3 (b) 5 (c) 0 (d) 4

Answer key, with the one-line reason

- | # | Key | Why |
|----|-----|--|
| 1 | B | $100 \bmod 12 = 4$; $7^4 \equiv 9 \pmod{13}$ — Fermat, not brute force |
| 2 | B | $3^{-1} \equiv 4 \pmod{11}$; $x = 4 \cdot 5 \equiv 9$ |
| 3 | D | $105 = 3 \cdot 5 \cdot 7$; two roots per prime factor $\rightarrow 2^3 = 8$ |
| 4 | B | $\varphi = 40$, $d = 27$; $M = 12^{27} \bmod 55 = 23$, coprime to 55 — a valid block |
| 5 | B | $y \equiv 2 \pmod{3}$ with $1 \leq y \leq 17$ keeps $x > 0$: 6 pairs; $y = 20$ forces $x = 0$, not positive |
| 6 | D | $(p-1)! \equiv -1 \equiv 12 \pmod{p}$, $p = 13$ |
| 7 | D | Exact binomial 0.4033 vs Poisson 0.4060: when $n = 100$ is given, choose the exact value |
| 8 | B | $\Phi(1.96) - \Phi(-1.96) = 0.9500$ — the confidence level behind every 95% interval |
| 9 | B | Perfect fit, negative slope $\rightarrow r = -1$; $ r = 1$, sign from the slope |
| 10 | C | $\tan \theta = (1 - 0.16)/(0.8 + 0.2) = 0.84 \rightarrow \theta = 40.03^\circ$ |
| 11 | D | $\pi_1 = 0.9\pi_1 + 0.2\pi_2 \rightarrow 0.1\pi_1 = 0.2\pi_2 \rightarrow \text{ratio } 2 : 1 \rightarrow \pi_1 = 0.6667$ |
| 12 | D | $df = 5 - 1 - 0 = 4$ |

C.2 · Generation log (verbatim)

Every option value and every letter of the key, as computed:

GATE-STYLE SET (computed)

```

-----
Q1  7^100 mod 13      = 9
Q2  3x=5 (mod 11): x  = 9

```

```

Q3 #  $x^2=1 \pmod{105}$  = 8 ( $105=3*5*7 \rightarrow 2^3$ )
Q4 RSA  $n=55$   $e=3$   $C=12$  :  $d = 27$ ,  $M = 23$ 
Q5  $3x+5y=100$ ,  $x>0, y>0$  = 6 solutions
Q6  $12! \pmod{13}$  = 12
Q7  $P(X \leq 1)$  binom = 0.4033 poisson(2) = 0.4060
Q8  $P(|Z| < 1.96)$  = 0.9500
Q9  $r$  on an exact line  $y=3-2x$  = -1
Q10 angle between lines = 40.03 deg ( $\tan = 0.84$ )
Q11 steady-state  $P(A)$  = 0.6667 ( $2/3$ )
Q12 dof =  $5 - 1$  = 4
answer key: B B D B B D D B B C C D
[gate tokens saved]

```

Appendix D — Reproducing Every Number in This Book

The book is a build artifact of a machine-verified fact bank: every number printed in Parts A and B and every cell of Appendix B is looked up at build time from the tokens the computation scripts emit. Nothing is a hand-typed figure: the text carries placeholders such as `{{m5.z7}}` that the builder fills from `out/tokens.json`. If a script changes, the affected numbers change on the next build.

D.1 · The pipeline

Command	What it does
<code>python3 src/collect.py</code>	Runs every script in <code>src/</code> (21 computations plus the table generator), captures each terminal transcript to <code>out/<stem>.txt</code> , and rebuilds <code>out/tokens.json</code> from the emitted tokens
<code>python3 build.py</code>	Fills tokens and tables into the HTML, renders the math images (<code>eqmath.py</code> → <code>build/eq/</code>), runs WeasyPrint twice, stamps roman/arabic footers, writes the PDF
<code>python3 qa.py</code>	Post-build audit: page count, TOC/page-map consistency, orphan-placeholder scan (an unfilled <code>{{token}}</code> fails the build), verbatim-syllabus spot checks, table-cell counts

D.2 · Where each part of the book comes from

Book content	Source
Modules 1–5 worked answers	<code>src/m1_modular.py ... m5_inference.py</code> → <code>out/m*.txt</code>
Sheets 1–14 solved & practice answers	<code>src/t01.py ... t14.py</code> → <code>out/t*.txt</code>
Prose, checkpoints, exercises	<code>content_m*.py</code> , <code>content_tut.py</code>
Appendix B tables	<code>src/gen_tables.py</code> → <code>out/tables.json</code>
GATE set and key	<code>src/gate.py</code>
Syllabus, COs, rubrics (verbatim)	<code>content_syllabus.py</code> – transcribed from the VTU 2025 scheme document for 1BMATCS301

D.3 · A one-command honesty check

After a build, search `build/book.html` for `{{` — any hit is an unfilled placeholder, which the builder is written to reject loudly. To verify a printed value, find its caption in the matching transcript (for instance “CK2” in `out/m2_stats.txt`) and compare — book and log must agree, because the builder pastes transcripts verbatim and draws every number from the same bank.

D.4 · Study route through this book

Read the module chapter, close it, redo its exercises on paper; then open the matching tutorial sheet and redo its solved items — the computation log sits under every answer, so a stuck step is one glance away. Finish with the checkpoint questions and the GATE set before the IA tests; the compliance map in the front matter says which sheet drills which module.

Environment used for this edition: Python 3.13 with `sympy`, `scipy`, `numpy`, `matplotlib` (math rendering) and WeasyPrint 70 (layout). A future `scipy` release can move a fourth decimal in a table — that is the point: the book prints what the machine says, not what a 1985 appendix says.

— end of the book —